Structure-Preserving Modeling, Complex-Representation Hamiltonian Neural Networks, and KAM Stability Analysis for Magnetically Actuated Flexible Catheters

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Abstract: This study develops a unified framework for the three-dimensional large-deformation dynamics of a magnetically actuated flexible catheter subjected to magnetic actuation, background flow, and vessel-wall contact in microgravity. The continuous model treats the centerline position and cross-sectional orientation as independent variables and retains extension, shear, bending, torsion, and rotary inertia. Energy and power identities are used to distinguish conservative effects, passive dissipation, and external inputs. A joint translational–rotational Galerkin discretization, local reduction on the constraint manifold, and a consistent Legendre transformation yield a finite-dimensional canonical Hamiltonian system together with a mapping from COMSOL field variables to canonical states, momenta, and generalized forces. The learning study uses 96 prospective COMSOL trajectories partitioned by whole trajectory into 64 training, 16 validation, and 16 independent-test trajectories. Four models are trained for 300 epochs with five random seeds each, and model selection uses validation data only. On the independent test set, Gray CHNN attains a normalized full-state one-step MSE of 1.62675×10^{-3} and an 800-step rollout state error of 4.57008, corresponding to reductions of approximately 98.20% and 99.07%, respectively, relative to the best Real HNN baseline under the same experimental protocol. This establishes an empirical advantage for the conservative task considered here, not a universal performance claim. For local stability of the conservative system, recentering the material-reference Hamiltonian and performing a fourth-order Birkhoff normalization give a full-rank twist matrix for the selected quartic conservative approximation. Together with the analytic local canonical transformation and the scaled $O(\varepsilon)$ perturbation, the classical Kolmogorov KAM conditions imply a local positive-measure Cantor family of Lagrangian invariant tori at sufficiently small nonzero amplitudes. This conclusion is restricted to the quartic approximation and does not constitute a rigorous KAM theorem for the full material-reference Hamiltonian or a numerical certification of a specific torus. Lean4 is additionally used to verify several finite-dimensional structural identities of the open-port formulation, including the real-valued Hamiltonian output, port-power decomposition, nonpositive passive dissipative power, exclusive accounting of magnetic actuation, and the explicit nonautonomous $\partial_t H$ contribution.

To examine the mechanical consistency between the engineering finite-element discretization and the canonical Hamiltonian representation, reduced inertia operators are also constructed independently from the scalar kinetic energy and from the solver-assembled equations. The kinetic-energy Hessian is symmetric positive definite and satisfies the canonical velocity relation of the physical Hamiltonian. Two independent extractions of the solver inertial operator agree to relative error below 4×10^{-17}, demonstrating strong internal consistency of that representation. Nevertheless, in two independent reduced subspaces the solver inertia differs from the kinetic-energy metric by approximately 2.21% and 2.31%, with corresponding canonical-inertia errors of approximately 3.85% and 4.27%. The tested simple coordinate/rate-carrier alternatives and solver scaling are insufficient to account for the discrepancy, while the available interface does not expose sufficient information to reconstruct the test-space inertia mapping uniquely. In a reference variational discretization for which the trial functions, test functions, quadrature, and kinetic energy are all explicit, the alternative inertia representations agree to approximately 10^{-16}. The remaining discrepancy is therefore more plausibly associated with an unresolved mapping between the engineering discretization and the canonical representation than with the Hamiltonian construction itself. Under this information limitation, the present study does

not report corrected open-port force labels, retraining based on such labels, or final open-port predictive performance.

Keywords: magnetically actuated flexible catheter; geometrically exact rod; Hamiltonian neural network; complex-valued neural network; symplectic algorithm; KAM theory

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Introduction

Magnetically actuated flexible catheters in microgravity are simultaneously subjected to magnetic actuation, background flow, vessel-wall contact, and boundary manipulation. Their dynamics therefore involve large rotations, strong geometric nonlinearity, bending–torsion coupling, and energy exchange with external inputs. If the continuous model, finite-dimensional reduction, data representation, and long-time stability analysis use inconsistent state variables or energy definitions, physical modeling errors, learning errors, and numerical discretization errors cannot be assessed within a common framework. A consistent mathematical description is therefore needed across continuum mechanics, structure-preserving reduction, and verifiable machine learning.

Geometrically exact beam and rod theories provide a unified kinematic and variational basis for three-dimensional flexible structures undergoing large deformation, while magnetic, fluid, and contact models describe the surrounding multiphysics environment[1–18]. To preserve the required mechanical structure after reduction, translational and rotational degrees of freedom must be retained consistently, conservative loads must be separated from dissipation and external work at the energy level, and the weak form, Galerkin discretization, constraint reduction, and Legendre transformation must use compatible variables.

Hamiltonian neural networks introduce conservative structure through the symplectic gradient of a real-valued energy function. Dissipative and port-based extensions allow forced and dissipative dynamics to be modeled in a structured manner, while complex-valued networks can provide useful representations for amplitude–phase coupling[19–24]. A complex internal representation, however, does not imply that the physical Hamiltonian should be complex-valued, nor can a single complex scalar represent dissipation and external forcing without additional structure. Likewise, a multi-component architecture is not identifiable by construction; its

physical components must be assessed through known physics, port-power information, paired operating conditions, and ablation studies.

KAM theory and symplectic numerical methods provide rigorous tools for quasiperiodic dynamics in nearly integrable Hamiltonian systems and their discrete maps[25–33]. Continuous-time KAM theory, KAM results for symplectic maps, two-dimensional twist-map theory, and infinite-dimensional KAM theory concern different mathematical objects. Neither a training loss nor a trajectory residual can replace the smallness conditions required in higher-order or analytic norms. Learning errors must therefore be considered together with physical perturbations, coordinate changes, Galerkin truncation, and time-discretization errors in compatible norms.

The main contributions of this work are as follows.

(1) A multiphysics geometrically exact catheter model is formulated with centerline translation, independent cross-sectional orientation, torsion, and rotary inertia. Magnetic, fluid, contact, and boundary effects are organized consistently through energy and power identities.

(2) A joint translational–rotational Galerkin discretization, local reduction of the Kirchhoff constraint, and a consistent Legendre transformation are used to construct a finite-dimensional canonical Hamiltonian system compatible with COMSOL field data.

(3) A structured neural model is developed in which complex-valued features enhance the representation while the physical Hamiltonian remains real-valued. Conservative corrections, passive dissipation, external inputs, and unmodeled residuals are represented separately.

(4) Statistical learning error, higher-order regularity error, symplectic-discretization error, and KAM assumptions are linked within a common analysis that distinguishes continuous Hamiltonian flows, symplectic maps, and a posteriori invariant-torus verification.

(5) The relation between the discrete inertia represented by an engineering finite-element solver and the inertia implied by the canonical Hamiltonian formulation is examined directly. The solver inertial operator can be extracted reproducibly, but a stable discrepancy remains between the assembled DAE inertia and the inertia metric obtained from scalar kinetic energy. The tested simple coordinate/rate-carrier alternatives and solver scaling are insufficient to explain the discrepancy. A reference discretization with fully explicit variational ingredients recovers agreement at machine precision, indicating that the unresolved issue lies in the mapping between the engineering discretization and the canonical representation, together with the limited observability of that mapping, rather than in the Hamiltonian construction itself.

The numerical study proceeds through a low-dimensional P0 architecture comparison, three-trajectory P1/P1.5/P1.6 ablations, validation of the COMSOL–Galerkin mapping, consistency checks for canonical states, momenta, and generalized forces, and a formal conservative learning task based on 96 prospective COMSOL trajectories. Three time-step levels and an extended 1 s case are used to assess the numerical consistency of the state, momentum, and force mappings. The 96 trajectories are then partitioned by whole trajectory into 64/16/16 training, validation, and independent-test sets, and four models are trained with five algorithmic seeds each for 300 epochs. Candidate models are selected using validation data only, and the independent test set is evaluated once after selection. Gray CHNN substantially outperforms the three trainable baselines on this fixed conservative task, but the result is not extrapolated to open-port conditions. The open-port analysis instead addresses the equivalence of the solver-discrete and canonical inertia representations and identifies an observability limitation in the test-space mapping. High-fidelity model comparison and finite-domain numerical KAM certification remain topics for future work.

The remainder of the paper is organized as follows. Sections 1 and 2 develop the continuous multiphysics model and its joint Galerkin discretization. Section 3 derives the finite-dimensional Hamiltonian structure and the canonical state mapping. Section 4 presents the complex-representation-enhanced structured HNN and its error analysis. Section 5 discusses symplectic discretization, Birkhoff normal forms, and invariant-torus analysis. Numerical results are reported in Section 6; limitations and scope are discussed in Section 7; and Section 8 summarizes the main findings.

1. Geometrically Exact Bending–Torsion Dynamics of a Flexible Catheter under Multiphysics Loading

1.1. **Modeling framework.** Consider the three-dimensional large-deformation motion of a magnetically actuated flexible catheter in a space-station microgravity environment under the combined action of magnetic fields, fluid flow, and vessel-wall contact. To retain the complete mechanical information required for the subsequent finite-dimensional Hamiltonian structure, complex-representation Hamiltonian neural network, and KAM stability analysis, the centerline position and cross-sectional orientation are treated as independent primary fields. The model retains axial extension, transverse shear, bending about two directions, independent torsion, cross-sectional rotary inertia, and externally distributed couples.

The continuous model is first formulated within a geometrically exact Cosserat-rod framework. For a slender catheter, a Kirchhoff shear-free constraint may subsequently be imposed; this constraint removes transverse shear strain but does not eliminate cross-sectional orientation, torsional degrees of freedom, angular velocity, or rotary inertia. The complete translational–rotational model is therefore not replaced by a centerline-only field equation.[1, 2, 17]

Let the reference material coordinate be

$$S \in [0, L], \qquad t \in [0, T], \tag{1.1}$$

where L is the catheter length in the reference configuration. Unless otherwise stated, all cross-sectional parameters may vary with S to represent nonuniform geometry, spatially varying material properties, or localized magnetic components.

Magnetic, fluid, and contact effects enter the rod equations as distributed forces and couples per unit reference length. A strictly coupled multiphysics PDE system would require simultaneous solution of the electromagnetic, fluid, and structural field equations. The formulation developed in this section is instead a geometrically exact rod-dynamics model driven by multiphysics load inputs.

1.1.1. *Model hierarchy and computational branches.* The overall route is as follows: a continuous model is constructed from Kirchhoff/geometrically exact rod mechanics, Maxwell magnetic actuation, Morison-type fluid loading, and catheter–vessel-wall contact; Galerkin discretization yields finite-dimensional dynamics; and the resulting system supports a Hamiltonian backbone, a complex-representation-enhanced structured Hamiltonian neural network, and KAM/structure-preserving stability analysis. To avoid conflating the continuous parent model, the COMSOL training model, the learned structured model, and the KAM analysis object, the following four-level hierarchy is used throughout.

1. **Geometrically exact physical parent model.** The coupled translational–cross-sectional-orientation Cosserat rod, independent torsion, rotary inertia, distributed forces, and distributed couples are retained. This level preserves the complete mechanical information and provides the common energy and virtual-work basis for all reduced branches[1, 2].
2. **Strict Kirchhoff theoretical branch.** For slender, low-shear catheters, the shear-free constraint is imposed without removing cross-sectional orientation, torsion, angular velocity, or rotary inertia. Before CHNN or KAM analysis, the constrained finite-dimensional system must be expressed in independent local coordinates on the constraint manifold or approximated by a regularized model whose convergence to the constrained system has been verified. A differential–algebraic system with Lagrange multipliers must not be treated directly as a canonical Hamiltonian system on the full coordinate space.
3. **COMSOL Kirchhoff-consistent training branch.** The primary training data preferentially use the three-dimensional Euler–Bernoulli Beam interface with geometric nonlinearity, dynamic inertia, and torsional degrees of freedom, which is the closest built-in COMSOL realization of the Kirchhoff shear-free assumptions. Because this interface permits large displacements and rotations while retaining a small-strain, element-local linearization, it is referred to as a "Kirchhoff-consistent engineering model rather than

being identified unconditionally with the geometrically exact parent model[4–6]. When term-by-term reproduction of the coupled weak form is required, custom Weak Form PDE or Weak Contribution equations are used[7, 8].

4. **High-fidelity validation and open-port branch.** Timoshenko beams, three-dimensional Solid Mechanics, or a custom Cosserat weak form are used to assess shear effects, local cross-sectional effects, and model discrepancy. Magnetic control, background-flow input, passive drag, contact dissipation, friction, and boundary actuation enter the finite-dimensional model through mutually exclusive power ports. High-fidelity data are used for external validation and model-discrepancy learning and are not mixed with conservative Kirchhoff samples without explicit labels. The complete open system is used for structured CHNN identification and safe-domain computation; only the autonomous conservative backbone obtained by closing the ports, or an independently proven exact symplectic extension, is a direct object of classical KAM theory.

Accordingly, in addition to the continuous equations and Galerkin system, the first two sections provide Sections 3 to 5 with mutually consistent conservative energies, constraint-reduced states, port decompositions, model-level labels, and numerical-representation labels. All subsequent engineering reductions are performed within this common physical–mathematical framework.

1.2. **Geometry and kinematics.**

1.2.1. *Centerline and cross-sectional frame.* Let the catheter centerline be described by the position vector

$$\mathbf{r} = \mathbf{r}(S,t) \in \mathbb{R}^3. \tag{1.2}$$

Introduce a right-handed orthonormal frame on each cross section,

$$\{\mathbf{d}_1(S,t), \mathbf{d}_2(S,t), \mathbf{d}_3(S,t)\}, \tag{1.3}$$

and define the rotation matrix

$$\mathbf{R}(S,t) = [\mathbf{d}_1\ \mathbf{d}_2\ \mathbf{d}_3] \in SO(3), \tag{1.4}$$

which satisfies

$$\mathbf{R}^T\mathbf{R} = \mathbf{I}, \qquad \det \mathbf{R} = 1. \tag{1.5}$$

The full model does not impose $\mathbf{d}_3$ to be parallel to the centerline tangent; transverse shear is therefore admissible. The centerline stretch and unit tangent are defined, respectively, by

$$\lambda(S,t) = \| \mathbf{r}_{,S} \|, \qquad \mathbf{t}(S,t) = \frac{\mathbf{r}_{,S}}{\lambda}. \tag{1.6}$$

1.2.2. *Skew-symmetric operator.* For any $\mathbf{a} \in \mathbb{R}^3$, define the skew-symmetric matrix $\widehat{\mathbf{a}} \in \mathfrak{so}(3)$ such that[9, 17]

$$\widehat{\mathbf{a}}\,\mathbf{b} = \mathbf{a} \times \mathbf{b}, \qquad \forall \mathbf{b} \in \mathbb{R}^3. \tag{1.7}$$

Its inverse mapping is denoted by $(\cdot)^\vee$. Hence, both spatial and temporal derivatives of the rotation matrix can be represented by three-dimensional vectors.

1.2.3. *Shear–extension strain.* Define the shear–extension strain vector in the material frame as

$$\boldsymbol{\nu}(S,t) = \mathbf{R}^T \mathbf{r}_{,S} = \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix}. \tag{1.8}$$

where ν_1 and ν_2 are the two transverse shear-strain components and ν_3 is the axial extension component along the third director. The reference strain is denoted by

$$\boldsymbol{\nu}_0(S) = \begin{bmatrix} \nu_{01} \\ \nu_{02} \\ \nu_{03} \end{bmatrix}. \tag{1.9}$$

For a straight rod with no initial shear and no reference-length extension, one may take

$$\boldsymbol{\nu}_0 = (0,0,1)^T. \tag{1.10}$$

In the fully shearable Cosserat parent model, the axial material-strain component along the third director must be distinguished from the centerline stretch strain. The former is defined by

$$\varepsilon_3 = \nu_3 - \nu_{03}. \tag{1.11}$$

On the other hand, $\mathbf{r}_{,S} = \mathbf{R}\boldsymbol{\nu}$ together with $\mathbf{R}^T\mathbf{R} = \mathbf{I}$ gives $\lambda = \| \mathbf{r}_{,S} \| = \| \boldsymbol{\nu} \|$. The centerline extension strain is therefore defined as

$$\varepsilon_\lambda = \lambda - \lambda_0 = \| \boldsymbol{\nu} \| - \| \boldsymbol{\nu}_0 \|. \tag{1.12}$$

Under the Kirchhoff shear-free condition $\nu_1 = \nu_2 = 0$, if the reference configuration also satisfies $\nu_{03} = \lambda_0$, then $\varepsilon_3 = \varepsilon_\lambda$. The constitutive matrices used below remain formulated in terms of the material strain $\boldsymbol{\nu} - \boldsymbol{\nu}_0$.

1.2.4. *Curvature–torsion kinematics.* Define the curvature–torsion vector in the material frame by

$$\widehat{\boldsymbol{\kappa}} = \mathbf{R}^T \mathbf{R}_{,S}, \qquad \boldsymbol{\kappa} = \begin{bmatrix} \kappa_1 \\ \kappa_2 \\ \kappa_3 \end{bmatrix}. \tag{1.13}$$

where κ_1 and κ_2 are the bending curvatures about the two principal directions and κ_3 is the independent twist rate. The variation of the local frame along the reference coordinate satisfies

$$\mathbf{d}_{i,S} = (\mathbf{R}\boldsymbol{\kappa}) \times \mathbf{d}_i, \qquad i = 1,2,3. \tag{1.14}$$

The reference curvature–torsion vector is denoted by

$$\boldsymbol{\kappa}_0(S) = (\kappa_{01}, \kappa_{02}, \kappa_{03})^T. \tag{1.15}$$

1.2.5. *Cross-sectional angular velocity and compatibility.* Define the cross-sectional angular velocity in the material frame by

$$\widehat{\boldsymbol{\omega}} = \mathbf{R}^T \mathbf{R}_{,t}. \tag{1.16}$$

The corresponding angular velocity in the spatial frame is

$$\boldsymbol{\omega}_s = \mathbf{R}\boldsymbol{\omega}. \tag{1.17}$$

The identities $\mathbf{r}_{,St} = \mathbf{r}_{,tS}$ and $\mathbf{R}_{,St} = \mathbf{R}_{,tS}$ yield the kinematic compatibility relations[1, 2, 9]

$$\dot{\boldsymbol{\nu}} = \mathbf{R}^T \dot{\mathbf{r}}_{,S} - \boldsymbol{\omega} \times \boldsymbol{\nu}, \tag{1.18}$$

$$\dot{\boldsymbol{\kappa}} = \boldsymbol{\omega}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\omega}. \tag{1.19}$$

The second relation enforces compatibility between the spatial and temporal evolution of the cross-sectional orientation and is indispensable to the complete bending–torsion dynamics.

1.2.6. *Motion of material points within a cross section.* Let the in-plane material coordinate be

$$\boldsymbol{\xi} = \xi_1 \mathbf{e}_1 + \xi_2 \mathbf{e}_2, \tag{1.20}$$

The spatial position of a material point in the rod is then

$$\mathbf{x}\left(S, \boldsymbol{\xi}, t\right) = \mathbf{r}(S,t) + \mathbf{R}(S,t)\boldsymbol{\xi}. \tag{1.21}$$

and its velocity is

$$\dot{\mathbf{x}} = \dot{\mathbf{r}} + \mathbf{R}\left(\boldsymbol{\omega} \times \boldsymbol{\xi}\right) = \dot{\mathbf{r}} - \mathbf{R}\widehat{\boldsymbol{\xi}}\boldsymbol{\omega}. \tag{1.22}$$

This expression contains both centerline translation and cross-sectional rotational velocity and is the starting point for the complete kinetic energy and the configuration-dependent mass matrix.

1.3. **Complete kinetic energy, elastic potential, and constitutive relations.**

1.3.1. *Cross-sectional mass parameters and the mass-centerline convention.* Let the material density be $\rho\left(S, \boldsymbol{\xi}\right)$. Define the mass per unit reference length as

$$\mu(S) = \int_{A(S)} \rho \, dA, \tag{1.23}$$

Define the first mass moment of the cross section by

$$\mathbf{c}_\rho(S) = \int_{A(S)} \rho \boldsymbol{\xi} \, dA, \tag{1.24}$$

and the mass-inertia tensor per unit reference length in the material frame by

$$\mathbf{J}_c(S) = \int_{A(S)} \rho \left(\| \boldsymbol{\xi} \|^2 \mathbf{I} - \boldsymbol{\xi}\boldsymbol{\xi}^T \right) dA. \tag{1.25}$$

The preceding definitions retain a general cross-sectional mass distribution. To ensure that the subsequent momentum balance, weak form, and discrete mass matrix use a common inertial reference, the centerline $\mathbf{r}(S,t)$ is taken, from Section 1.4 onward, to pass through the mass center of each cross section. The distributed rod therefore satisfies

$$\boxed{\mathbf{c}_\rho(S) = \mathbf{0}.} \tag{1.26}$$

This convention removes neither cross-sectional rotational degrees of freedom nor rotary inertia. A magnetic tip, sensor, or other localized component whose center of mass is offset from the centerline is not absorbed into $\mathbf{c}_\rho$; it is introduced as an attached rigid body with its own coupled translational–rotational kinetic energy. This treatment preserves both closure of the base-rod equations and the physical inertial coupling of eccentric attachments.

If the principal cross-sectional axes coincide with $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, then

$$\mathbf{J}_c = diag\left(\rho I_1, \rho I_2, \rho I_3\right), \tag{1.27}$$

where I_1, I_2 are the area moments of inertia about the two principal directions and $I_3 = I_1 + I_2$ is the polar moment. For a nonuniform density distribution, the area-integral definitions must be used directly; the density cannot in general be factored out of the integrals.

1.3.2. *Complete structural kinetic energy.* The three-dimensional kinetic energy of the base rod follows from the material-point velocity as

$$T_{str} = \frac{1}{2}\int_0^L \int_{A(S)} \rho \left|\dot{\mathbf{r}} + \mathbf{R}\left(\boldsymbol{\omega}\times\boldsymbol{\xi}\right)\right|^2 dA\, dS. \tag{1.28}$$

After integration over the cross section, the general form becomes

$$T_{str} = \frac{1}{2}\int_0^L \left[\mu\left|\dot{\mathbf{r}}\right|^2 + 2\dot{\mathbf{r}}^T\mathbf{R}\left(\boldsymbol{\omega}\times\mathbf{c}_\rho\right) + \boldsymbol{\omega}^T\mathbf{J}_c\boldsymbol{\omega}\right] dS. \tag{1.29}$$

Under the mass-centerline convention of Section 1.3.1, $\mathbf{c}_\rho = \mathbf{0}$, and the kinetic energy of the base rod reduces to

$$\boxed{T_{str} = \frac{1}{2}\int_0^L \left[\mu\left|\dot{\mathbf{r}}\right|^2 + \boldsymbol{\omega}^T\mathbf{J}_c\boldsymbol{\omega}\right] dS.} \tag{1.30}$$

If eccentric attached components are present, their kinetic energy T_e is added separately and generally reintroduces translational–rotational cross terms. Regardless of whether the base rod uses a mass-centerline convention, the kinetic energy retains rotary inertia about both transverse axes and torsional inertia about the tangent direction. The model must not subsequently be reduced to a centerline-only field by deleting the rotational terms.

1.3.3. *Elastic strain-energy density.* Let the shear–extension and bending–torsion stiffness matrices be, respectively,

$$\mathbf{K}_\nu(S) = diag\left(k_1GA, k_2GA, EA\right), \tag{1.31}$$

$$\mathbf{K}_\kappa(S) = diag\left(EI_1, EI_2, GJ\right), \tag{1.32}$$

where E is Young's modulus, G is the shear modulus, k_1, k_2 are shear correction factors, and J is the cross-sectional torsional constant. For asymmetric materials or cross sections, $\mathbf{K}_\nu$ and $\mathbf{K}_\kappa$ may be general symmetric positive-definite matrices so that directional coupling is retained.

The elastic strain-energy density per unit reference length is defined by

$$W\left(\boldsymbol{\nu},\boldsymbol{\kappa};S\right) = \frac{1}{2}\left(\boldsymbol{\nu}-\boldsymbol{\nu}_0\right)^T\mathbf{K}_\nu\left(\boldsymbol{\nu}-\boldsymbol{\nu}_0\right) + \frac{1}{2}\left(\boldsymbol{\kappa}-\boldsymbol{\kappa}_0\right)^T\mathbf{K}_\kappa\left(\boldsymbol{\kappa}-\boldsymbol{\kappa}_0\right). \tag{1.33}$$

The internal elastic potential energy is

$$V_{int} = \int_0^L W\, dS. \tag{1.34}$$

1.3.4. *Constitutive laws for internal force and couple.* The resultant internal force and couple in the material frame are obtained from the gradient of the strain-energy density:

$$\mathbf{n}_b = \frac{\partial W}{\partial \boldsymbol{\nu}} = \mathbf{K}_\nu\left(\boldsymbol{\nu}-\boldsymbol{\nu}_0\right), \tag{1.35}$$

$$\mathbf{m}_b = \frac{\partial W}{\partial \boldsymbol{\kappa}} = \mathbf{K}_\kappa\left(\boldsymbol{\kappa}-\boldsymbol{\kappa}_0\right). \tag{1.36}$$

Their counterparts in the spatial frame are

$$\mathbf{n} = \mathbf{R}\mathbf{n}_b, \qquad \mathbf{m} = \mathbf{R}\mathbf{m}_b. \tag{1.37}$$

These constitutive equations retain extension, shear in both transverse directions, bending about two axes, and independent torsion. For a general nonlinear hyperelastic material, the quadratic form W may be replaced by the corresponding objective strain-energy density, with the constitutive resultants still obtained from its derivatives.

1.4. **Momentum balance and multiphysics loading.**

1.4.1. *Translational balance.* The continuous balance laws below use the mass centerline specified in Section 1.3.1 as the centerline of the base catheter. The linear-momentum balance per unit reference length is

$$\mu\ddot{\mathbf{r}} = \mathbf{n}_{,S} + \mathbf{f}_{ext}, \tag{1.38}$$

where

$$\mathbf{f}_{ext} = \mathbf{f}_{mag} + \mathbf{f}_{hydro} + \mathbf{f}_{cont} + \mathbf{f}_{g}. \tag{1.39}$$

All terms represent spatially distributed forces per unit reference length. The inertia of eccentric attachments is not contained implicitly in $\mu\ddot{\mathbf{r}}$; it is introduced consistently at the discrete level through their independent kinetic energy. Under the energy-consistent fluid formulation of Section 1.4.4, the added-inertia operator is placed on the left-hand side, while the fluid-acceleration input on the right-hand side contains only the pure Froude–Krylov term and drag. The complete engineering Morison branch and this theoretical branch must not be activated simultaneously.

1.4.2. *Rotational balance.* The angular-momentum balance about the centerline is written as

$$\mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \frac{\partial}{\partial t}\left(\mathbf{R}\mathbf{J}_{c}\boldsymbol{\omega}\right), \tag{1.40}$$

where

$$\mathbf{l}_{ext} = \mathbf{l}_{mag} + \mathbf{l}_{hydro} + \mathbf{l}_{cont}. \tag{1.41}$$

Using $\dot{\mathbf{R}} = \mathbf{R}\widehat{\boldsymbol{\omega}}$, the expanded form is

$$\mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \mathbf{R}\left[\mathbf{J}_{c}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \left(\mathbf{J}_{c}\boldsymbol{\omega}\right)\right]. \tag{1.42}$$

where $\boldsymbol{\omega} \times \left(\mathbf{J}_{c}\boldsymbol{\omega}\right)$ is the gyroscopic inertia term. It follows from the exact time derivative of angular momentum and is not an empirical correction that may be added or removed arbitrarily.

1.4.3. *Magnetic force and couple from the Maxwell stress tensor.* In the air or vacuum region surrounding an infinitesimal magnetic catheter segment, the Maxwell stress tensor is[10, 11]

$$\mathbf{T}_{M} = \frac{1}{\mu_{0}}\left(\mathbf{B} \otimes \mathbf{B} - \frac{1}{2}\left|\mathbf{B}\right|^{2}\mathbf{I}\right), \tag{1.43}$$

where $\mathbf{B}$ is the magnetic flux density and μ_0 is the permeability of free space. For a segment of reference length ΔS, the magnetic force per unit reference length and the magnetic couple about the centerline are defined by

$$\mathbf{f}_{mag} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \int_{\partial\Omega_{\Delta S}} \mathbf{T}_{M}\mathbf{n}_{A}\, dA, \tag{1.44}$$

$$\mathbf{l}_{mag} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \int_{\partial\Omega_{\Delta S}} \left(\mathbf{x} - \mathbf{r}\right) \times \left(\mathbf{T}_{M}\mathbf{n}_{A}\right)\, dA, \tag{1.45}$$

where $\mathbf{n}_A$ is the outward unit normal to the closed integration surface.

When a magnetic segment can be represented by a dipole approximation, let $\boldsymbol{\mu}_{0}^{b}(S)$ be the magnetic moment per unit reference length in the material frame. The corresponding spatial magnetic moment is[10, 11]

$$\boldsymbol{\mu}(S,t) = \mathbf{R}(S,t)\boldsymbol{\mu}_{0}^{b}(S). \tag{1.46}$$

For a prescribed external magnetic field $\mathbf{B}\left(\mathbf{r},t\right)$, the magnetic potential energy is

$$V_{mag} = -\int_{0}^{L} \boldsymbol{\mu} \cdot \mathbf{B}\left(\mathbf{r},t\right)\, dS. \tag{1.47}$$

The associated magnetic force and couple are

$$\mathbf{f}_{mag} = \nabla_{\mathbf{r}} (\boldsymbol{\mu} \cdot \mathbf{B}), \qquad \mathbf{l}_{mag} = \boldsymbol{\mu} \times \mathbf{B}. \tag{1.48}$$

The Maxwell-stress integral and the magnetic-dipole potential are two scale-dependent representations of the same magnetic action. One should be selected as the primary load model according to the required fidelity; they must not be superposed and counted twice.

1.4.4. *Modified Morison loading and energy-consistent added mass.* Let the fluid velocity be $\mathbf{v}_f(\mathbf{r}, t)$, with material acceleration

$$\mathbf{a}_f = \frac{\partial \mathbf{v}_f}{\partial t} + (\mathbf{v}_f \cdot \nabla)\, \mathbf{v}_f. \tag{1.49}$$

Define the fluid velocity relative to the catheter by

$$\mathbf{u}_r = \mathbf{v}_f - \dot{\mathbf{r}}. \tag{1.50}$$

If only the component normal to the current centerline is retained, define

$$\mathbf{P}_\perp = \mathbf{I} - \mathbf{t} \otimes \mathbf{t}, \qquad \mathbf{u}_{r\perp} = \mathbf{P}_\perp \mathbf{u}_r. \tag{1.51}$$

An engineering Morison validation branch may be written as[3, 12, 13]

$$\mathbf{f}^M_{hydro} = \mathbf{f}^M_{in} + \mathbf{f}^M_{add} + \mathbf{f}_{drag}, \tag{1.52}$$

where

$$\mathbf{f}^M_{in} = \rho_f C_M A_d \mathbf{P}_\perp \mathbf{a}_f, \tag{1.53}$$

$$\mathbf{f}^M_{add} = -\rho_f C_A A_d \mathbf{P}_\perp \ddot{\mathbf{r}}, \tag{1.54}$$

$$\mathbf{f}_{drag} = \frac{1}{2} \rho_f C_D D \parallel \mathbf{u}_{r\perp} \parallel \mathbf{u}_{r\perp}. \tag{1.55}$$

Here, ρ_f is the fluid density, A_d is the characteristic displaced-fluid area per unit length, D is the catheter outer diameter, and C_A, C_M, C_D are the added-mass, inertia, and drag coefficients, respectively; under a common convention, $C_M = 1 + C_A$. This empirical branch must be used as a complete formulation and must not be combined with the functional T_A introduced below.

To incorporate configuration-dependent added mass into the full inertial structure, the theoretical branch is instead defined through a velocity-dependent added-fluid inertia functional. Let[12, 13]

$$m_A(S) = \rho_f C_A A_d, \qquad \mathbf{w}_\perp = \mathbf{P}_\perp (\dot{\mathbf{r}} - \mathbf{v}_f), \tag{1.56}$$

and take

$$T_A[\mathbf{r}, \dot{\mathbf{r}}; t] = \frac{1}{2} \int_0^L m_A\, \mathbf{w}_\perp^T \mathbf{w}_\perp \, dS. \tag{1.57}$$

The corresponding inertial action is generated consistently by the functional Euler–Lagrange derivative

$$\mathbf{I}_A = \frac{d}{dt} \frac{\delta T_A}{\delta \dot{\mathbf{r}}} - \frac{\delta T_A}{\delta \mathbf{r}} \tag{1.58}$$

When $\mathbf{P}_\perp$ is held fixed, the leading term is

$$\mathbf{I}_A = \rho_f C_A A_d \mathbf{P}_\perp (\ddot{\mathbf{r}} - \mathbf{a}_f), \tag{1.59}$$

Variations of $\mathbf{P}_\perp$ with configuration and spatial variations of the prescribed flow generate additional geometric terms. When the theoretical branch based on T_A is used, the explicit fluid-acceleration input on the right-hand side retains only the pure Froude–Krylov term

$$\boxed{\mathbf{f}_{FK} = \rho_f A_d \mathbf{P}_\perp \mathbf{a}_f}, \tag{1.60}$$

with

$$\mathbf{f}_{hydro,rem} = \mathbf{f}_{FK} + \mathbf{f}_{drag}. \tag{1.61}$$

In this formulation, T_A already supplies the complete added-mass action associated with $C_A(\mathbf{a}_f - \ddot{\mathbf{r}})$. Together with the Froude–Krylov term having coefficient 1, it recovers $C_M = 1 + C_A$; consequently, $\mathbf{f}_{in}^M$ must no longer retain its $C_A\mathbf{a}_f$ term, and no additional $\mathbf{f}_{add}^M$ may be included. When $\mathbf{v}_f = \mathbf{0}$ and m_A is time independent, T_A reduces to an added kinetic energy that can be included in the conservative inertial backbone. For a general prescribed flow, its cross terms and explicit time dependence are treated collectively as a flow port.

For quadratic drag, define the nonnegative instantaneous state-dependent coefficient

$$d_f = \frac{1}{2}\rho_f C_D D \parallel \mathbf{u}_{r\perp} \parallel \geq 0. \tag{1.62}$$

The drag force must then be decomposed exclusively into a passive dissipative kernel and a background-flow input:

$$\boxed{\mathbf{f}_{drag} = d_f \mathbf{P}_\perp (\mathbf{v}_f - \dot{\mathbf{r}}) = \underbrace{-d_f \mathbf{P}_\perp \dot{\mathbf{r}}}_{\mathbf{f}_{drag,diss}} + \underbrace{d_f \mathbf{P}_\perp \mathbf{v}_f}_{\mathbf{f}_{drag,flow}}.} \tag{1.63}$$

Only the passive kernel has a definite dissipative sign with respect to the catheter mechanical energy:

$$\dot{\mathbf{r}}^T \mathbf{f}_{drag,diss} = -d_f \parallel \mathbf{P}_\perp \dot{\mathbf{r}} \parallel^2 \leq 0, \tag{1.64}$$

The term $\dot{\mathbf{r}}^T \mathbf{f}_{drag,flow}$ is generally sign-indefinite and is therefore assigned to the flow-input port. The complete drag force also satisfies the relative-motion passivity identity

$$(\dot{\mathbf{r}} - \mathbf{v}_f)^T \mathbf{f}_{drag} = -d_f \parallel \mathbf{P}_\perp (\dot{\mathbf{r}} - \mathbf{v}_f) \parallel^2 \leq 0, \tag{1.65}$$

This identity, however, does not replace the port accounting in the structural mechanical-energy balance. Thus, when $\mathbf{v}_f \neq \mathbf{0}$, the complete quadratic drag must not be assigned entirely to $\mathbf{Q}_{diss}$. Because d_f depends on the relative velocity, the passive kernel is a statewise positive-semidefinite nonlinear operator rather than a constant damping matrix.

If damping of cross-sectional spin by the fluid is relevant, an additional fluid couple per unit reference length may be introduced, for example,

$$\mathbf{l}_{hydro} = -c_\omega(S) \left|\omega_3 - \omega_{f3}\right| (\omega_3 - \omega_{f3}) \mathbf{d}_3, \tag{1.66}$$

where ω_{f3} is the effective fluid angular velocity about the catheter tangent. If this effect is negligible compared with the magnetic and structural torsional moments, one may set $\mathbf{l}_{hydro} = \mathbf{0}$ for the operating condition under consideration, while retaining the cross-sectional rotational degree of freedom.

The baseline computational route assumes a stationary vessel wall and Morison loading associated with normal cross-flow, neglecting axial tangential skin friction; the effective fluid spin about the catheter tangent is set to $\omega_{f3} = 0$. These are medium-fidelity reductions of the environmental loads and do not remove the structural variables describing cross-sectional orientation, independent torsion, angular velocity, or rotary inertia. Moving walls, tangential friction, or fluid spin can be incorporated later through additional ports following the same input–dissipation accounting.

1.4.5. *Catheter–vessel-wall contact force and couple.* Let $g(S,t)$ be the gap function between the catheter surface and the vessel wall, and let g_0 be the contact-detection threshold. Define the penetration measure

$$\delta_c = \max(g_0 - g, 0). \tag{1.67}$$

To represent Hertzian elasticity, normal-impact dissipation, and regularized friction simultaneously, let $\dot{\delta}_c$ denote the material time derivative of the penetration at the contact point. When contact is active, $\delta_c > 0$, the normal force is taken as[14–16]

$$f_n = k_c \delta_c^{\alpha} + c_n \dot{\delta}_c, \qquad \delta_c > 0. \tag{1.68}$$

If the preceding expression is negative or if $\delta_c = 0$, set $f_n = 0$. This nonnegative truncation prevents nonphysical tensile contact forces during separation or noncontact. The tangential friction force and total contact force are, respectively,

$$\mathbf{f}_t = -\mu_c f_n \frac{\mathbf{v}_t}{\sqrt{\mathbf{v}_t \cdot \mathbf{v}_t + v_0^2}}, \tag{1.69}$$

$$\mathbf{f}_{cont} = f_n \mathbf{n}_w + \mathbf{f}_t. \tag{1.70}$$

Here, k_c is the contact stiffness, c_n is the normal contact-damping coefficient, α is the contact exponent, and Hertzian contact commonly uses $\alpha = 3/2$. The vector $\mathbf{n}_w$ is the unit normal directed away from the wall, $\mathbf{v}_t$ is the tangential velocity of the contact point relative to the wall, μ_c is the friction coefficient, and v_0 is a regularization parameter. For automatic differentiation or a consistent Newton iteration, the contact activation and nonnegative truncation may be replaced by smooth regularizations. For conservative contact, one may set $c_n = \mu_c = 0$, in which case the elastic contact term is derivable from a potential.

If the contact force acts at an offset vector $\boldsymbol{\rho}_c$ from the centerline, the contact couple is

$$\mathbf{l}_{cont} = \boldsymbol{\rho}_c \times \mathbf{f}_{cont} + \mathbf{l}_{spin}, \tag{1.71}$$

where $\mathbf{l}_{spin}$ may represent additional damping or frictional torque associated with cross-sectional spin in the contact region. Normal damping, tangential friction, and $\mathbf{l}_{spin}$ are all included in the nonconservative power.

1.4.6. *Microgravity term.* If a residual acceleration $\mathbf{g}_{res}(t)$ is present in the space station, the residual gravitational force per unit reference length is

$$\mathbf{f}_g = \mu \mathbf{g}_{res}. \tag{1.72}$$

Under the ideal microgravity approximation, set

$$\mathbf{g}_{res} = \mathbf{0}. \tag{1.73}$$

1.4.7. *Complete strong-form system.* Collecting the kinematics, balance laws, and constitutive equations gives the geometrically exact system for the base rod:

$$\begin{cases} \mathbf{r}_{,S} = \mathbf{R}\boldsymbol{\nu}, \\ \mathbf{R}_{,S} = \mathbf{R}\widehat{\boldsymbol{\kappa}}, \\ \mathbf{R}_{,t} = \mathbf{R}\widehat{\boldsymbol{\omega}}, \\ \mu \ddot{\mathbf{r}} = \mathbf{n}_{,S} + \mathbf{f}_{ext}, \\ \mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \mathbf{R}\left[\mathbf{J}_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c \boldsymbol{\omega})\right], \\ \mathbf{n} = \mathbf{R}\mathbf{K}_{\nu}\left(\boldsymbol{\nu} - \boldsymbol{\nu}_0\right), \\ \mathbf{m} = \mathbf{R}\mathbf{K}_{\kappa}\left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0\right). \end{cases} \tag{1.74}$$

with the compatibility relations

$$\dot{\boldsymbol{\nu}} = \mathbf{R}^T \dot{\mathbf{r}}_{,S} - \boldsymbol{\omega} \times \boldsymbol{\nu}, \qquad \dot{\boldsymbol{\kappa}} = \boldsymbol{\omega}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\omega}. \tag{1.75}$$

Under the energy-consistent added-mass formulation, $\mathbf{I}_A$ is placed on the left-hand side together with the structural inertia, and only $\mathbf{f}_{FK}$, drag, and the remaining fluid ports are retained on the right-hand side. Under the engineering Morison validation branch, the complete combination $\mathbf{f}_{in}^M + \mathbf{f}_{add}^M$ is retained and T_A is omitted. Exactly one of these two representations must be used.

The primary unknowns are $\mathbf{r}, \mathbf{R}, \boldsymbol{\nu}, \boldsymbol{\kappa}, \boldsymbol{\omega}, \mathbf{n}, \mathbf{m}$. The system is a geometrically nonlinear mixed partial differential–algebraic system with Lie-group-valued orientation variables.

1.5. **Variational formulation and complete weak form.**

1.5.1. *Virtual displacement and virtual rotation.* Let $\delta\mathbf{r}$ be the virtual centerline displacement and represent the orientation variation by the material virtual rotation $\boldsymbol{\eta}$:

$$\delta\mathbf{R} = \mathbf{R}\widehat{\boldsymbol{\eta}}. \tag{1.76}$$

The corresponding spatial virtual rotation is

$$\boldsymbol{\eta}_s = \mathbf{R}\boldsymbol{\eta}. \tag{1.77}$$

The strain definitions give

$$\delta\boldsymbol{\nu} = \mathbf{R}^T\delta\mathbf{r}_{,S} - \boldsymbol{\eta} \times \boldsymbol{\nu}, \tag{1.78}$$

$$\delta\boldsymbol{\kappa} = \boldsymbol{\eta}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\eta}. \tag{1.79}$$

1.5.2. *Internal virtual work.* The internal virtual work is

$$\delta W_{int} = \int_0^L (\mathbf{n}_b \cdot \delta\boldsymbol{\nu} + \mathbf{m}_b \cdot \delta\boldsymbol{\kappa})\, dS. \tag{1.80}$$

Substituting the strain variations and integrating the spatial derivatives by parts gives

$$\delta W_{int} = [\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m}_b \cdot \boldsymbol{\eta}]_0^L - I_r - I_R, \tag{1.81}$$

where

$$I_r = \int_0^L \mathbf{n}_{,S} \cdot \delta\mathbf{r}\, dS, \tag{1.82}$$

$$I_R = \int_0^L (\mathbf{m}_{b,S} + \boldsymbol{\kappa} \times \mathbf{m}_b + \boldsymbol{\nu} \times \mathbf{n}_b) \cdot \boldsymbol{\eta}\, dS. \tag{1.83}$$

The material-form rotational balance term satisfies

$$\mathbf{m}_{b,S} + \boldsymbol{\kappa} \times \mathbf{m}_b + \boldsymbol{\nu} \times \mathbf{n}_b + \mathbf{l}_b = \mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega}), \tag{1.84}$$

and $\mathbf{l}_b = \mathbf{R}^T\mathbf{l}_{ext}$.

1.5.3. *Inertial and external virtual work.* The inertial virtual work of the base rod is

$$\delta W_{in,str} = \int_0^L [\mu\ddot{\mathbf{r}} \cdot \delta\mathbf{r} + (\mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega})) \cdot \boldsymbol{\eta}]\, dS. \tag{1.85}$$

If the energy-consistent added-mass formulation of Section 1.4.4 is used, the added-inertia virtual work is

$$\delta W_{in,A} = \left\langle \frac{d}{dt}\frac{\delta T_A}{\delta\dot{\mathbf{r}}} - \frac{\delta T_A}{\delta\mathbf{r}},\, \delta\mathbf{r} \right\rangle, \tag{1.86}$$

where $\langle\cdot,\cdot\rangle$ denotes the duality pairing along the rod. If eccentric attachments are present, their inertial virtual work is generated by the Euler–Lagrange derivative of the corresponding rigid-body kinetic energy T_e. The total inertial virtual work is therefore

$$\delta W_{in} = \delta W_{in,str} + \delta W_{in,A} + \delta W_{in,e}. \tag{1.87}$$

The virtual work of distributed external loads is

$$\delta W_{ext} = \int_0^L (\mathbf{f}_{ext,rem} \cdot \delta\mathbf{r} + \mathbf{l}_{ext} \cdot \boldsymbol{\eta}_s)\, dS + \delta W_{\partial}, \tag{1.88}$$

where $\mathbf{f}_{ext,rem}$ no longer contains the added inertia already represented by T_A. In the energy-consistent fluid branch, the fluid contribution contains only $\mathbf{f}_{FK}$, drag, and other inputs not represented by T_A. Because $\boldsymbol{\eta}_s = \mathbf{R}\boldsymbol{\eta}$, it may equivalently be written as

$$\mathbf{l}_{ext} \cdot \boldsymbol{\eta}_s = \mathbf{l}_b \cdot \boldsymbol{\eta}. \tag{1.89}$$

1.5.4. *Coupled weak form.* By D'Alembert's principle, every admissible pair $\delta\mathbf{r}$ and $\boldsymbol{\eta}$ satisfying the essential boundary conditions obeys

$$\delta W_{in} + \delta W_{int} - \delta W_{ext} = 0. \tag{1.90}$$

Expansion yields the complete coupled translational–rotational weak form:

$$\begin{gathered} \int_0^L \mu \ddot{\mathbf{r}} \cdot \delta\mathbf{r}\, dS + \int_0^L \left[\mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega})\right] \cdot \boldsymbol{\eta}\, dS \\ + \delta W_{in,A} + \delta W_{in,e} + \int_0^L \left(\mathbf{n}_b \cdot \delta\boldsymbol{\nu} + \mathbf{m}_b \cdot \delta\boldsymbol{\kappa}\right) dS \\ - \int_0^L \left(\mathbf{f}_{ext,rem} \cdot \delta\mathbf{r} + \mathbf{l}_b \cdot \boldsymbol{\eta}\right) dS - \delta W_\partial = 0. \end{gathered} \tag{1.91}$$

If neither the added-fluid inertia functional nor eccentric attachments are included, setting $\delta W_{in,A} = \delta W_{in,e} = 0$ recovers the base-rod weak form. Because the weak form contains only first spatial derivatives of the centerline and orientation fields, C^0-continuous approximation spaces for both fields are sufficient; no fourth-order centerline equation need be constructed by eliminating the rotational equation.

In functional terms, the centerline and orientation may be viewed as mappings in $\mathbf{r} \in H^1\left(0, L; \mathbb{R}^3\right)$ and $\mathbf{R} \in H^1\left(0, L; SO(3)\right)$, respectively. The virtual displacement $\delta\mathbf{r}$ and material virtual rotation $\boldsymbol{\eta}$ belong to the corresponding H^1 test spaces satisfying homogeneous essential boundary conditions. Thus, the coupled weak form requires only first-order spatial regularity.

1.5.5. *Boundary terms and boundary conditions.* Integration by parts produces the natural boundary term

$$\left[\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m} \cdot \boldsymbol{\eta}_s\right]_0^L. \tag{1.92}$$

At an endpoint $S = S_e$, one of the following two classes of boundary conditions may be prescribed:

$$\mathbf{r}\left(S_e, t\right) = \overline{\mathbf{r}}(t) \quad \text{or} \quad \mathbf{n}\left(S_e, t\right) = \overline{\mathbf{n}}(t), \tag{1.93}$$

$$\mathbf{R}\left(S_e, t\right) = \overline{\mathbf{R}}(t) \quad \text{or} \quad \mathbf{m}\left(S_e, t\right) = \overline{\mathbf{m}}(t). \tag{1.94}$$

At a clamped end, $\delta\mathbf{r} = \mathbf{0}$ and $\boldsymbol{\eta} = \mathbf{0}$; at a free end without applied terminal loads, the natural boundary conditions are $\mathbf{n} = \mathbf{0}$ and $\mathbf{m} = \mathbf{0}$. Boundary forces and moments must remain in the weak form unless the boundary conditions have been specified explicitly.

The complete initial–boundary-value problem also requires the initial conditions

$$\mathbf{r}(S, 0) = \mathbf{r}^0(S), \qquad \dot{\mathbf{r}}(S, 0) = \mathbf{v}^0(S), \tag{1.95}$$

$$\mathbf{R}(S, 0) = \mathbf{R}^0(S) \in SO(3), \qquad \boldsymbol{\omega}(S, 0) = \boldsymbol{\omega}^0(S). \tag{1.96}$$

The initial fields must satisfy the geometric compatibility relations $\boldsymbol{\nu}^0 = \left(\mathbf{R}^0\right)^T \mathbf{r}^0_{,S}$ and $\widehat{\boldsymbol{\kappa}}^0 = \left(\mathbf{R}^0\right)^T \mathbf{R}^0_{,S}$, and they must be consistent with the essential boundary conditions. The strict Kirchhoff branch additionally requires the initial position constraint and its velocity-level compatibility condition.

1.6. **Kirchhoff shear-free constraint.** For a slender catheter, impose the extensible Kirchhoff shear-free constraint

$$\nu_1 = 0, \qquad \nu_2 = 0, \qquad \nu_3 = \lambda. \tag{1.97}$$

which gives

$$\mathbf{r}_{,S} = \lambda \mathbf{d}_3, \qquad \mathbf{d}_3 = \mathbf{t}. \tag{1.98}$$

This condition aligns the third director with the centerline tangent but does not determine the rotation of $\mathbf{d}_1, \mathbf{d}_2$ about $\mathbf{d}_3$. Consequently, κ_3, ω_3, GJ, the polar rotary inertia, and the magnetic couple must all be retained.

In the strong form, the transverse shear force is the reaction enforcing the shear-free constraint and does not vanish merely because $\nu_1 = \nu_2 = 0$. The constraint may be imposed exactly with Lagrange multipliers or regularized by retaining a large but finite shear stiffness. In either case, the cross-sectional orientation field must remain in the model.[17, 18]

1.7. **Continuous energy balance.**

1.7.1. *Structural energy and extended inertial energy.* The mechanical energy of the base rod is defined as

$$E_{str} = T_{str} + V_{int}. \tag{1.99}$$

If eccentric attachments are present, add their kinetic energy T_e. In the energy-consistent added-mass branch with a stationary flow $\mathbf{v}_f = \mathbf{0}$ and time-independent m_A, define the total kinetic energy as

$$T_{tot} = T_{str} + T_A + T_e. \tag{1.100}$$

If magnetic loading, frictionless contact, or residual gravity is derivable from potentials, define

$$V = V_{int} + V_{mag} + V_{cont} + V_g, \tag{1.101}$$

and the extended mechanical energy

$$\boxed{E = T_{tot} + V.} \tag{1.102}$$

For a general prescribed flow, T_A may depend explicitly on time and contain cross terms involving relative velocity. It is then a velocity-dependent added-fluid inertia functional in the Lagrangian rather than the ordinary mechanical kinetic energy of a closed system. Its contribution must remain explicit as an independent flow port in the power balance; it does not imply conservation of the mechanical energy of the combined structure–prescribed-flow system.

1.7.2. *Power balance.* Taking the inner product of the translational equation with $\dot{\mathbf{r}}$, the rotational equation with the spatial angular velocity $\boldsymbol{\omega}_s$, and integrating along the rod gives

$$\frac{d}{dt}\left(T_{tot} + V_{int}\right) = P_\partial + P_{mag} + P_{flow} + P_{cont} + P_g, \tag{1.103}$$

where the boundary power is

$$P_\partial = \left[\mathbf{n} \cdot \dot{\mathbf{r}} + \mathbf{m} \cdot \boldsymbol{\omega}_s\right]_0^L. \tag{1.104}$$

After all conservative external loads have been incorporated into V, one obtains

$$\boxed{\frac{dE}{dt} = P_\partial + P_{diss} + P_{in} + P_{flow} + \frac{\partial V_{ext}}{\partial t}.} \tag{1.105}$$

Here, P_{diss} contains only the passive components obtained after the input–dissipation split, including $P_{drag,diss}$, normal contact damping and friction for a fixed wall, and any retained spin damping, and it satisfies $P_{diss} \leq 0$. The term P_{flow} collects the background-flow drag power $P_{drag,flow}$, the Froude–Krylov input, and reactive-flow power assigned to the port under

the selected representation; it is generally sign-indefinite. The term P_{in} contains only nonfluid nonconservative body inputs and control inputs. Terminal actuation contributes only through P_{∂}. If a nonautonomous magnetic action has already been represented by $V_{mag}(\mathbf{r}, \mathbf{R}, t)$, its explicit-time power appears only through $\partial V_{ext}/\partial t$ and must not be counted again in P_{in}. The extended mechanical energy is conserved only when $P_{diss} = P_{in} = P_{flow} = P_{\partial} = 0$ and the potential is time independent.

1.8. **Section summary.** With the centerline $\mathbf{r}$ and cross-sectional orientation $\mathbf{R}$ as independent primary fields, this section established a geometrically exact rod-dynamics model retaining extension, shear, bending about two axes, independent torsion, cross-sectional rotary inertia, magnetic couples, fluid loading, and contact couples. The base catheter uses a mass-centerline convention, whereas eccentric magnetic tips and sensors retain translational–rotational coupling through independent rigid-body kinetic energies. In the energy-consistent fluid representation, T_A generates the added inertia and the pure Froude–Krylov term supplies the remaining fluid-acceleration input; this branch is mutually exclusive with the engineering Morison representation, thereby avoiding double counting of $C_A\mathbf{a}_f$.

The complete weak form retains translational and rotational test functions as well as terminal force and moment terms. The passive kernel of quadratic drag, normal contact damping, and friction are assigned to dissipative power, while background-flow drag input and prescribed-flow inertia are assigned to an independent flow port. The Kirchhoff shear-free condition is enforced as a constraint rather than by deleting cross-sectional orientation, independent torsion, or rotary inertia. This coupled weak form provides the theoretical basis for the joint translational–rotational Galerkin discretization in Section 2.

2. Joint Translational–Rotational Galerkin Discretization

2.1. **Discretization principles.** Section 1 established the complete coupled weak form with $\mathbf{r}(S,t)$ and $\mathbf{R}(S,t) \in SO(3)$ as the primary fields. Rather than reducing it to a fourth-order centerline-only equation, this section performs a joint Galerkin discretization directly in the translational space and in a local coordinate chart of the rotation group.

The discretization follows five principles. First, the centerline position and cross-sectional orientation retain independent generalized coordinates. Second, rotation matrices cannot be superposed linearly as ordinary vectors and must be updated through the exponential map or an equivalent multiplicative rule. Third, the mass–inertia matrix must be derived from the complete three-dimensional material-point kinetic energy rather than assumed a priori to be constant or configuration dependent. Fourth, geometric inertial terms must follow rigorously from the Euler–Lagrange derivative of the discrete kinetic energy. Fifth, internal forces are obtained from the gradient of the discrete potential, while boundary forces and moments enter the generalized forces naturally through the weak form.[9, 17, 18]

2.2. **Joint finite-dimensional approximation.**

2.2.1. *Translational approximation.* Let $\mathbf{N}(S)$ be matrix-valued basis functions satisfying the essential translational boundary conditions. Approximate the centerline by

$$\mathbf{r}_h(S,t) = \mathbf{r}_0(S) + \mathbf{N}(S)\mathbf{a}(t), \tag{2.1}$$

where $\mathbf{a}(t) \in \mathbb{R}^{n_a}$ contains the translational generalized coordinates. Hence,

$$\dot{\mathbf{r}}_h = \mathbf{N}\dot{\mathbf{a}}, \qquad \mathbf{r}_{h,S} = \mathbf{r}_{0,S} + \mathbf{N}_{,S}\mathbf{a}. \tag{2.2}$$

If the essential boundary conditions are time dependent, a known lifting function may be added to the approximation; its first and second time derivatives are treated as prescribed generalized loads. To emphasize the structural formulation, the boundary conditions are assumed homogeneous below.

2.2.2. *Approximation of cross-sectional orientation.* Let $\mathbf{R}_0(S)$ be the reference orientation. In a neighborhood of the reference state, use exponential coordinates for the rotation vector:

$$\mathbf{R}_h(S,t) = \mathbf{R}_0(S)\exp\left(\widehat{\boldsymbol{\theta}_h(S,t)}\right), \tag{2.3}$$

$$\boldsymbol{\theta}_h(S,t) = \boldsymbol{\Psi}(S)\mathbf{b}(t), \tag{2.4}$$

where $\mathbf{b}(t) \in \mathbb{R}^{n_b}$ contains the rotational generalized coordinates and $\boldsymbol{\Psi}(S)$ is a matrix-valued basis satisfying the essential rotational boundary conditions. Define

$$\mathbf{E}_h = \exp\left(\widehat{\boldsymbol{\theta}_h}\right). \tag{2.5}$$

so that

$$\mathbf{R}_h = \mathbf{R}_0\mathbf{E}_h. \tag{2.6}$$

Exponential coordinates form a local chart on $SO(3)$. Provided that the rotational neighborhood does not cross the singular set of the exponential map, this parameterization is smooth and one-to-one and is suitable for finite-dimensional modal truncation and local nonlinear analysis. Global time integration through large rotations may instead employ chart switching or unit quaternions without changing the underlying mechanical derivation.

2.2.3. *Right Jacobian and angular velocity.* Let $\theta = \| \boldsymbol{\theta}_h \|$. The right Jacobian of the rotation vector is defined by

$$\mathbf{J}_r(\boldsymbol{\theta}_h) = \mathbf{I} - \frac{1-\cos\theta}{\theta^2}\widehat{\boldsymbol{\theta}_h} + \frac{\theta-\sin\theta}{\theta^3}\widehat{\boldsymbol{\theta}_h}^2. \tag{2.7}$$

As $\theta \to 0$, continuous extension gives

$$\mathbf{J}_r(\boldsymbol{\theta}_h) = \mathbf{I} - \frac{1}{2}\widehat{\boldsymbol{\theta}_h} + \frac{1}{6}\widehat{\boldsymbol{\theta}_h}^2 + O\left(\theta^3\right). \tag{2.8}$$

The exact differential relation of the exponential map gives the material angular velocity

$$\boldsymbol{\omega}_h = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\dot{\boldsymbol{\theta}}_h = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\boldsymbol{\Psi}\dot{\mathbf{b}}. \tag{2.9}$$

Define the angular-velocity matrix

$$\mathbf{B}_\omega(S,\mathbf{b}) = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\boldsymbol{\Psi}(S), \tag{2.10}$$

Then

$$\boldsymbol{\omega}_h = \mathbf{B}_\omega(S,\mathbf{b})\,\dot{\mathbf{b}}. \tag{2.11}$$

Thus, $\boldsymbol{\omega}_h = \boldsymbol{\Psi}\dot{\mathbf{b}}$ cannot in general be assumed; that relation is only a zeroth-order small-rotation approximation.

2.2.4. *Joint generalized coordinates and virtual variables.* Define the complete generalized coordinate vector

$$\mathbf{q} = \begin{bmatrix}\mathbf{a}\\ \mathbf{b}\end{bmatrix} \in \mathbb{R}^n, \qquad n = n_a + n_b. \tag{2.12}$$

The translational virtual displacement is

$$\delta\mathbf{r}_h = \mathbf{N}\,\delta\mathbf{a}. \tag{2.13}$$

From $\delta\mathbf{R}_h = \mathbf{R}_h\widehat{\boldsymbol{\eta}_h}$, the material virtual rotation satisfies

$$\boldsymbol{\eta}_h = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\delta\boldsymbol{\theta}_h = \mathbf{B}_\omega(S,\mathbf{b})\,\delta\mathbf{b}. \tag{2.14}$$

Accordingly, the translational and rotational test functions are $\mathbf{N}$ and $\mathbf{B}_\omega$, respectively, and both enter the coupled Galerkin condition.

For $\mathbf{q} = \mathbf{0}$ to represent a stress-free reference configuration, the reference centerline, reference orientation, and reference strains must satisfy the compatibility relations

$$\boldsymbol{\nu}_0(S) = \mathbf{R}_0^T(S)\mathbf{r}_{0,S}(S), \tag{2.15}$$

$$\widehat{\boldsymbol{\kappa}}_0(S) = \mathbf{R}_0^T(S)\mathbf{R}_{0,S}(S). \tag{2.16}$$

If these relations are not satisfied, $\mathbf{q} = \mathbf{0}$ describes a pre-extended, pre-sheared, pre-bent, or pre-twisted configuration, with the corresponding initial force and moment generated automatically by $V_{int,h}$. The equilibrium state, tangent stiffness, and center of the KAM expansion must use the same reference convention.

2.3. **Discrete strains and potential energy.**

2.3.1. *Shear–extension strain.* The discrete shear–extension strain is

$$\boldsymbol{\nu}_h(\mathbf{q}) = \mathbf{R}_h^T\mathbf{r}_{h,S} = \mathbf{E}_h^T\mathbf{R}_0^T\left(\mathbf{r}_{0,S} + \mathbf{N}_{,S}\mathbf{a}\right). \tag{2.17}$$

Its virtual variation can be written as

$$\delta\boldsymbol{\nu}_h = \mathbf{R}_h^T\mathbf{N}_{,S}\delta\mathbf{a} - \boldsymbol{\eta}_h \times \boldsymbol{\nu}_h. \tag{2.18}$$

Thus, there exists a strain–generalized-coordinate matrix $\mathbf{B}_\nu(S, \mathbf{q})$ such that

$$\delta\boldsymbol{\nu}_h = \mathbf{B}_\nu(S, \mathbf{q})\,\delta\mathbf{q}. \tag{2.19}$$

Explicitly,

$$\mathbf{B}_\nu = \begin{bmatrix}\mathbf{R}_h^T\mathbf{N}_{,S} & \widehat{\boldsymbol{\nu}_h}\mathbf{B}_\omega\end{bmatrix}, \tag{2.20}$$

because $-\boldsymbol{\eta}_h \times \boldsymbol{\nu}_h = \widehat{\boldsymbol{\nu}_h}\boldsymbol{\eta}_h$.

2.3.2. *Curvature–torsion strain.* Define the curvature–torsion vector of the reference orientation by

$$\widehat{\boldsymbol{\kappa}_R} = \mathbf{R}_0^T\mathbf{R}_{0,S}. \tag{2.21}$$

From $\mathbf{R}_h = \mathbf{R}_0\mathbf{E}_h$, one obtains

$$\mathbf{R}_h^T\mathbf{R}_{h,S} = \mathbf{E}_h^T\widehat{\boldsymbol{\kappa}_R}\mathbf{E}_h + \mathbf{E}_h^T\mathbf{E}_{h,S}. \tag{2.22}$$

Hence, the discrete curvature–torsion vector is

$$\boxed{\boldsymbol{\kappa}_h(\mathbf{q}) = \mathbf{E}_h^T\boldsymbol{\kappa}_R + \mathbf{J}_r(\boldsymbol{\theta}_h)\,\boldsymbol{\theta}_{h,S}} \tag{2.23}$$

where

$$\boldsymbol{\theta}_{h,S} = \boldsymbol{\Psi}_{,S}\mathbf{b}. \tag{2.24}$$

The corresponding virtual variation satisfies

$$\delta\boldsymbol{\kappa}_h = \boldsymbol{\eta}_{h,S} + \boldsymbol{\kappa}_h \times \boldsymbol{\eta}_h. \tag{2.25}$$

Therefore, there exists $\mathbf{B}_\kappa(S, \mathbf{q})$ such that

$$\delta\boldsymbol{\kappa}_h = \mathbf{B}_\kappa(S, \mathbf{q})\,\delta\mathbf{q}. \tag{2.26}$$

In assembly, $\mathbf{B}_\kappa$ may be obtained by analytic differentiation of the preceding expression or by evaluating the Jacobian of $\boldsymbol{\kappa}_h(\mathbf{q})$ through automatic differentiation. In either implementation, the independent torsional contribution κ_{h3} must be retained.

2.3.3. *Discrete internal potential energy.* Substituting the discrete strains into the potential energy of Section 1, the discrete internal potential energy can be decomposed as

$$V_{int,h}(\mathbf{q}) = V_{\nu,h}(\mathbf{q}) + V_{\kappa,h}(\mathbf{q}), \tag{2.27}$$

where

$$V_{\nu,h}(\mathbf{q}) = \frac{1}{2}\int_0^L (\boldsymbol{\nu}_h - \boldsymbol{\nu}_0)^T \mathbf{K}_\nu (\boldsymbol{\nu}_h - \boldsymbol{\nu}_0)\, dS, \tag{2.28}$$

$$V_{\kappa,h}(\mathbf{q}) = \frac{1}{2}\int_0^L (\boldsymbol{\kappa}_h - \boldsymbol{\kappa}_0)^T \mathbf{K}_\kappa (\boldsymbol{\kappa}_h - \boldsymbol{\kappa}_0)\, dS. \tag{2.29}$$

The discrete internal generalized force is

$$\mathbf{f}_{int}(\mathbf{q}) = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.30}$$

Using the virtual-work expression, it can be written as

$$\mathbf{f}_{int} = \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b\right] dS. \tag{2.31}$$

This expression contains the restoring forces associated with extension, shear, bending about two axes, and torsion. In general, $\mathbf{f}_{int}$ is a nonlinear vector-valued function of $\mathbf{q}$ and cannot be written a priori as a constant stiffness matrix multiplied by $\mathbf{q}$.

2.4. Complete discrete kinetic energy and mass–inertia matrix.

2.4.1. *Material-point velocity matrix.* The discrete material-point position is

$$\mathbf{x}_h(S, \boldsymbol{\xi}, t) = \mathbf{r}_h(S,t) + \mathbf{R}_h(S,t)\boldsymbol{\xi}. \tag{2.32}$$

and its velocity is

$$\dot{\mathbf{x}}_h = \mathbf{N}\dot{\mathbf{a}} - \mathbf{R}_h\widehat{\boldsymbol{\xi}}\mathbf{B}_\omega\dot{\mathbf{b}}. \tag{2.33}$$

Define the material-point velocity matrix

$$\mathbf{B}_x(S, \boldsymbol{\xi}, \mathbf{q}) = \begin{bmatrix}\mathbf{N} & -\mathbf{R}_h\widehat{\boldsymbol{\xi}}\mathbf{B}_\omega\end{bmatrix}, \tag{2.34}$$

Then

$$\dot{\mathbf{x}}_h = \mathbf{B}_x(S, \boldsymbol{\xi}, \mathbf{q})\,\dot{\mathbf{q}}. \tag{2.35}$$

2.4.2. *Rigorous definition of the mass–inertia matrix* $\mathbf{M}(\mathbf{q})$. The discrete kinetic energy is

$$T_h(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2}\int_0^L \int_{A(S)} \rho \dot{\mathbf{x}}_h^T \dot{\mathbf{x}}_h \, dA\, dS. \tag{2.36}$$

Substitution of the velocity matrix gives

$$\boxed{T_h(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}}} \tag{2.37}$$

where

$$\boxed{\mathbf{M}(\mathbf{q}) = \int_0^L \int_{A(S)} \rho \mathbf{B}_x^T \mathbf{B}_x \, dA\, dS} \tag{2.38}$$

is the mass–inertia matrix derived rigorously from the complete three-dimensional translational–rotational kinetic energy.

2.4.3. *Block form of the mass–inertia matrix.* With $\mathbf{q} = \left(\mathbf{a}^T, \mathbf{b}^T\right)^T$, write

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{M}_{ab}(\mathbf{q}) \\ \mathbf{M}_{ba}(\mathbf{q}) & \mathbf{M}_{bb}(\mathbf{q}) \end{bmatrix}. \tag{2.39}$$

Using $\mu, \mathbf{c}_\rho, \mathbf{J}_c$ defined in Section 1, the blocks are

$$\mathbf{M}_{aa} = \int_0^L \mu \mathbf{N}^T \mathbf{N}\, dS, \tag{2.40}$$

$$\mathbf{M}_{ab}(\mathbf{q}) = -\int_0^L \mathbf{N}^T \mathbf{R}_h \widehat{\mathbf{c}}_\rho \mathbf{B}_\omega\, dS, \tag{2.41}$$

$$\mathbf{M}_{ba}(\mathbf{q}) = \mathbf{M}_{ab}^T(\mathbf{q}), \tag{2.42}$$

$$\boxed{\mathbf{M}_{bb}(\mathbf{q}) = \int_0^L \mathbf{B}_\omega^T \mathbf{J}_c \mathbf{B}_\omega\, dS} \tag{2.43}$$

Thus,

$$\mathbf{M}_{bb}(\mathbf{q}) = \int_0^L \mathbf{\Psi}^T \mathbf{J}_r^T(\boldsymbol{\theta}_h)\, \mathbf{J}_c \mathbf{J}_r(\boldsymbol{\theta}_h)\, \mathbf{\Psi}\, dS. \tag{2.44}$$

Whether the mass matrix depends on the configuration is therefore not an a priori assumption; it is determined jointly by the centerline definition, orientation parameterization, cross-sectional mass distribution, and attached components.

2.4.4. *Mass centerline and eccentric attachments.* Consistently with Section 1, the centerline of the base catheter is taken to pass through the mass center of each cross section, so the distributed rod satisfies

$$\mathbf{c}_\rho = \mathbf{0}, \tag{2.45}$$

and the translational–rotational cross block of the base rod vanishes:

$$\mathbf{M}_{ab} = \mathbf{M}_{ba} = \mathbf{0}. \tag{2.46}$$

In this case,

$$\mathbf{M}_{str}(\mathbf{q}) = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{bb}(\mathbf{b}) \end{bmatrix}. \tag{2.47}$$

However, $\mathbf{M}_{bb}$ still depends on the rotational generalized coordinates through $\mathbf{J}_r(\boldsymbol{\theta}_h)$, and the complete structural mass–inertia matrix therefore generally remains $\mathbf{M}_{str}(\mathbf{q})$.

If the center of mass of a magnetic tip, sensor, or other attached component is offset from the centerline, its kinetic energy must be added separately. Let the center-of-mass position of the attachment be

$$\mathbf{x}_e = \mathbf{r}_h(S_e) + \mathbf{R}_h(S_e)\, \boldsymbol{\rho}_e, \tag{2.48}$$

If its mass is m_e and its material-frame rotary inertia is $\mathbf{J}_e$, then

$$T_e = \frac{1}{2} m_e |\dot{\mathbf{x}}_e|^2 + \frac{1}{2} \boldsymbol{\omega}_e^T \mathbf{J}_e \boldsymbol{\omega}_e. \tag{2.49}$$

Let $\mathbf{B}_e(\mathbf{q})$ map $\dot{\mathbf{q}}$ to the attachment velocity. Then

$$T_e = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}_e(\mathbf{q})\, \dot{\mathbf{q}}, \qquad \mathbf{M}_e = \mathbf{B}_e^T \mathbf{M}_e^{rig} \mathbf{B}_e. \tag{2.50}$$

This contribution generally produces a nonzero translational–rotational coupling block. Thus, use of the mass centerline for the base rod does not imply that the cross-mass block of the complete system must vanish.

2.4.5. *Scope of the small-rotation constant-mass approximation.* When $\| \boldsymbol{\theta}_h \| \ll 1$, if only

$$\mathbf{J}_r(\boldsymbol{\theta}_h) \approx \mathbf{I}, \tag{2.51}$$

is retained, then

$$\mathbf{M}_{bb} \approx \int_0^L \boldsymbol{\Psi}^T \mathbf{J}_c \boldsymbol{\Psi}\, dS, \tag{2.52}$$

becomes constant. This is a zeroth-order small-rotation inertial approximation rather than a general property of the complete model. The principal model retains the exact $\mathbf{J}_r$ and $\mathbf{M}(\mathbf{q})$.

2.4.6. *Energy-consistent discretization of fluid added mass.* Define the joint translational velocity matrix

$$\mathbf{N}_q(S) = \begin{bmatrix} \mathbf{N}(S) & \mathbf{0} \end{bmatrix}, \tag{2.53}$$

such that $\dot{\mathbf{r}}_h = \mathbf{N}_q \dot{\mathbf{q}}$. Let

$$\mathbf{w}_{\perp,h} = \mathbf{P}_\perp(\mathbf{q}) \left[\mathbf{N}_q \dot{\mathbf{q}} - \mathbf{v}_f(\mathbf{r}_h, t) \right], \qquad m_A = \rho_f C_A A_d. \tag{2.54}$$

The added-fluid inertia functional of Section 1 is discretized as follows (and becomes an added kinetic energy for a stationary flow):

$$\boxed{T_{A,h} = \frac{1}{2} \int_0^L m_A\, \mathbf{w}_{\perp,h}^T \mathbf{w}_{\perp,h}\, dS.} \tag{2.55}$$

Expansion gives

$$T_{A,h} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}_A(\mathbf{q})\, \dot{\mathbf{q}} - \dot{\mathbf{q}}^T \mathbf{b}_A(\mathbf{q}, t) + c_A(\mathbf{q}, t), \tag{2.56}$$

where

$$\boxed{\mathbf{M}_A(\mathbf{q}) = \int_0^L m_A\, \mathbf{N}_q^T \mathbf{P}_\perp(\mathbf{q})\, \mathbf{N}_q\, dS,} \tag{2.57}$$

$$\mathbf{b}_A(\mathbf{q}, t) = \int_0^L m_A\, \mathbf{N}_q^T \mathbf{P}_\perp(\mathbf{q})\, \mathbf{v}_f\, dS, \tag{2.58}$$

$$c_A(\mathbf{q}, t) = \frac{1}{2} \int_0^L m_A\, \mathbf{v}_f^T \mathbf{P}_\perp(\mathbf{q})\, \mathbf{v}_f\, dS. \tag{2.59}$$

The structural kinetic energy, eccentric-attachment kinetic energy, and added-fluid contribution are therefore collected into the velocity-dependent inertial functional

$$T_h = T_{str,h} + \sum_e T_e + T_{A,h}. \tag{2.60}$$

Its Hessian with respect to the generalized velocity gives the total effective mass–inertia matrix

$$\boxed{\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_e \mathbf{M}_e(\mathbf{q}) + \mathbf{M}_A(\mathbf{q}).} \tag{2.61}$$

Only when $\mathbf{v}_f = \mathbf{0}$ is it physically valid to set $\mathbf{b}_A = \mathbf{0}$ and $c_A = 0$. For a general prescribed flow, moving the cross terms to an external port to construct an autonomous conservative backbone merely removes $\mathbf{b}_A$ and c_A from the selected Hamiltonian backbone and assigns their complete Euler–Lagrange action to $\mathbf{Q}_{flow}$; it does not assume that these quantities vanish in the original system. For a general flow, one cannot retain only $\mathbf{M}_A \dot{\mathbf{q}}$. The constant added mass obtained by freezing $\mathbf{P}_\perp$ is used only as an engineering validation branch.

2.4.7. *Symmetry and positive definiteness.* By definition of the kinetic-energy Hessian,

$$\mathbf{M}^T(\mathbf{q}) = \mathbf{M}(\mathbf{q}). \tag{2.62}$$

If the material and added-mass parameters are positive, the orientation coordinates remain in a local chart where $\mathbf{J}_r$ is nonsingular, and the joint velocity map does not degenerate almost everywhere for any nonzero generalized velocity, namely,

$$\dot{\mathbf{q}} \neq \mathbf{0} \quad \Rightarrow \quad T_h^{(2)}(\mathbf{q}, \dot{\mathbf{q}}) > 0, \tag{2.63}$$

then

$$\mathbf{y}^T \mathbf{M}(\mathbf{q})\, \mathbf{y} > 0, \qquad \forall \mathbf{y} \neq \mathbf{0}. \tag{2.64}$$

Therefore, $\mathbf{M}(\mathbf{q})$ is symmetric positive definite. What must be excluded is a redundant generalized coordinate or a degenerate velocity map, not a rigid-body mode under free boundary conditions. A rigid-body mode may give a zero eigenvalue of the stiffness matrix, while a valid rigid-body motion still has positive kinetic energy.

2.5. **Joint Galerkin projection.**

2.5.1. *Inertial term.* Substituting the finite-dimensional approximation into the total inertial virtual work of Section 1 gives

$$\delta W_{in,h} = \delta \mathbf{q}^T \left[\frac{d}{dt} \left(\frac{\partial T_h}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial T_h}{\partial \mathbf{q}} \right]. \tag{2.65}$$

For a stationary flow, or for an autonomous conservative inertial backbone selected after assigning the complete Euler–Lagrange action of the prescribed-flow cross terms to $\mathbf{Q}_{flow}$,

$$T_h = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\, \dot{\mathbf{q}}, \tag{2.66}$$

the inertial generalized force can be arranged as

$$\mathbf{M}(\mathbf{q})\, \ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}). \tag{2.67}$$

For a general prescribed flow, T_h also contains $-\dot{\mathbf{q}}^T \mathbf{b}_A + c_A$, whose additional contribution is denoted by $\mathbf{g}_f(\mathbf{q}, \dot{\mathbf{q}}, t)$. The unified form is therefore

$$\boxed{\mathbf{M}(\mathbf{q})\, \ddot{\mathbf{q}} + \mathbf{g}_T(\mathbf{q}, \dot{\mathbf{q}}, t), \qquad \mathbf{g}_T = \mathbf{g}_M + \mathbf{g}_f.} \tag{2.68}$$

The terms are derived rigorously in Section 2.6.

2.5.2. *Internal-force term.* The discrete internal virtual work is

$$\delta W_{int,h} = \delta \mathbf{q}^T \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b \right] dS. \tag{2.69}$$

Therefore,

$$\mathbf{f}_{int}(\mathbf{q}) = \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b \right] dS = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.70}$$

2.5.3. *Joint projection of distributed forces and couples.* First, decompose the distributed loads that remain in the external-force ports through mutually exclusive accounting:

$$\mathbf{f}_{ext,rem} = \mathbf{f}_{cons} + \mathbf{f}_{nc}, \qquad \mathbf{l}_{ext} = \mathbf{l}_{cons} + \mathbf{l}_{nc}. \tag{2.71}$$

The conservative part is represented through an external potential in Section 2.5.5; only the nonconservative part is projected directly onto the right-hand side. A spatially distributed couple is first transformed to the material frame, $\mathbf{l}_{nc}^{b} = \mathbf{R}_h^T \mathbf{l}_{nc}$, and the nonconservative distributed generalized force is

$$\boxed{\mathbf{Q}_{nc,dist}\left(\mathbf{q}, \dot{\mathbf{q}}, t\right) = \int_0^L \begin{bmatrix} \mathbf{N}^T \mathbf{f}_{nc} \\ \mathbf{B}_\omega^T \mathbf{R}_h^T \mathbf{l}_{nc} \end{bmatrix} dS.} \tag{2.72}$$

This expression allows magnetic, contact, and fluid couples to enter the rotational generalized coordinates directly when they belong to nonconservative ports. If a load has already been represented by a potential, it must not be included again in $\mathbf{Q}_{nc,dist}$.

For normal quadratic drag acting only on the centerline translational velocity, define the joint velocity matrix

$$\boxed{\mathbf{N}_q = \begin{bmatrix} \mathbf{N} & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{3\times n}, \qquad \dot{\mathbf{r}}_h = \mathbf{N}_q \dot{\mathbf{q}}.} \tag{2.73}$$

Its consistent generalized-force projection is

$$\mathbf{Q}_{drag} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp \left(\mathbf{v}_f - \mathbf{N}_q \dot{\mathbf{q}}\right) dS = -\mathbf{D}_{drag}\left(\mathbf{q}, \dot{\mathbf{q}}, t\right) \dot{\mathbf{q}} + \mathbf{Q}_{drag,flow}\left(\mathbf{q}, \dot{\mathbf{q}}, t\right), \tag{2.74}$$

where

$$\boxed{\mathbf{D}_{drag} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp \mathbf{N}_q \, dS, \qquad \mathbf{Q}_{drag,flow} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp \mathbf{v}_f \, dS.} \tag{2.75}$$

For every $\mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{y}^T \mathbf{D}_{drag} \mathbf{y} = \int_0^L d_f \parallel \mathbf{P}_\perp \mathbf{N}_q \mathbf{y} \parallel^2 \, dS \geq 0, \tag{2.76}$$

Hence, $\mathbf{D}_{drag} = \mathbf{D}_{drag}^T \succcurlyeq 0$. The matrix $\mathbf{N}_q^T$, rather than $\mathbf{N}^T$, must be used so that $\mathbf{Q}_{drag} \in \mathbb{R}^n$ and $\mathbf{D}_{drag} \in \mathbb{R}^{n\times n}$. Because $\mathbf{N}_q = [\mathbf{N}\ \mathbf{0}]$, the ab, ba, and bb blocks associated with centerline drag vanish. Eccentric fluid couples and cross-sectional spin damping still enter the orientation channels through independent distributed-couple projections. Since d_f depends on the relative velocity, $\mathbf{D}_{drag}$ is a statewise positive-semidefinite nonlinear matrix.

2.5.4. *Boundary generalized force.* The boundary virtual work in Section 1 is

$$\left[\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m}_b \cdot \boldsymbol{\eta}\right]_0^L. \tag{2.77}$$

Substitution of the finite-dimensional virtual variables gives

$$\boxed{\mathbf{Q}_\partial\left(\mathbf{q}, t\right) = \begin{bmatrix} \mathbf{N}^T \overline{\mathbf{n}} \\ \mathbf{B}_\omega^T\left(\mathbf{q}\right) \overline{\mathbf{m}}_b \end{bmatrix}_0^L.} \tag{2.78}$$

The boundary generalized force generally depends on the configuration because the rotational test matrix $\mathbf{B}_\omega$ and the transformation from a spatial moment to a material moment may both vary with orientation. At an essential boundary, the corresponding test function vanishes and the boundary term disappears automatically. At a natural boundary, the prescribed terminal force and moment must be retained.

2.5.5. *Conservative and nonconservative external loads.* If the conservative distributed and boundary loads admit a discrete external potential $V_{ext,h}(\mathbf{q},t)$, their total conservative generalized force is

$$\mathbf{Q}_{cons} = -\nabla_{\mathbf{q}} V_{ext,h}, \tag{2.79}$$

and define

$$V_h(\mathbf{q},t) = V_{int,h}(\mathbf{q}) + V_{ext,h}(\mathbf{q},t). \tag{2.80}$$

The nonconservative generalized force is denoted collectively by

$$\mathbf{Q}_{nc} = \mathbf{Q}_{nc,dist} + \mathbf{Q}_{control} + \mathbf{Q}_{disturbance} + \mathbf{Q}_{other,nc}. \tag{2.81}$$

The final equations use only $\nabla_{\mathbf{q}} V_h$ on the left-hand side and $\mathbf{Q}_{nc}$ on the right-hand side. The original load projection containing conservative components must not be added again in its entirety to the right-hand side. If a time-dependent magnetic action has already been included in $V_{ext,h}(\mathbf{q},t)$, its explicit-time power is represented only by $\partial V_{ext,h}/\partial t$.

2.6. **Rigorous derivation of generalized momentum and geometric inertial terms.**

2.6.1. *Generalized momentum.* For the velocity-dependent inertial functional in Section 2.4.6,

$$T_h = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \dot{\mathbf{q}}^T \mathbf{b}_A(\mathbf{q},t) + c_A(\mathbf{q},t), \tag{2.82}$$

define the generalized momentum

$$\boxed{\mathbf{p} = \frac{\partial T_h}{\partial \dot{\mathbf{q}}} = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \mathbf{b}_A(\mathbf{q},t).} \tag{2.83}$$

For a general prescribed flow, one therefore cannot unconditionally write $\mathbf{p} = \mathbf{M}\dot{\mathbf{q}}$. The backbone momentum is defined only for a stationary flow or after the complete action of $\mathbf{b}_A$ has been moved to $\mathbf{Q}_{flow}$ in a deliberately selected autonomous conservative backbone:

$$\mathbf{p}_0 = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}}. \tag{2.84}$$

2.6.2. *Geometric inertia generated by the mass metric.* Taking the Euler–Lagrange derivative of the purely quadratic velocity term gives

$$\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q},\dot{\mathbf{q}}), \tag{2.85}$$

where

$$\boxed{\mathbf{g}_M = \dot{\mathbf{M}}\dot{\mathbf{q}} - \frac{1}{2}\nabla_{\mathbf{q}}\left(\dot{\mathbf{q}}^T \mathbf{M}\dot{\mathbf{q}}\right).} \tag{2.86}$$

Define

$$\Gamma_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i}\right), \tag{2.87}$$

Then

$$g_{M,i} = \sum_{j,k=1}^{n} \Gamma_{ijk}\dot{q}_j\dot{q}_k. \tag{2.88}$$

Let $C_{ij} = \sum_k \Gamma_{ijk}\dot{q}_k$. Then $\mathbf{g}_M = \mathbf{C}\dot{\mathbf{q}}$, and $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew symmetric, which guarantees the kinetic-energy power identity.

2.6.3. *Additional action generated by prescribed-flow cross terms.* The term $-\dot{\mathbf{q}}^T\mathbf{b}_A + c_A$ gives

$$\boxed{\mathbf{g}_f = -\dot{\mathbf{b}}_A + \left(\nabla_{\mathbf{q}}\mathbf{b}_A\right)^T \dot{\mathbf{q}} - \nabla_{\mathbf{q}} c_A.} \tag{2.89}$$

Here, $\dot{\mathbf{b}}_A$ denotes the total derivative along the trajectory. The total inertial action is therefore

$$\boxed{\frac{d}{dt}\frac{\partial T_h}{\partial \dot{\mathbf{q}}} - \frac{\partial T_h}{\partial \mathbf{q}} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \mathbf{g}_f.} \tag{2.90}$$

Equivalently, $\mathbf{g}_f$ may be moved to the right-hand side and denoted as a flow input $\mathbf{Q}_{flow}$, but the entire term must be moved; retaining only selected components is inconsistent.

2.6.4. *Constant-mass special case.* If, under a specific small-rotation, zero-eccentricity, fixed-normal-projection, and fixed-added-mass approximation,

$$\frac{\partial \mathbf{M}}{\partial q_i} = \mathbf{0}, \qquad i = 1, \ldots, n, \tag{2.91}$$

then $\mathbf{g}_M = \mathbf{0}$. If $\mathbf{b}_A = \mathbf{0}$ as well, all geometric inertial terms vanish. This conclusion applies only to the corresponding local approximation and must not be used in reverse to deny configuration-dependent inertia in the complete model.

2.7. **Discrete treatment of the Kirchhoff shear-free constraint.**

2.7.1. *Discrete constraints.* The strict Kirchhoff model requires[17, 18]

$$\nu_{h1}\left(\mathbf{q}\right) = 0, \qquad \nu_{h2}\left(\mathbf{q}\right) = 0. \tag{2.92}$$

Collect all discrete constraints in

$$\mathbf{c}\left(\mathbf{q}\right) = \mathbf{0}. \tag{2.93}$$

Define the constraint Jacobian

$$\mathbf{G}\left(\mathbf{q}\right) = \frac{\partial \mathbf{c}}{\partial \mathbf{q}}. \tag{2.94}$$

2.7.2. *Lagrange-multiplier form.* Introduce the Lagrange multiplier $\boldsymbol{\lambda}_c$. The generalized constraint reaction is

$$\mathbf{Q}_c = \mathbf{G}^T\left(\mathbf{q}\right)\boldsymbol{\lambda}_c. \tag{2.95}$$

The complete constrained discrete system is

$$\begin{cases} \mathbf{M}\left(\mathbf{q}\right)\ddot{\mathbf{q}} + \mathbf{g}\left(\mathbf{q}, \dot{\mathbf{q}}\right) + \nabla_{\mathbf{q}} V_h\left(\mathbf{q}, t\right) + \mathbf{G}^T\left(\mathbf{q}\right)\boldsymbol{\lambda}_c = \mathbf{Q}_{nc}\left(\mathbf{q}, \dot{\mathbf{q}}, t\right), \\ \mathbf{c}\left(\mathbf{q}\right) = \mathbf{0}. \end{cases} \tag{2.96}$$

This is a nonlinear differential–algebraic system. The constraint eliminates shear deformation only; it does not remove rotational generalized coordinates or torsional modes.

2.7.3. *Finite-shear-stiffness form.* If the fully shearable Cosserat rod is used, or if the Kirchhoff constraint is approximated by a large but finite shear stiffness, no Lagrange multiplier is required because the shear energy is already included in $V_{int,h}$. The resulting system is an ordinary second-order nonlinear ODE. The two treatments have different numerical properties but share the same complete translational–rotational discretization.

2.8. **Global finite-dimensional equations of motion.**

2.8.1. *Form without explicit constraints.* Assembling the inertial term, internal conservative force, conservative external potential, nonconservative distributed loads, and boundary loads gives

$$\boxed{\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_T(\mathbf{q},\dot{\mathbf{q}},t) + \nabla_{\mathbf{q}} V_h(\mathbf{q},t) = \mathbf{Q}_{nc}(\mathbf{q},\dot{\mathbf{q}},t) + \mathbf{Q}_{\partial}(\mathbf{q},t)\,.} \tag{2.97}$$

Here, $\mathbf{q}$ contains both translational and rotational generalized coordinates; $\mathbf{M}$ is generated by the velocity-quadratic terms of the base rod, eccentric attachments, and the energy-consistent added mass; $\mathbf{g}_T = \mathbf{g}_M + \mathbf{g}_f$; and V_h contains extension, shear, bending, torsion, and external potentials satisfying the potential condition.

If the prescribed-flow cross terms are moved collectively to the input port and dissipation is separated from control inputs, the system becomes

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \nabla_{\mathbf{q}} V_h = \mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}. \tag{2.98}$$

The passively dissipative drag kernel obtained by exclusive splitting, normal contact damping and friction for a fixed wall, and any retained spin damping constitute $\mathbf{Q}_{diss}$, with $\dot{\mathbf{q}}^T\mathbf{Q}_{diss} \le 0$. The background-flow drag term $\mathbf{Q}_{drag,flow}$ must be assigned to $\mathbf{Q}_{flow}$, and its power is generally sign-indefinite; therefore, no global dissipativity claim may be made for the complete relative drag. Nor should $\mathbf{Q}_{diss}$ be forced into the form of a constant damping matrix times velocity, because the quadratic-drag matrix $\mathbf{D}_{drag}(\mathbf{q},\dot{\mathbf{q}},t)$ itself depends on state and relative velocity. The same fluid action may be counted only once in any selected representation.

2.8.2. *Constrained form.* The strict Kirchhoff shear-free model is finally written as

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_T + \nabla_{\mathbf{q}} V_h + \mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}_c = \mathbf{Q}_{nc} + \mathbf{Q}_{\partial}, \tag{2.99}$$

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}. \tag{2.100}$$

The system retains complete bending–torsion dynamics, cross-sectional orientation, and rotary inertia while enforcing the shear-free condition exactly.

2.9. **Discrete energy and power identities.**

2.9.1. *Discrete mechanical energy of the conservative inertial backbone.* For a stationary flow with $\mathbf{b}_A = \mathbf{0}$ and $c_A = 0$, or for an autonomous conservative backbone selected after the prescribed-flow cross terms have been moved completely to $\mathbf{Q}_{flow}$, define

$$E_h(\mathbf{q},\dot{\mathbf{q}},t) = \frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} + V_h(\mathbf{q},t)\,. \tag{2.101}$$

Because $\mathbf{g}_M$ is generated by the Euler–Lagrange derivative of the complete quadratic kinetic energy,

$$\dot{\mathbf{q}}^T\left[\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M\right] = \frac{d}{dt}\left(\frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}\dot{\mathbf{q}}\right). \tag{2.102}$$

2.9.2. *General port-power balance.* Left-multiplication of the global equation by $\dot{\mathbf{q}}^T$ gives

$$\frac{dE_h}{dt} = \dot{\mathbf{q}}^T\left(\mathbf{Q}_{nc} + \mathbf{Q}_{\partial} - \mathbf{g}_f\right) + \frac{\partial V_h}{\partial t}. \tag{2.103}$$

If the complete Euler–Lagrange contribution of the prescribed-flow cross action is treated as part of the flow input, then

$$\boxed{\frac{dE_h}{dt} = \dot{\mathbf{q}}^T\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}\right) + \frac{\partial V_h}{\partial t}.} \tag{2.104}$$

Here, $\mathbf{Q}_{diss}$ contains only the passive dissipative kernel after splitting and satisfies $\dot{\mathbf{q}}^T\mathbf{Q}_{diss} \le 0$. The term $\mathbf{Q}_{flow}$ collects $\mathbf{Q}_{drag,flow}$, the Froude–Krylov input, and the reactive-flow contribution determined by the selected representation; its power is not assigned a fixed sign. The

relative-motion identity verifies physical passivity of the drag law, whereas the dissipative structural constraint of the CHNN must be imposed on the preceding $\mathbf{Q}_{diss}$; the two must not be conflated. For an ideal constraint,

$$\mathbf{G}(\mathbf{q})\,\dot{\mathbf{q}} = \mathbf{0}, \tag{2.105}$$

Therefore,

$$\dot{\mathbf{q}}^T \mathbf{G}^T \boldsymbol{\lambda}_c = 0. \tag{2.106}$$

Only in the absence of nonconservative loads, flow inputs, boundary-power inputs, and explicit time dependence of the potential does $dE_h/dt = 0$.

2.9.3. *Consistent tangent matrix.* For nonlinear solution or local linearization, the consistent internal tangent matrix is defined by

$$\mathbf{K}_T(\mathbf{q}) = \frac{\partial \mathbf{f}_{int}}{\partial \mathbf{q}} = \nabla^2_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.107}$$

This matrix contains both material stiffness and the geometric stiffness generated by variations in orientation, extension, and curvature. It is not the secant matrix obtained by multiplying a state-dependent scalar stiffness by an integral of fixed basis functions.

2.10. **Local linearization and engineering reductions.**

2.10.1. *Local linearization about an equilibrium configuration.* Use the dissipation–input decomposition

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \nabla V_h + \mathbf{G}^T \boldsymbol{\lambda}_c = \mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}. \tag{2.108}$$

Let $\mathbf{q}_e$ be a static equilibrium configuration with $\dot{\mathbf{q}}_e = \mathbf{0}$. Then

Here, a "static equilibrium configuration" requires the explicit time-dependent input, control parameters, and prescribed flow to be held fixed at $t = t_e$, or that the system itself is autonomous. If the magnetic field or flow is periodically driven, a Floquet linearization about a periodic orbit must be used instead of a static equilibrium matrix.

$$\nabla V_h(\mathbf{q}_e) + \mathbf{G}^T(\mathbf{q}_e)\,\boldsymbol{\lambda}_{ce} = \mathbf{Q}_{diss}(\mathbf{q}_e, \mathbf{0}) + \mathbf{Q}_{in,e} + \mathbf{Q}_{flow,e} + \mathbf{Q}_{\partial,e}. \tag{2.109}$$

Usually, $\mathbf{Q}_{diss}(\mathbf{q}, \mathbf{0}) = \mathbf{0}$. Let $\mathbf{q} = \mathbf{q}_e + \boldsymbol{\eta}$ and neglect second- and higher-order perturbation terms. For the unconstrained system,

$$\mathbf{M}_e \ddot{\boldsymbol{\eta}} + \mathbf{C}_e \dot{\boldsymbol{\eta}} + \mathbf{K}_e \boldsymbol{\eta} = \delta\mathbf{Q}(t), \tag{2.110}$$

where

$$\mathbf{M}_e = \mathbf{M}(\mathbf{q}_e), \tag{2.111}$$

$$\mathbf{C}_e = -\left.\frac{\partial \mathbf{Q}_{diss}}{\partial \dot{\mathbf{q}}}\right|_e, \tag{2.112}$$

$$\boxed{\mathbf{K}_e = \left.\left[\nabla^2_{\mathbf{q}} V_h - \frac{\partial \mathbf{Q}_{diss}}{\partial \mathbf{q}} - \frac{\partial\left(\mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}\right)}{\partial \mathbf{q}}\right]\right|_e.} \tag{2.113}$$

Here, $\delta\mathbf{Q}(t)$ collects explicit time-dependent input perturbations not absorbed by the state Jacobian. Under the actual dissipative-force convention above, passive dissipation gives $\mathbf{C}_e \succcurlyeq \mathbf{0}$ near the equilibrium. If the load Jacobian contains nonsymmetric follower terms, $\mathbf{K}_e$ need not be symmetric.

If contact is active near the equilibrium, the preceding Jacobian and consistent tangent assume that the contact active set remains fixed. If the equilibrium lies on a contact opening/closing switching surface, a smooth contact regularization, semismooth Newton method,

or piecewise linearization must be used; the max function cannot be treated as differentiable everywhere.

For a purely conservative system without boundary follower loads, the expression reduces directly to

$$\boxed{\mathbf{K}_e = \nabla_{\mathbf{q}}^2 V_h\left(\mathbf{q}_e\right),} \tag{2.114}$$

No derivative of a conservative generalized force already included in V_h is subtracted again. If strict constraints are present, the system must be projected onto the constraint tangent space or first expressed in independent local coordinates satisfying $\mathbf{c}\left(\mathbf{q}\right) = 0$. The tangent stiffness at a constrained equilibrium must also include the second-order contribution of the constraint Lagrangian. A constant-matrix linear system is only a local approximation of the complete coupled nonlinear model near the specified equilibrium configuration.[17, 18]

2.10.2. *Scope of engineering reductions.* Small rotations, constant mass, a fixed normal projection, and low-order modal truncation are used only for numerical checks, parameter identification, or ablation experiments. The Euler–Bernoulli branch has two roles: theoretically, it is the Kirchhoff shear-free reduction of the geometrically exact parent model; within COMSOL, it is the Kirchhoff-consistent engineering realization used for the primary training data. It may be used as the primary training branch only after the slenderness ratio, the fraction of shear energy, and the comparison error relative to Timoshenko/solid models satisfy prescribed tolerances. The corotational Beam geometric-nonlinearity formulation in COMSOL permits large rotations but assumes locally small element strains; the mesh must therefore keep the rotation increment between adjacent nodes sufficiently small[4, 5].

2.10.3. *Modal truncation and selection of the principal computational order.* The CHNN and KAM analyses must use a finite-dimensional system of fixed dimension and cannot remain indefinitely at an abstract modal order n. For candidate truncation orders $n = 1, 2, \ldots, n_{max}$, compute at least the leading natural frequencies, terminal displacement, structural energy, torsional response, and principal port powers. Define the adjacent-order convergence indicators

$$e_n^{freq} = \max_{1\le j\le m} \frac{\left|\omega_j^{(n+1)} - \omega_j^{(n)}\right|}{\left|\omega_j^{(n+1)}\right|}, \qquad e_n^{traj} = \frac{\| \mathbf{q}^{(n+1)} - \Pi_{n+1\to n}\mathbf{q}^{(n+1)} \|}{\| \mathbf{q}^{(n+1)} \|}, \tag{2.115}$$

and define analogous indicators for energy and torsion. The principal computational order n_* is the first order at which all prescribed indicators simultaneously fall below their tolerances. Until this computation has been completed, the manuscript may retain n_* only as an undetermined parameter and must not describe an empirically selected order as proven sufficient.

The canonical state mapping is examined using a fixed $N = 2$ basis. The basis is extracted from the mesh-80 near-frequency cluster, orthonormalized with respect to the consistent finite-element mass metric, and fixed in sign, normalization, and degree-of-freedom ordering. It defines the paired canonical coordinates used in the P1 series and in the subsequent consistency analysis. The full COMSOL–Galerkin mapping still retains the first 12 production modes and the mesh-40/80 comparison, so $N = 2$ is a diagnostic choice rather than a replacement for the convergence criterion for n_* or a claim of high-fidelity modal sufficiency.

2.11. **Numerical implementation and finite-dimensional model outputs.**

2.11.1. *COMSOL model hierarchy, projection, and validation.* The COMSOL data use a three-level hierarchy: “primary training–strict reproduction–high-fidelity comparison.

1. **Kirchhoff-consistent primary training model.** Use the Euler–Bernoulli formulation of the three-dimensional Beam interface with Time Dependent dynamics and geometric nonlinearity, retaining axial deformation, bending about two axes, three-dimensional cross-sectional rotations, and torsion. The three-dimensional Euler–Bernoulli section stiffness in COMSOL includes axial force, torque, and two bending

moments; shear-free therefore does not mean torsion-free[4, 6]. This model generates data under assumptions consistent with the strict Kirchhoff branch.

2. **Strict weak-form reproduction model.** When a term-by-term check of the coupled translational–orientation weak form, nonstandard constraints, or custom distributed couples is required, Weak Form PDE or Weak Contribution is used to specify test functions, boundary terms, and auxiliary variables[7, 8]. This branch checks discrepancies between the built-in Beam interface and the present equations; it is not the default model for every operating condition.
3. **High-fidelity comparison model.** A Timoshenko Beam model quantifies corrections from transverse shear and rotary inertia, while three-dimensional Solid Mechanics checks local cross-sectional stresses, contact regions, and attachment effects. These models output X_{HF}. Their discrepancy from the Kirchhoff-consistent model X_K is used for validation and model-discrepancy learning rather than directly replacing the primary KAM branch.

Every model should output the centerline position $\mathbf{r}_C(S,t)$, cross-sectional orientation $\mathbf{R}_C(S,t)$ or an equivalent rotation representation, centerline velocity, cross-sectional angular velocity, resultant section forces and moments, structural kinetic energy, and strain energy. The primary training branch should additionally output the separate powers associated with magnetic, fluid, contact, and boundary actions.

Define the translational Gram matrix

$$\mathbf{G}_a = \int_0^L \mathbf{N}^T \mathbf{W}_a \mathbf{N}\, dS, \tag{2.116}$$

Then the projection of the COMSOL centerline field is

$$\mathbf{a}(t) = \mathbf{G}_a^{-1} \int_0^L \mathbf{N}^T \mathbf{W}_a \left[\mathbf{r}_C(S,t) - \mathbf{r}_0(S)\right] dS. \tag{2.117}$$

Orientation must not be projected linearly by its Euler-angle components. First compute the logarithmic coordinates of the relative rotation

$$\boldsymbol{\theta}_C(S,t) = \log\left(\mathbf{R}_0^T(S)\mathbf{R}_C(S,t)\right)^\vee, \tag{2.118}$$

and then set

$$\mathbf{G}_b = \int_0^L \boldsymbol{\Psi}^T \mathbf{W}_b \boldsymbol{\Psi}\, dS, \qquad \mathbf{b}(t) = \mathbf{G}_b^{-1} \int_0^L \boldsymbol{\Psi}^T \mathbf{W}_b \boldsymbol{\theta}_C(S,t)\, dS. \tag{2.119}$$

This gives $\mathbf{q} = \left(\mathbf{a}^T, \mathbf{b}^T\right)^T$. The angular velocity should be checked through $\boldsymbol{\omega}_h = \mathbf{B}_\omega(\mathbf{q})\,\dot{\mathbf{b}}$, rather than by setting $\boldsymbol{\omega} = \boldsymbol{\Psi}\dot{\mathbf{b}}$.

The COMSOL–Galerkin validation compares stationary energy and field exports, mesh-40/80 modal consistency, the first 12 frequency clusters, and the mass-metric basis. The maximum relative frequency difference between the two meshes is 4.15920×10^{-5}, and the maximum off-diagonal entry of the modal mass Gram matrix is at most 1.32445×10^{-15}. The three 1 s trajectories do not all satisfy the prescribed 0.5% perturbation-energy criterion, whereas all three satisfy an independent 2% generalized-force closure criterion on the fixed common temporal support and anchor. These criteria assess different physical quantities; their numerical values are reported in Section 6.2.

The fluid-loading implementation must match the selected theoretical branch. The theoretical route uses T_A (or its discrete equivalent mass and geometric terms), the pure Froude–Krylov term, and drag. The engineering Morison comparison uses the complete $C_M\mathbf{a}_f - C_A\ddot{\mathbf{r}}$ form but must not activate T_A simultaneously. The same physical action may be counted only once in any data sample.

2.11.2. *Finite-dimensional state, energy, port, and structure-learning outputs.* After the joint Galerkin discretization and the consistent Legendre postprocessing of Section 3, the numerical implementation should output unified finite-dimensional states, energies, ports, and representation labels:

$$\mathcal{O}_h = \left(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \mathbf{p}_{rep}, \dot{\mathbf{p}}_{rep}, \mathbf{M}\left(\mathbf{q}\right), V_h, E_h, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \mathbf{Q}_{\partial}, \mathbf{c}, \mathbf{G}, representation\right). \tag{2.120}$$

Here, $\mathbf{p}_{rep}$ and $\dot{\mathbf{p}}_{rep}$ must be consistent with the selected Lagrangian representation. The *representation* label must distinguish at least the autonomous-backbone, exact-nonautonomous, and engineering-reduction representations. The strict Kirchhoff branch also stores the discrete constraint $\mathbf{c}\left(\mathbf{q}\right)$ and its Jacobian $\mathbf{G}\left(\mathbf{q}\right)$; these two components may be marked as empty in an unconstrained branch.

On this basis, Section 3 constructs two data objects that must not be mixed. The autonomous-backbone data are

$$\mathcal{D}_{sk} = \left(\mathbf{z}_0, \dot{\mathbf{z}}_0, H_{cons}, \nabla_{\mathbf{z}_0} H_{cons}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}^{sk}, \mathbf{Q}_{\partial}\right), \qquad \mathbf{z}_0 = \left(\mathbf{q}, \mathbf{p}_0\right), \tag{2.121}$$

and the exact nonautonomous data are

$$\mathcal{D}_{na} = \left(\mathbf{z}, \dot{\mathbf{z}}, t, H, \nabla_{\mathbf{z}} H, \partial_t H, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}^{H}, \mathbf{Q}_{\partial}\right), \qquad \mathbf{z} = \left(\mathbf{q}, \mathbf{p}\right). \tag{2.122}$$

If $\dot{\mathbf{p}}_{rep}$ or $\dot{\mathbf{z}}_{rep}$ is obtained by numerical differentiation rather than from a solver residual, analytic equations, or automatic differentiation, the differentiation order, time step, and derivative-error estimate must also be recorded. CHNN derivative supervision, energy-balance supervision, and the subsequent KAM error budget may use only representation-consistent data with traceable errors.

2.11.3. *Training-data hierarchy and model labels.* Each sample must store

$$model_level \in \{K, WF, T, S\}, \qquad representation \in \{sk, na, red\}, \tag{2.123}$$

Here, K denotes the Kirchhoff-consistent Beam primary training model, WF the strict weak-form model, T the Timoshenko comparison, and S the three-dimensional solid comparison; sk denotes the autonomous-backbone representation, na the exact nonautonomous representation, and red an engineering reduction.

The data set used for the conservative CHNN and the statistical-learning–KAM interface is restricted to

$$\mathcal{D}_K^{cons} = \{(\mathbf{z}_0, \dot{\mathbf{z}}_0, H_{cons}, \nabla H_{cons}) : model_level = K,\ representation = sk,\ \mathbf{Q}_{port} = 0\}, \tag{2.124}$$

The open-port training set is

$$\mathcal{D}_K^{port} = \{(\mathbf{z}_{rep}, \dot{\mathbf{z}}_{rep}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \mathbf{Q}_{\partial}) : model_level = K\}, \tag{2.125}$$

The high-fidelity comparison set is

$$\mathcal{D}_{HF} = \{X_{HF} - X_K : model_level \in \{WF, T, S\}\}. \tag{2.126}$$

Only after closure, power-sign, and regularity tests may $\mathcal{D}_{HF}$ enter a conservative correction, dissipative correction, or general residual, respectively; the entire high-fidelity discrepancy must not be interpreted as a Hamiltonian perturbation. Samples with different *model_level* and *representation* labels must not be mixed directly without labels and pairing.

2.12. **Section summary.** Starting directly from the complete translational–rotational weak form, this section constructed a joint Galerkin approximation using a centerline expansion and an exponential-map representation of cross-sectional orientation. The compatibility conditions for the reference configuration and strains, the initial conditions, the contact active set, and the mutually exclusive accounting of conservative and nonconservative loads were specified explicitly. The base rod, eccentric attachments, and energy-consistent fluid added mass were collected in a velocity-dependent inertial functional, whose Hessian with respect to generalized velocity gives

$$\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_{e} \mathbf{M}_{e}(\mathbf{q}) + \mathbf{M}_{A}(\mathbf{q}). \tag{2.127}$$

For a general prescribed flow, the generalized momentum is rigorously

$$\mathbf{p} = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \mathbf{b}_{A}(\mathbf{q}, t), \tag{2.128}$$

Both the geometric inertia and the prescribed-flow cross terms are generated by the complete Euler–Lagrange derivative rather than added empirically. The internal restoring force is obtained from the gradient of the full extension–shear–bending–torsion potential, while magnetic and other distributed couples are projected directly through the rotational test functions. The local linearization eliminates double counting of conservative external loads and explicitly includes boundary follower-load and constraint-tangent contributions.

The resulting finite-dimensional nonlinear dynamical systems are provided in unconstrained and Kirchhoff-constrained forms, together with a projection from COMSOL full-order fields to the joint generalized coordinates and a unified output specification. The energy-consistent fluid branch uses $T_A + \mathbf{f}_{FK} + \mathbf{f}_{drag}$, with the drag further split, under a general background flow, into $\mathbf{f}_{drag,diss} + \mathbf{f}_{drag,flow}$; the engineering Morison formulation is used only as a mutually exclusive validation branch. Small rotations, constant mass, and fixed added mass are retained only as reduced approximations. Euler–Bernoulli serves as the COMSOL Kirchhoff-consistent primary training branch after the slenderness and shear-error tolerances are satisfied. These discrete objects support not only the canonical momentum and Hamiltonian construction of Section 3, but also provide the sole upstream interface to the CHNN training in Section 4 and the structure-preservation and KAM validation in Section 5 through $\mathcal{D}_K^{cons}$, $\mathcal{D}_K^{port}$, $\mathcal{D}_{HF}$, port labels, model-level labels, and constraint labels.

3. Finite-Dimensional Hamiltonian Structure and Canonicalization

3.1. Finite-Dimensional Phase-Space Variables and Hamiltonian Formulation.

3.1.1. *Joint Generalized Coordinates.* The joint Galerkin projection of centerline position and cross-sectional orientation developed in Section 2 yields

$$\boxed{\mathbf{q} = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \in \mathbb{R}^{n}, \qquad n = n_a + n_b,} \tag{3.1}$$

where $\mathbf{a}$ denotes the generalized coordinates of centerline translation and $\mathbf{b}$ contains the finite-dimensional coefficients of the local rotational coordinates for cross-sectional orientation. The vector $\mathbf{b}$ is an independent configuration variable describing the bending orientation of the cross section, torsion about the tangent, angular velocity, and rotary inertia; it is not an auxiliary variable derived from the centerline displacement. Thus, $\mathbf{q}$ is the complete joint generalized configuration coordinate for translation and cross-sectional orientation.

The discrete dynamics of the finite-shear-stiffness geometrically exact Cosserat parent model can be written as

$$\boxed{\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_{M}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}_{f}(\mathbf{q}, \dot{\mathbf{q}}, t) + \nabla_{\mathbf{q}} V_{h}(\mathbf{q}, t) = \mathbf{Q}_{port}.} \tag{3.2}$$

The mass–inertia matrix

$$\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_{e} \mathbf{M}_{e}(\mathbf{q}) + \mathbf{M}_{A}(\mathbf{q}) \tag{3.3}$$

contains the translational kinetic energy of the rod, cross-sectional rotational kinetic energy, translational–rotational coupling of eccentric attachments, and energy-consistent added mass. The Hamiltonian structure developed below is therefore always based on the complete bending–torsion model; it does not use a constant-mass, small-rotation, centerline-only, or torsion-deleted surrogate.

3.1.2. *Basic Framework of the Hamiltonian Formulation.* The finite-dimensional Hamiltonian formulation follows the chain[17, 29]

$$\mathbf{q} \to \mathbf{p} \to H(\mathbf{q}, \mathbf{p}) \to \begin{cases} \dot{\mathbf{q}} = \nabla_{\mathbf{p}} H, \\ \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} H + \mathbf{Q}, \end{cases} \to \frac{dH}{dt}. \tag{3.4}$$

This derivation applies to the autonomous conservative backbone and its forced–dissipative extension. A general prescribed flow generates a velocity-linear term in the Lagrangian and consequently shifts the canonical momentum; Section 3.5 gives the complete nonautonomous Hamiltonian formulation.

3.2. **Lagrangian, Momentum, and Hamiltonian of the Autonomous Conservative Backbone.**

3.2.1. *Selection of the Autonomous Potential.* Let the external actuation be controlled by parameters $\boldsymbol{\lambda}(t)$, and choose a reference value $\bar{\boldsymbol{\lambda}}$. Write the discrete potential as

$$V_h(\mathbf{q}, t; \boldsymbol{\lambda}(t)) = V_{cons}\left(\mathbf{q}; \bar{\boldsymbol{\lambda}}\right) + V_{drv}\left(\mathbf{q}, t; \boldsymbol{\lambda}(t), \bar{\boldsymbol{\lambda}}\right), \tag{3.5}$$

and impose

$$V_{cons}\left(\mathbf{q}; \bar{\boldsymbol{\lambda}}\right) := V_h(\mathbf{q}, t; \boldsymbol{\lambda}(t))\big|_{\boldsymbol{\lambda}(t) \equiv \bar{\boldsymbol{\lambda}}}, \qquad V_{drv} := V_h - V_{cons}. \tag{3.6}$$

When the reference parameters are held fixed, no explicit time-dependent term remains; hence

$$\boxed{V_{drv}\left(\mathbf{q}, t; \bar{\boldsymbol{\lambda}}, \bar{\boldsymbol{\lambda}}\right) = 0, \qquad \partial_t V_{cons} = 0.} \tag{3.7}$$

The potential V_{cons} contains the discrete Cosserat strain energy and the conservative magnetic potential, elastic contact potential, and other conservative external potentials selected at the reference operating condition. This definition freezes the actuation parameters rather than arbitrarily selecting a time slice.

3.2.2. *Autonomous-Backbone Lagrangian and Generalized Momentum.* For the autonomous port-closed backbone, the Lagrangian is

$$\boxed{L_{cons}(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\, \dot{\mathbf{q}} - V_{cons}(\mathbf{q}).} \tag{3.8}$$

The canonical momentum conjugate to $\mathbf{q}$ is defined by[17, 29]

$$\mathbf{p}_0 = \frac{\partial L_{cons}}{\partial \dot{\mathbf{q}}}, \tag{3.9}$$

which gives the relation between canonical momentum and generalized velocity

$$\boxed{\mathbf{p}_0 = \mathbf{M}(\mathbf{q})\, \dot{\mathbf{q}}, \qquad \dot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{q})\, \mathbf{p}_0.} \tag{3.10}$$

The subscript 0 distinguishes this quantity from the shifted canonical momentum $\mathbf{p}$ arising under a general prescribed flow in Section 3.5. The $\mathbf{p}_0$ orientation component $\mathbf{p}_{0b}$ is the discrete momentum conjugate to the local coordinate $\mathbf{b}$; it is not, in general, equal to a component of the material angular momentum of any individual cross section.

3.2.3. *Legendre Regularity.* The velocity Hessian is[17, 29]

$$\frac{\partial^2 L_{cons}}{\partial \dot{\mathbf{q}}\, \partial \dot{\mathbf{q}}} = \mathbf{M}(\mathbf{q}). \tag{3.11}$$

If the joint basis contains no velocity redundancy, the material density, rotary inertia, and added-mass parameters satisfy the required physical positivity conditions, and throughout the configuration domain of interest

$$\boxed{\mathbf{y}^T \mathbf{M}(\mathbf{q})\, \mathbf{y} > 0 \quad (\forall \mathbf{y} \neq \mathbf{0}),} \tag{3.12}$$

then $\mathbf{M}(\mathbf{q}) = \mathbf{M}^T(\mathbf{q}) \succ 0$, and the Legendre map is locally invertible. Geometrically, $\mathbf{M}$ is the mass metric on the tangent space of configuration space, while $\mathbf{M}^{-1}$ is the corresponding dual metric on the cotangent space.

For the fixed N-dimensional reduced manifold $\mathcal{Q}_N$, the configuration coordinate satisfies $q_N \in \mathcal{Q}_N$, the generalized velocity satisfies $\dot{q}_N \in T_{q_N}\mathcal{Q}_N$, and the conjugate momentum satisfies $p_N \in T^*_{q_N}\mathcal{Q}_N$. In the same paired-coordinate trivialization, $\dot{p}_N$ is stored as the momentum-rate component paired with p_N; comparisons across charts additionally require the associated connection or coordinate transformation. The mapping from the COMSOL full-order outputs to these canonical cotangent variables is defined by

$$\boxed{q_N = \Pi_{\mathrm{conf},N}(U), \qquad p_N = B_N(q_N)^T P_{\mathrm{full}}, \qquad \dot{q}_N = M_N(q_N)^{-1} p_N, \qquad \dot{p}_N = B_N(q_N)^T L.} \tag{3.13}$$

Here, $B_N(q_N)$ is the tangent lift of the fixed translational–$SO(3)$ basis at the current configuration; P_{full} and L denote the full-order momentum and the unscaled load at the same solver time step; and $M_N = B_N^T E B_N$. The quantity $p_{\mathrm{mass}} = M_N \dot{q}_{\mathrm{native}}$ is used only as a diagnostic. It agrees with $B_N^T P_{\mathrm{full}}$ only when the full-order velocity lies in the selected tangent space and temporal storage is consistent with the projection. Offline comparisons show better closure for the cotangent projection of the full-order momentum, and the solver-time analysis together with the three-level time-step study further supports the momentum definition in Equation (3.13).

3.2.4. *Construction of the Hamiltonian.* The Hamiltonian is defined as

$$H_{cons}(\mathbf{q}, \mathbf{p}_0) = \mathbf{p}_0^T \dot{\mathbf{q}} - L_{cons}(\mathbf{q}, \dot{\mathbf{q}}). \tag{3.14}$$

Substituting $\dot{\mathbf{q}} = \mathbf{M}^{-1}\mathbf{p}_0$ gives

$$\boxed{H_{cons}(\mathbf{q}, \mathbf{p}_0) = \frac{1}{2}\mathbf{p}_0^T \mathbf{M}^{-1}(\mathbf{q})\, \mathbf{p}_0 + V_{cons}(\mathbf{q}).} \tag{3.15}$$

This Hamiltonian has the standard form $H = T + V$ and contains both translational and cross-sectional rotational degrees of freedom. Configuration dependence of the mass matrix does not invalidate the Legendre transformation; its contribution enters the geometric inertial terms through the derivative of the Hamiltonian with respect to $\mathbf{q}$.

3.2.5. *Nonlinear Potential and Restoring Force.* The discrete internal potential of the Cosserat rod is determined by the strain-energy functional in Section 2. Define

$$\boxed{\mathbf{f}_{int}(\mathbf{q}) = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}), \qquad \mathbf{K}_{tan}(\mathbf{q}) = \frac{\partial \mathbf{f}_{int}}{\partial \mathbf{q}} = \nabla^2_{\mathbf{q}} V_{int,h}(\mathbf{q}).} \tag{3.16}$$

Relative to a reference configuration $\mathbf{q}_*$, the conservative potential may also be written as the path-independent line integral

$$V_{int,h}(\mathbf{q}) - V_{int,h}(\mathbf{q}_*) = \int_0^1 \mathbf{f}_{int}(\mathbf{q}_* + s(\mathbf{q} - \mathbf{q}_*))^T (\mathbf{q} - \mathbf{q}_*)\, ds. \tag{3.17}$$

Thus, the conservative potential can be represented by the line integral of the restoring force along a path in configuration space. In general, one cannot assume $\mathbf{f}_{int} = \mathbf{K}_{tan}(\mathbf{q})\, \mathbf{q}$. A

quadratic approximation is justified only in a linearized neighborhood or when the restoring force genuinely admits that factorization:

$$V_{int,h}\left(\mathbf{q}_{*}+\boldsymbol{\eta}\right) \approx V_{int,h}\left(\mathbf{q}_{*}\right)+\frac{1}{2}\boldsymbol{\eta}^{T}\mathbf{K}_{tan}\left(\mathbf{q}_{*}\right)\boldsymbol{\eta}. \tag{3.18}$$

A global quadratic form is generally invalid because both the potential and the restoring force are geometrically nonlinear.

3.3. **Canonical Hamilton Equations and Geometric Inertia.**

3.3.1. *Standard Symplectic Structure.* Define the phase-space variable and the standard symplectic matrix

$$\mathbf{z}_{0}=\begin{bmatrix}\mathbf{q}\\ \mathbf{p}_{0}\end{bmatrix}, \qquad \mathbf{J}=\begin{bmatrix}\mathbf{0} & \mathbf{I}\\ -\mathbf{I} & \mathbf{0}\end{bmatrix}, \qquad \mathbf{J}^{T}=-\mathbf{J}. \tag{3.19}$$

The autonomous conservative backbone satisfies

$$\boxed{\dot{\mathbf{z}}_{0}=\mathbf{J}\nabla_{\mathbf{z}_{0}}H_{cons},} \tag{3.20}$$

that is,

$$\boxed{\dot{\mathbf{q}}=\nabla_{\mathbf{p}_{0}}H_{cons}, \qquad \dot{\mathbf{p}}_{0}=-\nabla_{\mathbf{q}}H_{cons}.} \tag{3.21}$$

The corresponding standard Poisson bracket is

$$\{F,G\}=\nabla_{\mathbf{q}}F^{T}\nabla_{\mathbf{p}_{0}}G-\nabla_{\mathbf{p}_{0}}F^{T}\nabla_{\mathbf{q}}G. \tag{3.22}$$

Because local orientation coordinates and their conjugate momenta are used, the standard canonical bracket applies. If global rotation matrices and material angular momenta were used instead, a Lie–Poisson or constrained symplectic structure would have to be constructed separately.

3.3.2. *Terms Generated by the Configuration-Dependent Mass Matrix.* For the i-th configuration component,

$$\frac{\partial H_{cons}}{\partial q_{i}}=\frac{1}{2}\mathbf{p}_{0}^{T}\frac{\partial\mathbf{M}^{-1}}{\partial q_{i}}\mathbf{p}_{0}+\frac{\partial V_{cons}}{\partial q_{i}}. \tag{3.23}$$

Therefore, the momentum equation is

$$\boxed{\dot{p}_{0i}=-\frac{1}{2}\mathbf{p}_{0}^{T}\frac{\partial\mathbf{M}^{-1}}{\partial q_{i}}\mathbf{p}_{0}-\frac{\partial V_{cons}}{\partial q_{i}}.} \tag{3.24}$$

Using the derivative identity for an inverse matrix,

$$\frac{\partial\mathbf{M}^{-1}}{\partial q_{i}}=-\mathbf{M}^{-1}\mathbf{M}_{,i}\mathbf{M}^{-1} \tag{3.25}$$

one obtains

$$-\frac{1}{2}\mathbf{p}_{0}^{T}\frac{\partial\mathbf{M}^{-1}}{\partial q_{i}}\mathbf{p}_{0}=\frac{1}{2}\dot{\mathbf{q}}^{T}\mathbf{M}_{,i}\dot{\mathbf{q}}. \tag{3.26}$$

This term fully represents the contribution of the configuration-dependent mass matrix in the Hamiltonian momentum equation. Only when the momentum equation is converted back to a second-order acceleration equation must it be combined with $\dot{\mathbf{p}}_{0}$, specifically with the contribution $\dot{\mathbf{M}}\dot{\mathbf{q}}$ contained therein.

3.3.3. *Equivalence with the Euler–Lagrange Equations.* From

$$\dot{\mathbf{p}}_0 = \mathbf{M}\ddot{\mathbf{q}} + \dot{\mathbf{M}}\dot{\mathbf{q}} \tag{3.27}$$

and the preceding Hamiltonian momentum equation, one obtains

$$\boxed{\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M\left(\mathbf{q}, \dot{\mathbf{q}}\right) + \nabla_{\mathbf{q}} V_{cons}\left(\mathbf{q}\right) = \mathbf{0},} \tag{3.28}$$

where

$$\boxed{\mathbf{g}_M = \dot{\mathbf{M}}\dot{\mathbf{q}} - \frac{1}{2}\nabla_{\mathbf{q}}\left(\dot{\mathbf{q}}^T \mathbf{M}\dot{\mathbf{q}}\right).} \tag{3.29}$$

This result agrees with the autonomous conservative Euler–Lagrange equation in Section 2, establishing the equivalence between the geometric-inertia representations in canonical momentum form and in second-order generalized-coordinate form.

3.3.4. *Energy Conservation.* Because $\mathbf{J}$ is skew-symmetric and H_{cons} has no explicit time dependence,

$$\frac{dH_{cons}}{dt} = \nabla H_{cons}^T \mathbf{J} \nabla H_{cons} = 0. \tag{3.30}$$

and hence

$$\boxed{H_{cons}\left(\mathbf{q}(t), \mathbf{p}_0(t)\right) = const.} \tag{3.31}$$

3.4. **Forced–Dissipative Hamiltonian Structure.**

3.4.1. *Unified Entry Point for Nonconservative Generalized Forces.* Nonconservative generalized forces enter uniformly through the right-hand side of the Hamiltonian momentum equation:

$$\boxed{\dot{\mathbf{q}} = \nabla_{\mathbf{p}_0} H_{cons}, \qquad \dot{\mathbf{p}}_0 = -\nabla_{\mathbf{q}} H_{cons} + \mathbf{Q}_{port}.} \tag{3.32}$$

Following the mutually exclusive power accounting established in Sections 1 and 2, write

$$\boxed{\mathbf{Q}_{port} = \mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}.} \tag{3.33}$$

where:

- $\mathbf{Q}_{diss}$ contains only passive components whose power with respect to the structural mechanical energy is nonpositive;
- $\mathbf{Q}_{in}$ contains magnetic control, actuation, and other nonfluid inputs that have not been included in the potential;
- $\mathbf{Q}_{flow}$ contains background-flow drag input, Froude–Krylov input, and representation-dependent reactive-flow effects;
- $\mathbf{Q}_{boundary}$ separately collects terminal forces, terminal moments, and boundary actuation;
- $\mathbf{Q}_{contact}$ denotes contact reactions and the contact port;
- $\mathbf{r}_{\eta}$ denotes only residual model error after known model terms and ports have been removed.

The joint virtual-work projection of these forces and moments was derived in Section 2; the corresponding finite-dimensional generalized forces are used directly here.

3.4.2. *Equivalent Viscous-Damping Special Case.* If, for a particular operating condition, the nonconservative force can be represented equivalently as

$$\mathbf{Q}_{port} = \mathbf{F}_{ext}\left(\mathbf{q}, t\right) - \mathbf{C}\left(\mathbf{q}, \dot{\mathbf{q}}, t\right)\dot{\mathbf{q}}, \qquad \mathbf{C} = \mathbf{C}^T \succcurlyeq 0, \tag{3.34}$$

then $\dot{\mathbf{q}} = \nabla_{\mathbf{p}_0} H_{cons}$ gives

$$\boxed{\dot{\mathbf{p}}_0 = -\nabla_{\mathbf{q}} H_{cons} + \mathbf{F}_{ext} - \mathbf{C}\nabla_{\mathbf{p}_0} H_{cons}.} \tag{3.35}$$

These equations form a forced–damped Hamiltonian special case on the autonomous backbone. The operator $\mathbf{C}$ may be a state-dependent effective dissipation operator and need not be constant. For quadratic drag, only the passive dissipative component enters $\mathbf{C}\dot{\mathbf{q}}$; the background-flow input remains in $\mathbf{Q}_{flow}$. Coulomb friction, contact switching, and impact may be treated as separate generalized-force ports without requiring a single globally smooth dissipation matrix.

If the magnetic and elastic contact forces have not been included in the potential, one may set

$$\mathbf{F}_{ext} = \mathbf{F}_{mag} + \mathbf{F}_{cont} \tag{3.36}$$

If the same actions have already been generated by the gradients of $V_{mag,cons}$ or $V_{cont,el}$, they must not be counted again in $\mathbf{F}_{ext}$.

3.4.3. *Energy Rate.* For an autonomous Hamiltonian,[17, 20, 21]

$$\frac{dH_{cons}}{dt} = \nabla_{\mathbf{q}} H_{cons}^T \dot{\mathbf{q}} + \nabla_{\mathbf{p}_0} H_{cons}^T \dot{\mathbf{p}}_0. \tag{3.37}$$

Substitution of the forced canonical equations causes the symplectic terms to cancel, yielding

$$\boxed{\frac{dH_{cons}}{dt} = \dot{\mathbf{q}}^T \mathbf{Q}_{port}.} \tag{3.38}$$

Under the equivalent-damping condition of Section 3.4.2,

$$\boxed{\frac{dH_{cons}}{dt} = \nabla_{\mathbf{p}_0} H_{cons}^T \left(\mathbf{F}_{ext} - \mathbf{C}\nabla_{\mathbf{p}_0} H_{cons}\right).} \tag{3.39}$$

where

$$-\nabla_{\mathbf{p}_0} H_{cons}^T \mathbf{C} \nabla_{\mathbf{p}_0} H_{cons} \leq 0, \tag{3.40}$$

The sign of the external input power depends on the operating condition. This energy-rate expression clearly distinguishes external input from passive dissipation and is consistent with the port classification.

3.4.4. *Phase-Space Form of the Complete Forced System.* Define

$$\mathbf{B} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix}, \tag{3.41}$$

The complete forced–dissipative system on the autonomous backbone is then

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla_{\mathbf{z}_0} H_{cons} + \mathbf{B}\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_\eta\right).} \tag{3.42}$$

The system is therefore a forced–dissipative Hamiltonian system, or equivalently an open system with a canonical Hamiltonian backbone. It reduces to a classical closed conservative Hamiltonian system only when every port is closed and the Hamiltonian is autonomous. A closed Hamiltonian representation of the dissipative environment would require additional energy-storage or heat-bath degrees of freedom, which are not part of the finite-dimensional variable set constructed in Sections 1 and 2.

3.5. **Canonical Momentum and Hamiltonian under a General Prescribed Flow.**

3.5.1. *General Discrete Lagrangian.* The energy-consistent prescribed-flow formulation in Sections 1 and 2 generates a velocity-linear term. The general discrete Lagrangian is[17, 29]

$$\boxed{L_h(\mathbf{q},\dot{\mathbf{q}},t) = \frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \dot{\mathbf{q}}^T\mathbf{b}_A(\mathbf{q},t) + c_A(\mathbf{q},t) - V_h(\mathbf{q},t)\,.} \tag{3.43}$$

Here, $\mathbf{M}$ still contains the velocity-quadratic added-mass contribution, while $\mathbf{b}_A$ and c_A represent the cross term and the pure-flow term induced by the prescribed flow. Added inertia already included in $\mathbf{M}_A, \mathbf{b}_A, c_A$ must not be reintroduced on the right-hand side as a complete Morison added-mass generalized force.

3.5.2. *General Canonical Momentum and Hamiltonian.* By definition,

$$\mathbf{p} = \frac{\partial L_h}{\partial \dot{\mathbf{q}}} \tag{3.44}$$

one obtains

$$\boxed{\mathbf{p} = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \mathbf{b}_A(\mathbf{q},t)\,, \qquad \dot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{q})\,(\mathbf{p} + \mathbf{b}_A)\,.} \tag{3.45}$$

When $\mathbf{b}_A = \mathbf{0}$, this relation reduces to $\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}$ for the autonomous backbone or the backbone–port representation. When the velocity-linear prescribed-flow term is retained explicitly, the general canonical momentum $\mathbf{p}$ must be used.

The Legendre transformation gives

$$\boxed{H(\mathbf{q},\mathbf{p},t) = \frac{1}{2}(\mathbf{p} + \mathbf{b}_A)^T\mathbf{M}^{-1}(\mathbf{q})\,(\mathbf{p} + \mathbf{b}_A) + V_h(\mathbf{q},t) - c_A(\mathbf{q},t)\,.} \tag{3.46}$$

where $-c_A$ follows directly from the Legendre transformation. If $\mathbf{M} = \mathbf{M}(\mathbf{q})$ has no explicit time dependence, then

$$\boxed{\nabla_{\mathbf{p}} H = \mathbf{M}^{-1}(\mathbf{p} + \mathbf{b}_A) = \dot{\mathbf{q}},} \tag{3.47}$$

$$\boxed{\frac{\partial H}{\partial q_i} = -\frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}_{,i}\dot{\mathbf{q}} + \mathbf{b}_{A,i}^T\dot{\mathbf{q}} + \frac{\partial V_h}{\partial q_i} - \frac{\partial c_A}{\partial q_i},} \tag{3.48}$$

and, at fixed $(\mathbf{q},\mathbf{p})$,

$$\boxed{\left.\frac{\partial H}{\partial t}\right|_{\mathbf{q},\mathbf{p}} = \dot{\mathbf{q}}^T\partial_t\mathbf{b}_A + \partial_t V_h - \partial_t c_A.} \tag{3.49}$$

3.5.3. *Equivalence with the General Euler–Lagrange Equations.* In the absence of external ports, the standard nonautonomous canonical equations still hold:

$$\dot{\mathbf{q}} = \nabla_{\mathbf{p}} H, \qquad \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} H. \tag{3.50}$$

From

$$\dot{\mathbf{p}} = \mathbf{M}\ddot{\mathbf{q}} + \dot{\mathbf{M}}\dot{\mathbf{q}} - \dot{\mathbf{b}}_A \tag{3.51}$$

one recovers

$$\boxed{\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \mathbf{g}_f + \nabla_{\mathbf{q}} V_h = \mathbf{0},} \tag{3.52}$$

where

$$\boxed{\mathbf{g}_f = -\dot{\mathbf{b}}_A + (\nabla_{\mathbf{q}}\mathbf{b}_A)^T\dot{\mathbf{q}} - \nabla_{\mathbf{q}} c_A.} \tag{3.53}$$

This result agrees with the prescribed-flow Euler–Lagrange equation in Section 2. Both $\mathbf{g}_M$ and $\mathbf{g}_f$ are generated automatically by the Legendre transformation and therefore must not be added again as separate generalized forces.

3.5.4. *Exact Representation and Backbone–Port Representation.* When $\mathbf{b}_A, c_A$ are retained, define[17, 29]

$$\mathbf{z} = \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix}, \tag{3.54}$$

The complete forced system is

$$\boxed{\dot{\mathbf{z}} = \mathbf{J}\nabla_{\mathbf{z}} H(\mathbf{z}, t) + \mathbf{B}\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}\right).} \tag{3.55}$$

In the exact representation, $\mathbf{g}_f$ is already contained in the Hamiltonian, so $\mathbf{Q}_{flow}$ does not include an additional $-\mathbf{g}_f$.

The magnetic actuation is represented in one of two mutually exclusive ways. If the conservative magnetic potential is included in

$$H_{sk} = H_{cons} + V_{drv},$$

the same generalized force $-\nabla_q V_{drv}$ must not also enter $\mathbf{Q}_{in}$. If instead

$$H_{sk} = H_{cons}, \qquad \mathbf{Q}_{drv} = -\nabla_q V_{drv},$$

the force enters the input/magnetic port exactly once. When the magnetic action is represented by a potential, its generalized force must equal the gradient-induced force from that potential; when it is represented as an input, only the port force is included. These two descriptions are not used simultaneously, so the same conservative magnetic contribution is counted exactly once.

In the backbone–port representation, take

$$\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}} \tag{3.56}$$

and define

$$H_{sk}(\mathbf{q}, \mathbf{p}_0, t) = \frac{1}{2}\mathbf{p}_0^T \mathbf{M}^{-1} \mathbf{p}_0 + V_h(\mathbf{q}, t), \tag{3.57}$$

The prescribed-flow cross term is then transferred in its entirety to the port:

$$\boxed{\mathbf{Q}_{flow}^{sk} = \mathbf{Q}_{flow} - \mathbf{g}_f.} \tag{3.58}$$

The two representations generate the same configuration trajectories, but each must be used as a complete and internally consistent set:

$$\boxed{(\mathbf{p}, H, \mathbf{Q}_{flow}) \neq \left(\mathbf{p}_0, H_{sk}, \mathbf{Q}_{flow}^{sk}\right).} \tag{3.59}$$

The two variable sets and generating functions must remain paired. The shifted momentum $\mathbf{p}$ is used with the exact Hamiltonian H, whereas $\mathbf{p}_0$ is used with H_{sk}. The term $-\mathbf{g}_f$ must not be added again to the exact Hamiltonian representation.

3.5.5. *General Energy Rate.* The exact representation satisfies

$$\boxed{\frac{dH}{dt} = \frac{\partial H}{\partial t} + \dot{\mathbf{q}}^T\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}\right).} \tag{3.60}$$

The backbone–port representation satisfies

$$\boxed{\frac{dH_{sk}}{dt} = \partial_t V_h + \dot{\mathbf{q}}^T\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow}^{sk} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}\right).} \tag{3.61}$$

These two expressions give the energy rates under nonautonomous potentials, prescribed flow, and multiple ports, and remain dynamically equivalent in their respective variable representations.

3.6. **Range of Applicability and the Strict Kirchhoff Constraint Branch.**

3.6.1. *Local Orientation Coordinates.* Exponential coordinates form a local chart on $SO(3)$. To use standard canonical variables within the current chart, require, for every $S \in [0, L]$,

$$\| \mathbf{\Psi}(S)\mathbf{b} \| < \pi. \tag{3.62}$$

This is a coordinate-chart restriction rather than a small-rotation mechanical approximation. When a trajectory crosses the chart boundary, the orientation coordinates must be switched and the conjugate momentum updated consistently through the corresponding cotangent transformation.

3.6.2. *Strict Kirchhoff Primary Branch and Finite-Shear Validation Branch.* The geometrically exact parent model in Section 1 is a finite-shear Cosserat rod. To remain consistent with the Kirchhoff slender-catheter assumptions and to provide Sections 4 and 5 with a nonredundant canonical state, the primary structured-learning and KAM branches preferentially use a locally reduced representation of the strict shear-free constraint. Write the constraint as[17, 18]

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}, \qquad \mathbf{G}(\mathbf{q}) = \frac{\partial \mathbf{c}}{\partial \mathbf{q}}, \tag{3.63}$$

If $\mathbf{G}$ has constant rank throughout the domain of interest, the constraint set is locally a smooth manifold. In the full coordinates, the equations are

$$\boxed{\dot{\mathbf{q}} = \nabla_{\mathbf{p}} H, \qquad \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} H + \mathbf{Q}_{port} - \mathbf{G}^T \boldsymbol{\lambda}_c, \qquad \mathbf{c}(\mathbf{q}) = \mathbf{0}.} \tag{3.64}$$

The ideal constraint reaction satisfies $\dot{\mathbf{q}}^T \mathbf{G}^T \boldsymbol{\lambda}_c = 0$. To obtain a canonical system that can be used directly by the CHNN and standard symplectic analysis, introduce independent local coordinates $\mathbf{q} = \boldsymbol{\chi}(\mathbf{y})$ and define

$$\mathbf{C}(\mathbf{y}) = \frac{\partial \boldsymbol{\chi}}{\partial \mathbf{y}}, \qquad \mathbf{M}_r(\mathbf{y}) = \mathbf{C}^T \mathbf{M}(\boldsymbol{\chi}(\mathbf{y}))\, \mathbf{C}, \tag{3.65}$$

When $\mathbf{M}_r \succ 0$, construct the reduced Hamiltonian. The reduced momentum of the autonomous backbone is $\mathbf{p}_r = \mathbf{M}_r \dot{\mathbf{y}}$. If the reduced prescribed-flow model still contains a velocity-linear term, use instead $\mathbf{p}_r = \mathbf{M}_r \dot{\mathbf{y}} - \mathbf{b}_{A,r}$. Sections 4 and 5 use only this independent-coordinate canonical system; the differential–algebraic equations containing the multiplier $\boldsymbol{\lambda}_c$ cannot be treated directly as a standard Hamiltonian system on the full space.

The finite-shear-stiffness Cosserat ordinary differential equations are retained for COMSOL comparison, numerical regularization, and constraint-convergence analysis. Data from this regularized branch may replace locally reduced data in CHNN training only when the shear strains, trajectories, energies, and principal frequencies converge to the Kirchhoff results as the shear stiffness increases. This strategy preserves the advantages of the complete parent model while keeping the primary computational object consistent with the original Kirchhoff research objective.

3.7. **Interfaces to CHNN, KAM, and Structure-Preserving Discretization.** Section 3 does not anticipate the neural-network construction or KAM theorem proofs developed in Sections 4 and 5. Instead, based on the HNN, dissipative/port-Hamiltonian network, statistical-learning–KAM, symplectic-algorithm, and parameterization-method literature used there, it specifies the unique and verifiable mathematical objects that connect the preceding and subsequent parts.[19–28]

Second-order BDF with `tsteps` and `tstepsstore=1` is used only to store states at a common solver time step and to generate the vector-field quantities required by Equation (3.13). The three-level time-step comparison assesses temporal consistency and convergence of the canonical-variable mapping, not symplecticity of BDF. Any symplectic or KAM inference requires a separate symplectic map satisfying the assumptions in Section 5.

3.7.1. *Canonical Nondimensionalization, Complex Coordinates, and the Physical Hamiltonian Baseline.* A subsequent complex representation cannot add $\mathbf{q}$ and $\mathbf{p}$ directly when they have different physical dimensions and scales. Choose a stable reference state $(\mathbf{q}_*, \mathbf{p}_*)$, coordinate and momentum scale matrices $\mathbf{S}_q$ and $\mathbf{S}_p$, a time scale t_*, and an energy scale E_*, and define the action scale $\mathcal{A}_* = E_* t_*$. Require

$$\boxed{\mathbf{S}_q^T \mathbf{S}_p = \mathcal{A}_* \mathbf{I}.} \tag{3.66}$$

Define

$$\mathbf{q} = \mathbf{q}_* + \mathbf{S}_q \mathbf{Q}, \qquad \mathbf{p} = \mathbf{p}_* + \mathbf{S}_p \mathbf{P}, \qquad t = t_* \tau, \qquad H = E_* \widehat{H}. \tag{3.67}$$

The scale-compatibility condition then gives

$$\frac{d\mathbf{Q}}{d\tau} = \nabla_{\mathbf{P}} \widehat{H}, \qquad \frac{d\mathbf{P}}{d\tau} = -\nabla_{\mathbf{Q}} \widehat{H}, \tag{3.68}$$

so the nondimensional variables still satisfy the standard canonical Hamilton equations. Define the complex coordinates[17, 29]

$$\boxed{\boldsymbol{\xi} = \frac{\mathbf{Q} + i\mathbf{P}}{\sqrt{2}}, \qquad \overline{\boldsymbol{\xi}} = \frac{\mathbf{Q} - i\mathbf{P}}{\sqrt{2}}.} \tag{3.69}$$

The real and imaginary parts of $\boldsymbol{\xi}$ are canonically paired configuration and momentum variables with compatible scales. A nontrivial real-valued Hamiltonian is generally written as $\widehat{H}\left(\boldsymbol{\xi}, \overline{\boldsymbol{\xi}}\right) \in \mathbb{R}$, rather than as a real-valued holomorphic function of $\boldsymbol{\xi}$ alone.[23, 24]

Section 3 supplies Section 4 with the analytical physical baseline

$$\widehat{H}_{phys} = \widehat{H}_{cons}, \tag{3.70}$$

The real-valued readout in Section 4 should use the physical-residual form

$$\boxed{\widehat{H}_\lambda = \widehat{H}_{phys} + \Delta \widehat{H}_\lambda.} \tag{3.71}$$

The known mass matrix, Cosserat strain energy, and identified conservative magnetic and elastic-contact potentials are not relearned by the network. The term $\Delta \widehat{H}_\lambda$ represents only parameter corrections, conservative corrections induced by Galerkin truncation, and other unmodeled conservative effects. Because a Hamiltonian inferred from vector-field data is identifiable only up to an additive constant, a gauge must be fixed, for example $\widehat{H}_\lambda(\mathbf{z}_*) = 0$ or zero mean over the training domain.[19, 21]

3.7.2. *Backbone–Port Structure and Consistent Data Representations.* In the autonomous representation with time-independent actuation parameters $\boldsymbol{\lambda}(t) \equiv \overline{\boldsymbol{\lambda}}$,

$$V_{drv} = 0, \qquad H_{sk} = H_{cons}, \qquad \partial_t H_{sk} = 0, \tag{3.72}$$

and

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J} \nabla_{\mathbf{z}_0} H_{cons} + \mathbf{B} \left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow}^{sk} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_\eta \right).} \tag{3.73}$$

Section 4 should parameterize or supervise the ports separately rather than allowing a single unstructured residual to absorb every physical contribution:

$$\mathbf{Q}_{port,\theta} = \mathbf{Q}_{diss,\mu} + \mathbf{Q}_{in,\nu} + \mathbf{Q}_{flow,\kappa} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_\eta. \tag{3.74}$$

Here, $\mathbf{Q}_{diss,\mu}$ represents only the passive component with nonpositive power; magnetic control, background-flow input, and boundary actuation remain separate ports; and $\mathbf{r}_\eta$ represents only the residual remaining after the known model and known ports have been removed. This hierarchy is consistent with the structured modeling principles of dissipative SymODEN and port-Hamiltonian neural networks.[20, 21]

For nonfrozen or pulsatile operating conditions, $H_{sk} = H_{cons} + V_{drv}$. The following two representations are mutually exclusive:

1. If the complete H_{sk} is used as the generator, the same conservative actuation force must not also enter $\mathbf{Q}_{in}$;
2. If the autonomous H_{cons} is used as the generator, set $\mathbf{Q}_{drv} = -\nabla_{\mathbf{q}} V_{drv}$ and include it exactly once in $\mathbf{Q}_{in}$.

Under a general prescribed flow, use $\mathbf{p} = \mathbf{M}\dot{\mathbf{q}} - \mathbf{b}_A$ and the corresponding nonautonomous Hamiltonian; the autonomous backbone uses $\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}$. The two data types must be distinguished by a *representation* label. The backbone and nonautonomous data supplied to Section 4 are, respectively,

$$\mathcal{D}_{sk} = \left(\mathbf{Q}, \mathbf{P}, \boldsymbol{\xi}, \dot{\mathbf{z}}_0, \widehat{H}_{cons}, \nabla \widehat{H}_{cons}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}^{sk}_{flow}, \mathbf{Q}_{boundary}, \mathbf{Q}_{contact}, \mathbf{r}_\eta\right), \tag{3.75}$$

$$\mathcal{D}_{na} = \left(\mathbf{z}, \dot{\mathbf{z}}, t, H, \nabla H, \partial_t H, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}^{H}_{flow}, \mathbf{Q}_{boundary}, \mathbf{Q}_{contact}, \mathbf{r}_\eta\right). \tag{3.76}$$

The statistical-learning–KAM interface preferentially uses derivative-supervised autonomous conservative samples and does not interpret the sum of trajectory loss, dissipation loss, and regularization terms directly as the KAM perturbation size.[22]

3.7.3. *Stable Equilibrium, Modal Canonicalization, and the Integrable Reference System.* The subsequent KAM analysis starts from the autonomous conservative backbone $H_{cons}(\mathbf{q}, \mathbf{p}_0)$, without assuming that it is globally integrable. Choose an equilibrium $(\mathbf{q}_*, \mathbf{0})$ satisfying

$$\nabla_{\mathbf{q}} V_{cons}(\mathbf{q}_*) = \mathbf{0}, \qquad \mathbf{M}_* = \mathbf{M}(\mathbf{q}_*) \succ 0, \qquad \mathbf{K}_* = \nabla^2_{\mathbf{q}} V_{cons}(\mathbf{q}_*) \succ 0. \tag{3.77}$$

Solve the generalized eigenvalue problem

$$\mathbf{K}_* \boldsymbol{\phi}_j = \omega_j^2 \mathbf{M}_* \boldsymbol{\phi}_j, \tag{3.78}$$

and choose the modal matrix $\boldsymbol{\Phi}$ such that

$$\boldsymbol{\Phi}^T \mathbf{M}_* \boldsymbol{\Phi} = \mathbf{I}, \qquad \boldsymbol{\Phi}^T \mathbf{K}_* \boldsymbol{\Phi} = \boldsymbol{\Omega}^2. \tag{3.79}$$

The linear canonical transformation

$$\mathbf{q} - \mathbf{q}_* = \boldsymbol{\Phi} \mathbf{Q}_m, \qquad \mathbf{p}_0 = \boldsymbol{\Phi}^{-T} \mathbf{P}_m \tag{3.80}$$

reduces the quadratic Hamiltonian to

$$H_2 = \frac{1}{2}\left(\mathbf{P}_m^T \mathbf{P}_m + \mathbf{Q}_m^T \boldsymbol{\Omega}^2 \mathbf{Q}_m\right) = \sum_{j=1}^{n} \omega_j I_j, \tag{3.81}$$

where

$$x_j = \sqrt{\omega_j} Q_{m,j}, \qquad y_j = \frac{P_{m,j}}{\sqrt{\omega_j}}, \qquad I_j = \frac{x_j^2 + y_j^2}{2}. \tag{3.82}$$

With quadratic terms alone, the frequencies are independent of the actions, and hence

$$\frac{\partial^2 H_2}{\partial \mathbf{I}^2} = \mathbf{0}, \tag{3.83}$$

Thus the linear harmonic oscillator is isochronously degenerate. The condition $\mathbf{K}_* \succ 0$ cannot replace the Kolmogorov nondegeneracy condition; nondegeneracy must be supplied by geometric or material nonlinearities, conservative magnetic potentials, or other higher-order conservative terms.

If H_{cons} is analytic in the neighborhood of interest, or has the regularity required by the selected finitely differentiable KAM theorem, and the linear frequencies avoid low-order resonances to the required order, a finite-order Birkhoff normal form can be constructed:[18, 25, 29]

$$H_{cons} \circ \mathbf{\Psi} = H_* + \boldsymbol{\omega} \cdot \mathbf{I} + Z_4(\mathbf{I}) + \cdots + Z_r(\mathbf{I}) + R_{r+1}(\mathbf{I}, \boldsymbol{\theta}). \tag{3.84}$$

Define

$$\boxed{H_0(\mathbf{I}) = H_* + \boldsymbol{\omega} \cdot \mathbf{I} + Z_4(\mathbf{I}) + \cdots + Z_r(\mathbf{I}), \qquad \varepsilon H_1 = R_{r+1}.} \tag{3.85}$$

The lowest nontrivial case commonly has

$$Z_4(\mathbf{I}) = \frac{1}{2}\mathbf{I}^T \mathbf{A} \mathbf{I}, \tag{3.86}$$

The Kolmogorov nondegeneracy condition over the action domain under consideration then requires

$$\boxed{\det\left(\frac{\partial^2 H_0}{\partial \mathbf{I}^2}\right) \neq 0,} \tag{3.87}$$

Near the origin, this reduces to $\det \mathbf{A} \neq 0$. If this matrix is singular, classical Kolmogorov KAM cannot be claimed; one must increase the normal-form order, choose another reference family, or invoke and verify an isoenergetic or degenerate KAM theorem.

On a fixed complex neighborhood or Sobolev domain $\mathcal{D}$, with a common function norm $\|\cdot\|_{\mathcal{X}(\mathcal{D})}$, write the nondimensional Hamiltonian as

$$\widehat{H}_{cons} = \widehat{H}_0 + \varepsilon P, \qquad \| P \|_{\mathcal{X}(\mathcal{D})} = 1, \tag{3.88}$$

and define

$$\boxed{\varepsilon = \| \widehat{H}_{cons} - \widehat{H}_0 \|_{\mathcal{X}(\mathcal{D})}.} \tag{3.89}$$

The frequency vector must also satisfy the corresponding Diophantine condition. If a low-order resonance is present, the analysis must use a resonant normal form and an appropriate theorem; the resonant term cannot simply be absorbed into an ordinary small perturbation.

3.7.4. *Learning Error, Discretization Error, and the KAM Smallness Budget.* To place the learning error from Section 4 and the symplectic-discretization error from Section 5 in one mathematical chain, all errors must be measured on the same domain and in compatible function norms. At the continuous Hamiltonian level, first define

$$\delta_{model} = \| H_{cons} - H_0 \|_{\mathcal{X}}, \qquad \delta_{learn} = \| H_\lambda - H_{cons} \|_{\mathcal{X}}, \tag{3.90}$$

so that

$$\boxed{\| H_\lambda - H_0 \|_{\mathcal{X}} \leq \delta_{model} + \delta_{learn}.} \tag{3.91}$$

At the discrete level, do not assume unconditionally that a globally exact "discrete Hamiltonian" exists. Let Φ_h^H denote the exact flow of Hamiltonian H over the time interval h, let $F_{\lambda,h}$ denote the conservative symplectic one-step map actually used in Section 5, and set the unperturbed reference map to $F_{0,h} = \Phi_h^{H_0}$. On a fixed compact domain, define in the C^r or analytic norm appropriate to the selected symplectic-map KAM theorem[18, 25]

$$\delta_{alg} = \| F_{\lambda,h} - \Phi_h^{H_\lambda} \|, \qquad \delta_{map} = \| F_{\lambda,h} - F_{0,h} \|. \tag{3.92}$$

The triangle inequality gives

$$\boxed{\delta_{map} \leq \delta_{alg} + \| \Phi_h^{H_\lambda} - \Phi_h^{H_{cons}} \| + \| \Phi_h^{H_{cons}} - \Phi_h^{H_0} \|.} \tag{3.93}$$

If the vector field depends locally Lipschitz-continuously on Hamiltonian perturbations and the flow remains in the same domain over the time interval considered, the last two terms can be bounded by $C_{learn}(h)\delta_{learn}$ and $C_{model}(h)\delta_{model}$, respectively. Hence,

$$\boxed{\delta_{map} \leq \delta_{alg} + C_{learn}(h)\delta_{learn} + C_{model}(h)\delta_{model}.} \tag{3.94}$$

If backward error analysis additionally constructs a modified Hamiltonian for a specified step size and finite time scale, an equivalent modified-Hamiltonian budget may be used. The domain of validity, truncation order, and remainder bound must accompany this object; global exact existence cannot be inferred solely from the statement that the integrator is symplectic.

Persistence of invariant tori can be concluded only when the continuous or discrete budget is below the concrete threshold of the selected KAM theorem, which depends on the analytic domain, Diophantine constants, nondegeneracy margin, dimension, and step size. The statistical-learning result of Chen–Matsubara–Yaguchi directly controls only the conservative HNN branch under its assumptions. Transferring a comparable probabilistic bound to the CHNN requires a real-valued readout and a corresponding complexity analysis of the function class.[22]

Quadratic drag, contact damping and friction, nonpotential magnetic actuation, background-flow input, boundary input, and contact activation or release are not included in the preceding classical autonomous-Hamiltonian perturbation. Their contributions to the complete safe operating domain must instead be treated through energy balance, port gains, structural defects, Lyapunov indicators, or a specialized theory of dissipative invariant tori.

3.7.5. *Symplectic Maps, Birkhoff Structure, and the Torus Equation.* For the autonomous conservative branch, a structure-preserving integrator produces the one-step map

$$F_h = \Phi_h^{cons}. \tag{3.95}$$

Section 5 first verifies whether F_h is an exact symplectic map and then applies the KAM conditions for symplectic maps and symplectic algorithms due to Shang together with the parameterization method of Haro et al.[25–28] The torus embedding $K : \mathbb{T}^d \to \mathcal{M}$ satisfies[28]

$$\boxed{F_h\left(K\left(\boldsymbol{\theta}\right)\right) = K\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right).} \tag{3.96}$$

The invariance residual of a numerical approximation K_N is

$$E_N\left(\boldsymbol{\theta}\right) = F_h\left(K_N\left(\boldsymbol{\theta}\right)\right) - K_N\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right). \tag{3.97}$$

For the complete data vector field, the known conservative backbone and known ports must first be removed:

$$\boxed{X_{res} = X_{data} - \mathbf{J}\nabla H_\lambda - \mathbf{B}\left(\mathbf{Q}_{diss,\mu} + \mathbf{Q}_{in,\nu} + \mathbf{Q}_{flow,\kappa} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact}\right).} \tag{3.98}$$

The "self-adjoint Birkhoff-system representation" used here and the "Hamiltonian Birkhoff normal form" used to construct H_0 in Section 3.7.3 are distinct concepts. The former concerns the variational self-adjointness and $K\left(\mathbf{z}, t\right)$-symplectic structure of a general first-order system, whereas the latter is the Hamiltonian normal form near an elliptic equilibrium. Section 5 tests only whether X_{res} admits a regular self-adjoint Birkhoff-system representation. If the closure, time-compatibility, or nondegeneracy conditions fail, X_{res} remains a general residual and the complete open system cannot be called a strict $K\left(\mathbf{z}, t\right)$-symplectic system. Explicit time dependence alone also does not necessarily imply a nonstandard symplectic structure: in appropriate canonical variables and extended phase space, the exact nonautonomous Hamiltonian branch should still be treated preferentially with standard symplectic theory.[18, 25]

3.7.6. *Structural Decomposition of the Difference between the Primary COMSOL Training Model and High-Fidelity Models.* Section 3 treats the Kirchhoff-consistent COMSOL branch as the primary training dynamics X_K, and the Weak Form, Timoshenko, or solid model as the high-fidelity reference X_{HF}. Define the paired-condition model discrepancy

$$\Delta X_{HF} = X_{HF} - X_K. \tag{3.99}$$

This discrepancy cannot be inserted wholesale into the Hamiltonian. First test its conservative closure in canonical variables. If a real-valued ΔH_{HF} exists such that

$$\Delta X_{HF}^{cons} = \mathbf{J}\nabla\Delta H_{HF}, \tag{3.100}$$

then that component may enter the conservative correction head in Section 4. If a momentum port satisfies

$$\nabla_{\mathbf{P}}\widehat{H}^{T}\Delta\mathbf{Q}_{HF}^{diss} \leq 0, \tag{3.101}$$

it may enter the dissipative correction head. Components that do not satisfy closure, power-sign, or smoothness conditions remain in the general model residual $\mathbf{r}_{HF}$. Thus,

$$\boxed{\Delta X_{HF} = \mathbf{J}\nabla\Delta H_{HF} + \mathbf{B}\Delta\mathbf{Q}_{HF}^{diss} + \mathbf{B}\Delta\mathbf{Q}_{HF}^{port} + \mathbf{r}_{HF}.} \tag{3.102}$$

This decomposition is a structural projection to be identified and validated, not an identity that holds automatically for an arbitrary high-fidelity discrepancy. The model error used in KAM analysis consists only of the conservatively closed component and its bound in the common function norm. Shear dissipation, contact switching, and local solid effects must not be included in H_1 without proof.

3.7.7. *Interfaces for Structured Learning and Stability Analysis.* The preceding derivations yield three interfaces required by the subsequent structured-learning and stability analyses:

$$\mathcal{I}_{CHNN} = \left\{ \begin{matrix} \mathbf{Q}, \mathbf{P}, \boldsymbol{\xi}, \widehat{H}_{phys}, \nabla\widehat{H}_{phys}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \\ \mathbf{Q}_{boundary}, \mathbf{Q}_{contact}, \mathbf{r}_{\eta}, model_level, representation \end{matrix} \right\}, \tag{3.103}$$

$$\mathcal{I}_{KAM} = \left(H_0, H_1, \varepsilon, \boldsymbol{\omega}\left(\mathbf{I}\right), \partial_{\mathbf{I}}\boldsymbol{\omega}, \gamma, \tau, \mathcal{D}, \mathcal{X}\right), \tag{3.104}$$

$$\mathcal{I}_{geom} = \left(\mathbf{J}, F_{0,h}, F_{\lambda,h}, X_{res}, E_N, \delta_{model}, \delta_{learn}, \delta_{alg}, \delta_{map}\right). \tag{3.105}$$

Each object is traceable to the physical model and Galerkin outputs in Sections 1 and 2 and to the canonicalization performed in this section. If an object has not yet been constructed or verified numerically, subsequent sections may list it only as a condition to be tested and must not present the conditional interface as an established theorem conclusion.

3.8. **Section Summary.** Using the complete joint coordinates for translation and cross-sectional orientation, the canonical momentum and Hamiltonian of the autonomous conservative backbone are

$$\boxed{\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}, \qquad H_{cons} = \frac{1}{2}\mathbf{p}_0^{T}\mathbf{M}^{-1}\mathbf{p}_0 + V_{cons},} \tag{3.106}$$

and the forced–dissipative canonical equations are

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla H_{cons} + \mathbf{B}\mathbf{Q}_{port}, \qquad \frac{dH_{cons}}{dt} = \dot{\mathbf{q}}^{T}\mathbf{Q}_{port}.} \tag{3.107}$$

The contribution of the configuration-dependent mass matrix in the Hamiltonian momentum equation is equivalent to the geometric-inertia vector $\mathbf{g}_M$ in Section 2. A nonlinear conservative potential can be represented as a path-independent line integral of the restoring force; under equivalent viscous damping, the energy rate reduces to the sum of external-input power and dissipative power.

Under a general prescribed flow, the canonical momentum and Hamiltonian become

$$\boxed{\mathbf{p} = \mathbf{M}\dot{\mathbf{q}} - \mathbf{b}_A, \qquad H = \frac{1}{2}\left(\mathbf{p} + \mathbf{b}_A\right)^{T}\mathbf{M}^{-1}\left(\mathbf{p} + \mathbf{b}_A\right) + V_h - c_A.} \tag{3.108}$$

This general representation retains cross-sectional orientation, independent torsion, rotary inertia, added mass, and external ports while unifying the autonomous backbone, prescribed-flow dynamics, and forced–dissipative dynamics. Canonical nondimensionalization, complex coordinates, a physical-residual Hamiltonian, Birkhoff normal form, a unified KAM error budget, and

the three interfaces $\mathcal{I}_{CHNN}$, $\mathcal{I}_{KAM}$, and $\mathcal{I}_{geom}$ then enable Section 3 to provide the subsequent analyses with mutually consistent and traceable mathematical objects for structured CHNN learning, symplectic discretization, parameterized-torus validation, and safe-domain computation.

4. Structured Hamiltonian Neural Networks with Complex-Valued Representations

4.1. **Learning Object and Upstream Interfaces.** This section constructs a Hamiltonian neural network with complex-valued representations on the reduced canonical finite-dimensional system and connects it to KAM stability analysis. To keep the physical objects unambiguous, a single complex-valued scalar is not used to generate conservative, dissipative, and driven dynamics simultaneously. Section 3 has already provided the independent canonical variables after local reduction of the strict Kirchhoff constraint, canonical nondimensionalization, the physical Hamiltonian baseline, and mutually exclusive port accounting. The CHNN is defined only on these objects.

Let the reduced nondimensional canonical state be

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Q} \\ \mathbf{P} \end{bmatrix} \in \mathbb{R}^{2d}, \qquad \mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix}, \tag{4.1}$$

and define the complex coordinate

$$\boxed{\boldsymbol{\xi} = \frac{\mathbf{Q} + i\mathbf{P}}{\sqrt{2}}, \qquad \overline{\boldsymbol{\xi}} = \frac{\mathbf{Q} - i\mathbf{P}}{\sqrt{2}}.} \tag{4.2}$$

Here, $\mathbf{Q}$ and $\mathbf{P}$ have undergone a canonical scaling satisfying $\mathbf{S}_q^T \mathbf{S}_p = E_{ref} t_{ref} \mathbf{I}$, so they may be combined as the real and imaginary parts of a complex variable. The dimensional variables $\mathbf{q}$ and $\mathbf{p}$ are not added directly before canonical nondimensionalization.

The COMSOL data also carry

$$model_level \in \{K, WF, T, S\}, \qquad representation \in \{sk, na, red\}, \tag{4.3}$$

which identify, respectively, the Kirchhoff-consistent Beam model, strict weak-form model, Timoshenko model, and solid model, together with the autonomous-backbone, exact nonautonomous, and engineering-reduction representations. Data with different labels are not mixed without a documented compatibility analysis.

4.2. **Conservative Hamiltonian Head and Complex-Valued Internal Representation.**

4.2.1. *Real-Valued Physical Hamiltonian.* A standard HNN generates the conservative vector field from the gradient of a real-valued scalar Hamiltonian.[19] We use

$$\boxed{\widehat{H}_\lambda(\mathbf{Z}) = \widehat{H}_{phys}(\mathbf{Z}) + \Delta\widehat{H}_\lambda(\mathbf{Z}) \in \mathbb{R},} \tag{4.4}$$

where $\widehat{H}_{phys}$ is supplied by the mass matrix, Cosserat/Kirchhoff strain energy, identified conservative magnetic potential, and elastic contact potential for a fixed active set from Section 3. The network learns only material-parameter discrepancies, conservative corrections caused by Galerkin truncation, and other conservatively closed model discrepancies. To remove the additive gauge freedom, impose

$$\widehat{H}_\lambda(\mathbf{Z}_{ref}) = 0 \tag{4.5}$$

or require zero mean over the training domain.

4.2.2. *Complex-Valued Representation and Real-Valued Hamiltonian.* For the open-port formulation, complex-valued features are used only as internal representations, whereas the Hamiltonian supplied to the equations of motion is constrained to be real valued. Specifically,

$$\boxed{h_\theta = \mathcal{N}_\theta^{\mathbb{C}}(\boldsymbol{\xi}, \overline{\boldsymbol{\xi}}), \qquad \Delta H_\theta = \mathcal{R}_{H,\theta}(h_\theta) \in \mathbb{R}, \qquad H_\theta(\mathbf{Z}, \mu) = H_{\text{phys}}(\mathbf{Z}, \mu) + \Delta H_\theta(\mathbf{Z}, \mu) \in \mathbb{R}.} \tag{4.6}$$

The complex layer therefore has no independent interpretation as an energy channel, and its imaginary component is not assigned a physical Hamiltonian meaning. The Lean4 verification addresses the real-valued construction in Equation (4.6); alternative complex-to-scalar mappings lie outside the present analysis. Consequently, the formal result establishes a finite-dimensional representation property but does not address optimization behavior, the physical origin of nonlinear corrections, or the validity of the complete CHNN model.

Let

$$\mathbf{h}_\theta = \mathcal{N}_\theta^{\mathbb{C}}\left(\boldsymbol{\xi}, \overline{\boldsymbol{\xi}}\right), \qquad \Delta \widehat{H}_\lambda = \mathcal{R}_{H,\lambda}\left(\mathbf{h}_\theta\right) \in \mathbb{R}. \tag{4.7}$$

A complex linear layer represents amplitude scaling and phase rotation simultaneously through its real–imaginary block structure. For noncircular phase-space data, both $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$ are retained as inputs, or an equivalent augmented representation is used. A practical CVNN may be treated as a real-differentiable mapping $\mathbb{R}^{2d} \to \mathbb{R}$; global holomorphy of the network is not required.[23, 24] A nonconstant real-valued Hamiltonian generally cannot be a purely holomorphic function on a connected open set, so the physical readout may depend on both $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$.

For the theoretical analysis, the activation functions are chosen to be sufficiently smooth in their real and imaginary arguments, for example

$$\phi_{split}(z) = \tanh(Rez) + i\tanh(Imz). \tag{4.8}$$

modReLU, CReLU, and zReLU may be examined in numerical ablations, but their piecewise smoothness or boundary nondifferentiability prevents their direct use in statistical-learning–KAM arguments that require higher-order regularity. Complex weights use complex Glorot or semi-unitary initialization; the Hamiltonian backbone does not use BatchNorm based on batch statistics by default.[23, 24]

4.2.3. *Widely Linear Complex Layers and Real-Affine Completeness.* To retain an amplitude–phase inductive bias while preserving expressive compatibility with general real networks, the complex backbone uses a widely linear layer. Let

$$\mathbf{z}_{out} = \mathbf{W}\mathbf{z} + \mathbf{V}\overline{\mathbf{z}} + \mathbf{b} \tag{4.9}$$

where $\mathbf{z} = \mathbf{x} + i\mathbf{y}$, $\mathbf{W} = \mathbf{A} + i\mathbf{B}$, $\mathbf{V} = \mathbf{C} + i\mathbf{D}$, and $\mathbf{b} = \mathbf{b}_R + i\mathbf{b}_I$. Then

$$\mathbf{x}_{out} = (\mathbf{A} + \mathbf{C})\,\mathbf{x} + (-\mathbf{B} + \mathbf{D})\,\mathbf{y} + \mathbf{b}_R \tag{4.10}$$

$$\mathbf{y}_{out} = (\mathbf{B} + \mathbf{D})\,\mathbf{x} + (\mathbf{A} - \mathbf{C})\,\mathbf{y} + \mathbf{b}_I \tag{4.11}$$

Proposition 4.1 (Real-affine completeness)**.** *For any real affine mapping*

$$\mathbf{x}_{out} = \mathbf{M}_{11}\mathbf{x} + \mathbf{M}_{12}\mathbf{y} + \mathbf{c}_1 \tag{4.12}$$

$$\mathbf{y}_{out} = \mathbf{M}_{21}\mathbf{x} + \mathbf{M}_{22}\mathbf{y} + \mathbf{c}_2 \tag{4.13}$$

there exist unique $\mathbf{W}, \mathbf{V}, \mathbf{b}$ *for which the widely linear complex layer is exactly equivalent, with the explicit construction*

$$\mathbf{A} = \frac{\mathbf{M}_{11} + \mathbf{M}_{22}}{2}, \qquad \mathbf{C} = \frac{\mathbf{M}_{11} - \mathbf{M}_{22}}{2} \tag{4.14}$$

$$\mathbf{B} = \frac{\mathbf{M}_{21} - \mathbf{M}_{12}}{2}, \qquad \mathbf{D} = \frac{\mathbf{M}_{21} + \mathbf{M}_{12}}{2} \tag{4.15}$$

$$\mathbf{b} = \mathbf{c}_1 + i\mathbf{c}_2 \tag{4.16}$$

Proof. Substituting the stated construction into the real and imaginary equations recovers the four blocks $\mathbf{M}_{11}, \mathbf{M}_{12}, \mathbf{M}_{21}, \mathbf{M}_{22}$. Conversely, solving the four linear block equations gives uniqueness. □

When $\mathbf{V} = \mathbf{0}$, the real block matrix associated with a standard complex-linear layer must commute with the standard complex structure and is therefore a proper subclass of general real linear mappings. We use standard complex-linear layers as a stronger amplitude–phase structural bias and widely linear layers as the default theoretical model when general real-affine expressivity must be retained. This result establishes only a function-class inclusion and does not by itself imply fewer parameters or higher sample efficiency for the complex-valued model.

4.2.4. *Embedding of Real MLPs and Conditional Inheritance of Sobolev Approximation.*

Theorem 4.1 (Layerwise embedding of the real MLP class)**.** *Let a real-valued multilayer perceptron take* $(\mathbf{Q}, \mathbf{P})$ *as input and use a smooth hidden-layer activation satisfying* $\sigma(0) = 0$*. Then there exists a complex network taking* $\boldsymbol{\xi} = (\mathbf{Q} + i\mathbf{P})/\sqrt{2}$ *and* $\overline{\boldsymbol{\xi}}$ *as inputs and using widely linear layers and split activations, whose real-valued output is identical to that of the real MLP.*

Proof. The first layer treats $(Re\boldsymbol{\xi}, Im\boldsymbol{\xi})$ as the realified input and uses Proposition 4.1 to realize the target real affine mapping while setting the target imaginary preactivation to zero. Since $\sigma(0) = 0$, the imaginary channel remains zero after the split activation. Starting from the second layer, the block-diagonal special case of Proposition 4.1 realizes each affine layer of the original real network. The final real-valued readout is therefore identical to that of the original MLP. □

Corollary 4.1 (Conditional inheritance of Sobolev approximation)**.** *Let* $\Omega \subset \mathbb{R}^{2d}$ *be a bounded Lipschitz domain,* $s \in \mathbb{N}$*, and* $1 \leq r < \infty$*. If the class of real MLPs with the chosen activation is dense in* $W^{s,r}(\Omega)$*, then the class of real-valued readouts of the preceding widely linear complex networks is also dense in* $W^{s,r}(\Omega)$*.*

This corollary transfers an existing real-network density result and does not remove the original assumptions on the activation, domain, network width, or derivative approximation.[35] The standard complex-linear subclass is constrained by its block structure and does not automatically inherit the same Sobolev-density conclusion.

The theoretical advantages established by Zhang et al. for complex-reaction networks depend on their specific complex reaction structure, activation, and problem setting.[34] They cannot be generalized directly into the claim that an arbitrary complex-valued CHNN must outperform a real-valued HNN. Here, that work motivates only the possibility of a task-relevant inductive bias from complex structure; actual sample-efficiency and error advantages are evaluated through parameter-matched ablation studies.

4.2.5. *Physical Limits of the Complex Representation.* Consider the complex scalar representation

$$\widetilde{\mathcal{H}} = H + iG \tag{4.17}$$

This is retained only as a special representation layer. A residual vector field may be represented by a real auxiliary potential G only if the corresponding one-form is closed and locally integrable. General quadratic drag, contact friction, magnetic control input, and background-flow effects do not satisfy this condition. The imaginary part is not interpreted directly as physical dissipation; physical energy is always given by the real-valued $\widehat{H}_\lambda$.

4.3. **Open-Port Dynamics and Structured Components.** The metriplectic framework of Coquinot–Morrison requires a Poisson bracket, a symmetric positive-semidefinite metric bracket, energy and entropy variables, and the corresponding degeneracy conditions and entropy-production relation.[30] The current catheter model does not establish an independent thermodynamic entropy field or these degeneracy structures. Therefore, the open-port model is not formulated metriplectically; that framework instead delineates the conditions required by future thermo-viscoelastic extensions and a rigorous entropy-production mechanism.

The complete structured model on the autonomous backbone is

$$\boxed{\dot{\mathbf{Z}} = \mathbf{J}\nabla H_\theta(\mathbf{Z}, \mu) + \mathbf{B}\left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}} + \mathbf{r}_\eta\right),} \tag{4.18}$$

where

$$\mathbf{B} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix}. \tag{4.19}$$

The additive terms in Equation (4.18) have distinct physical roles.

1. $\mathbf{Q}_{\text{diss}}$ represents only the passive component whose power with respect to the structural mechanical energy is nonpositive. For the equivalent-viscous subclass, one may take

$$\mathbf{Q}_{\text{diss}} = -\mathbf{D}_\theta \nabla_{\mathbf{P}} H_\theta, \qquad \mathbf{D}_\theta = \mathbf{L}_\theta \mathbf{L}_\theta^T \succcurlyeq 0, \tag{4.20}$$

- so that $P_{\text{diss}} = \nabla_{\mathbf{P}} H_\theta^T \mathbf{Q}_{\text{diss}} \leq 0$. This relation guarantees nonpositive instantaneous dissipative power for the adopted factorization. It does not imply monotone decay of the total open-system energy and is not, by itself, a dissipative-KAM hypothesis. Contact switching, Coulomb friction, and impact may remain separate ports rather than being forced into a single smooth damping matrix.[20]

2. $\mathbf{Q}_{\text{in}}$ represents magnetic control and other nonfluid inputs not included in the potential. Magnetic actions derivable from a potential at fixed magnetic parameters enter H_{phys}. A time-dependent conservative magnetic potential may either enter the nonautonomous Hamiltonian or enter the input port once as $-\nabla V_{drv}$; the two representations are mutually exclusive.
3. $\mathbf{Q}_{\text{flow}}$ represents the Froude–Krylov input, background-flow drag input, and reactive effects induced by the prescribed flow. The background-flow power is generally sign-indefinite and cannot be absorbed wholesale into a dissipation matrix.
4. $\mathbf{Q}_{\text{boundary}}$ is supplied by the known terminal force, terminal moment, and boundary actuation; $\mathbf{Q}_{\text{contact}}$ separately records contact and contact-reaction ports. Neither is relearned by the network in principle.
5. $\mathbf{r}_\eta$ represents only the residual remaining after the known physical backbone and all known ports have been removed.

This decomposition follows the separation of Hamiltonian, dissipation, and control in Dissipative SymODEN and the treatment of explicit time-dependent forcing and identifiability in port-Hamiltonian neural networks.[20, 21]

Numerical studies of pHNNs show that simultaneous identification of the Hamiltonian, time-dependent forcing, and damping from state trajectories alone may be non-identifiable, with one component compensating for another.[21] We therefore fix the physical Hamiltonian and known ports established in Sections 1 to 3; the network is allowed to learn only conservative corrections, sign-constrained dissipative corrections, and the residual after known terms have been removed. Staged training, port-power supervision, and sparsity or magnitude regularization are used to reduce mutual compensation among these components. These measures are structural constraints designed to mitigate leakage, but uniqueness and stability must still be tested through paired operating conditions and ablation experiments.

4.3.1. *Perturbation Types, Damping Classes, and Their KAM Interpretation.* To prevent distinct physical objects from being collapsed into one εH_1, we use the following type separation:

$$\boxed{\dot{\mathbf{z}} = X_{H_0} + \varepsilon X_{H_1} + \delta\, \mathbf{f}_{\text{diss}} + \mathbf{f}_{\text{ext}} + \mathbf{r}_\eta, \qquad X_{H_1} = \mathbf{J}\nabla H_1.} \tag{4.21}$$

Here X_{H_1} is a Hamiltonian perturbation, $\mathbf{f}_{\text{diss}}$ is a dissipative port, $\mathbf{f}_{\text{ext}}$ is an external/general port, and $\mathbf{r}_\eta$ is residual/model error. Only the Hamiltonian perturbation falls directly within classical Hamiltonian KAM theory. Dissipative terms, external inputs, and model residuals cannot be identified with H_1, nor can their KAM norm be inferred from the training loss alone. The Lean4 formalization represents these four classes separately so that this distinction is preserved in the finite-dimensional structural identities.

General learned damping may be written as $\mathbf{D}_\theta(\mathbf{Z}) \succeq 0$, but the present Lean result guarantees only nonpositive instantaneous dissipative power for the corresponding structure. It neither implies a conformally symplectic identity nor supplies the hypotheses required for dissipative KAM theory. The special choice $\mathbf{D} = \gamma\mathbf{I}$, $\gamma \geq 0$, may be examined as a weak-damping model in future dissipative or conformally symplectic analyses, but no such KAM result is claimed here.

4.3.2. *Lean4 Verification of Structural Identities.* Lean4 is used to verify finite-dimensional structural identities; it is not a claim that the complete physical model or the KAM analysis has been formally proved. The verified relations include the canonical skew structure, cancellation of internal Hamiltonian power, canonical port power, decomposition of the named-port powers, nonpositive passive dissipative power, exclusive accounting of the magnetic representation, and separation of open-port perturbation types. Statements that still depend on derivative existence, integrability, a nonautonomous chain-rule instantiation, or specific physical provenance remain conditional. The complete formal-status matrix is given in Appendix Section C.

4.4. **COMSOL Data Hierarchy and Training Objects.**

4.4.1. *Kirchhoff-Consistent Primary Training Set.* The primary COMSOL training model uses the three-dimensional Euler–Bernoulli Beam interface with geometric nonlinearity, dynamic inertia, and torsional degrees of freedom. In COMSOL 6.3, the Beam interface applies the Euler–Bernoulli formulation to slender beams and neglects transverse shear while retaining three-dimensional displacement, cross-sectional rotation, and torque; geometric nonlinearity is represented in a corotational large-rotation, small-strain form.[4–6] The model is therefore consistent with the strict Kirchhoff branch in the shear-free assumption but is not unconditionally identical to the geometrically exact Cosserat equations.

The conservative training set is restricted to

$$\mathcal{D}_K^{cons} = \left\{ \left(\mathbf{Z}_i, \dot{\mathbf{Z}}_i, \widehat{H}_{phys,i} \right) : model_level = K,\ representation = sk,\ \mathbf{Q}_{port} = 0 \right\}. \tag{4.22}$$

The open-port training set is

$$\mathcal{D}_K^{port} = \left\{ \left(\mathbf{Z}_i, \dot{\mathbf{Z}}_i, \mathbf{Q}_{\text{diss,i}}, \mathbf{Q}_{\text{in,i}}, \mathbf{Q}_{\text{flow,i}}, \mathbf{Q}_{\text{boundary,i}}, \mathbf{Q}_{\text{contact,i}}, \mathbf{r}_{\eta,i} \right) \right\}. \tag{4.23}$$

Only samples in $\mathcal{D}_K^{cons}$ are used when relating a learning-error bound to the small Hamiltonian perturbation required by KAM theory. The total loss of the open-port model has no such interpretation.

The P1 series uses three representative trajectories in the fixed $N = 2$ basis, corresponding to excitation dominated by the first mode, the second mode, and a mixed excitation of the two. Whole-trajectory leave-one-out partitioning prevents time samples from the same physical trajectory from appearing in both training and test data. P1 compares four models under derivative supervision in 60 training runs. P1.5 compares the prescribed physical model without a learned correction against two residual-correction models, also in 60 runs, and P1.6 compares models with and without exact local-equilibrium constraints in 60 runs. Five random initializations are used for each setting to quantify optimization variability; they do not increase

the number of independent physical samples. These experiments isolate the effects of structural assumptions in the small-sample regime, whereas P2 uses an independent 96-trajectory data-generation, partitioning, and model-selection procedure.

4.4.2. *Strict Weak-Form and High-Fidelity References.* When a term-by-term reproduction of the coupled weak form is required, Weak Form PDE or Weak Contribution is used to enter test functions, boundary contributions, and auxiliary variables.[7, 8] Timoshenko Beam and three-dimensional Solid Mechanics models are used to assess shear, local contact, and cross-sectional effects. For paired operating conditions, define

$$\Delta X_{HF} = X_{HF} - X_K. \tag{4.24}$$

Only the component that passes the closure test enters the conservative correction $\mathbf{J}\nabla\Delta H_{HF}$; a component with nonpositive power enters the dissipative correction; and all remaining components remain general residuals. The complete ΔX_{HF} must not be interpreted as a Hamiltonian perturbation or as H_1 in KAM theory.

4.5. **Loss Functions and Staged Training.**

4.5.1. *Conservative Derivative Supervision.* For $\mathcal{D}_K^{cons}$, define

$$\mathcal{L}_H = \frac{1}{N_c}\sum_{i=1}^{N_c} \parallel \dot{\mathbf{Z}}_i - \mathbf{J}\nabla\widehat{H}_\lambda\left(\mathbf{Z}_i\right) \parallel_2^2 . \tag{4.25}$$

If the physical Hamiltonian is known, add supervision of the energy correction

$$\mathcal{L}_{\Delta H} = \frac{1}{N_c}\sum_{i=1}^{N_c} \left|\Delta\widehat{H}_\lambda\left(\mathbf{Z}_i\right) - \Delta\widehat{H}_{ref,i}\right|^2 . \tag{4.26}$$

4.5.2. *Port and Energy-Balance Supervision.* Define the port residual

$$\mathcal{L}_{port} = \frac{1}{N}\sum_{i=1}^{N} \parallel \dot{\mathbf{Z}}_i - \mathbf{J}\nabla H_\theta - \mathbf{B}\left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}} + \mathbf{r}_\eta\right) \parallel_2^2 . \tag{4.27}$$

The energy-balance loss is

$$\mathcal{L}_E = \frac{1}{N}\sum_{i=1}^{N} \left|\frac{dH_\theta}{d\tau} - \nabla_{\mathbf{P}} H_\theta^T\left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}} + \mathbf{r}_\eta\right) - \partial_\tau H_\theta\right|^2 . \tag{4.28}$$

The dissipation-sign constraint is imposed only on $\mathbf{Q}_{\text{diss}}$:

$$\mathcal{L}_{pass} = \frac{1}{N}\sum_{i=1}^{N} \left[\max\left\{0, \nabla_{\mathbf{P}} H_\theta^T \mathbf{Q}_{\text{diss}}\right\}\right]^2 . \tag{4.29}$$

4.5.3. *Trajectory, Gauge, and Symplectic-Structure Losses.* The trajectory loss is

$$\mathcal{L}_{traj} = \frac{1}{N}\sum_{i=1}^{N} \parallel \mathbf{Z}_{i,n+1} - Integrate\left(X_\theta, \mathbf{Z}_{i,n}, \Delta\tau\right) \parallel_2^2 . \tag{4.30}$$

The additive-gauge loss is

$$\mathcal{L}_{gauge} = \left|\widehat{H}_\lambda\left(\mathbf{Z}_{ref}\right)\right|^2 . \tag{4.31}$$

The symplectic-structure loss is applied only to the conservative submap $F_{h,\lambda}^{cons}$:

$$\mathcal{L}_{symp} = \frac{1}{N}\sum_{i=1}^{N} \; \| \left(DF_{h,\lambda}^{cons}\right)^T \mathbf{J} DF_{h,\lambda}^{cons} - \mathbf{J} \; \|_F^2 \, . \tag{4.32}$$

The complete dissipative map is generally not symplectic. The loss must therefore not be imposed on the complete open map and then interpreted as evidence of a strictly symplectic flow.

Training proceeds in three stages. The parameters of $\widehat{H}_\lambda$ are first estimated from the conservative data; the dissipative and external-input terms are then identified while the conservative component is held fixed or updated only weakly; finally, all components are optimized jointly. The connection to statistical-learning estimates and KAM theory uses only the conservative derivative error associated with $\mathcal{L}_H$ and its function-space generalization bound.

The P1/P1.5/P1.6 experiments examine the effects of supervision and structural constraints in the conservative setting. P1.5 compares a baseline containing only the prescribed physical terms with Real-HNN and CHNN residual-correction models, including their computational cost. P1.6 compares models with and without exact enforcement of $f(0) = 0$ and the zero-state Jacobian condition. These small-sample results and the 96-trajectory P2 comparison are reported in Section 6.3. Joint learning under open-port conditions, higher-order function-space error bounds, and their implications for KAM perturbation estimates remain topics for future work.

4.5.4. *CHNN Implementation and Structural Validation.* To realize the preceding formulation numerically, we developed an independent CHNN implementation. The complex-valued operators were verified by unit tests and combined according to the canonical state representation, the gray-box Hamiltonian, the dissipative terms, and the external inputs defined above. The resulting dynamics are

$$H_\theta\left(\mathbf{x}, \boldsymbol{\mu}\right) = H_{phys}\left(\mathbf{x}, \boldsymbol{\mu}\right) + \Delta H_\theta\left(\mathbf{x}, \boldsymbol{\mu}\right), \tag{4.33}$$

$$\dot{\mathbf{x}} = \mathbf{J}\nabla_{\mathbf{x}} H_\theta + \mathbf{B}\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial} + \mathbf{Q}_{contact} + \mathbf{r}_\eta\right), \qquad \mathbf{B} = \begin{bmatrix}\mathbf{0}\\ \mathbf{I}\end{bmatrix}. \tag{4.34}$$

The passive dissipative term is parameterized as

$$\mathbf{Q}_{diss} = -\mathbf{D}_\theta \nabla_{\mathbf{p}} H_\theta, \qquad \mathbf{D}_\theta = \mathbf{L}_\theta \mathbf{L}_\theta^T \succcurlyeq \mathbf{0}, \tag{4.35}$$

which gives

$$\left(\nabla_{\mathbf{p}} H_\theta\right)^T \mathbf{Q}_{diss} \leq 0. \tag{4.36}$$

The numerical implementation follows the preceding decomposition directly. The dimensional canonical variables $(\mathbf{q}, \mathbf{p})$ are first mapped to the nondimensional pair $(\mathbf{Q}, \mathbf{P})$, after which the complex coordinates are formed. The complex-valued network permits both standard complex-linear maps and widely linear maps of the form $\mathbf{W}\boldsymbol{\xi} + \mathbf{V}\overline{\boldsymbol{\xi}} + \mathbf{b}$. The Hamiltonian component always returns a real scalar and uses the gray-box form $H_{phys} + \Delta H_\theta$. Dissipation, magnetic actuation, flow, boundary effects, contact, and unresolved residual dynamics are represented as separate generalized-force terms that enter only the momentum equation through $\mathbf{B}$. Samples generated from different physical models or momentum definitions retain their model origin and canonical representation before training so that mechanically distinct quantities are not combined in the same loss.

The implementation was checked against the structural identities used in the analysis, including the equivalence between widely linear complex layers and real block-linear maps, round-trip consistency of canonical scaling, recovery of the prescribed Galerkin subspace, the additive decomposition of the Hamiltonian, energy orthogonality of the continuous conservative vector field, and nonpositive passive dissipative power. External inputs act only through the momentum equation and do not directly modify the $\dot{\mathbf{q}}$ channel. These checks establish consistency

between the numerical implementation and the mathematical formulation; they do not constitute independent evidence for identification accuracy on the COMSOL catheter data or for KAM stability. The physical Hamiltonian $\widehat{H}_{phys}$, configuration-dependent mass–inertia matrix, nonautonomous terms, and COMSOL–Galerkin projection are determined from the physical model and numerical data considered below.

4.6. **From CHNN Learning Error to the KAM Smallness Condition.**

4.6.1. *The Integrable Reference System Is Not an Automatic Consequence of the Network Real Part.* Section 3 constructs, through a stable equilibrium, modal canonicalization, and a Hamiltonian Birkhoff normal form,

$$H_{cons} = H_0 + \varepsilon H_1. \tag{4.37}$$

The Hamiltonian H_0 must be verified to be integrable and nondegenerate. A real-valued H_λ is not integrable merely because it is real. The linear harmonic oscillator satisfies $\partial_{\mathbf{I}}^2 H_2 = 0$, so nondegeneracy must arise from higher-order conservative terms.

4.6.2. *Conservative Training Data, Function Class, and Additive Gauge.* The following statistical-learning statements apply only to the conservative training set $\mathcal{D}_K^{cons}$. Let $\Omega \subset \mathbb{R}^{2d}$ be a bounded connected Lipschitz domain, let the true conservative Hamiltonian be $H_\star$, and let the samples $\mathbf{Z}_1, \ldots, \mathbf{Z}_N$ be independent and identically distributed according to a probability measure $\mathbb{P}_Z$. Assume noiseless derivative labels satisfy

$$\dot{\mathbf{Z}}_i = \mathbf{J}\nabla H_\star\left(\mathbf{Z}_i\right). \tag{4.38}$$

Represent the complex parameters of the Hamiltonian network by the real parameter vector $\boldsymbol{\vartheta} \in \Theta = [-B, B]^P$, and denote the learning error by

$$e_{\boldsymbol{\vartheta}} = \widehat{H}_{\boldsymbol{\vartheta}} - H_\star. \tag{4.39}$$

State derivatives alone do not identify the absolute constant of the Hamiltonian. Therefore, impose on the theoretical domain

$$\int_\Omega e_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right)\, d\mathbf{Z} = 0, \tag{4.40}$$

or use a known energy anchor to select the same representative of the additive equivalence class. Choose a statistical exponent $r > 2d$ and define the scalar gradient loss

$$g_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) = \parallel \dot{\mathbf{Z}} - \mathbf{J}\nabla \widehat{H}_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) \parallel_{\ell^r}^r = \parallel \nabla e_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) \parallel_{\ell^r}^r . \tag{4.41}$$

To apply a uniform generalization bound for bounded losses, also assume $0 \le g_{\boldsymbol{\vartheta}} \le c$. This boundedness is an independent regularity assumption and does not follow automatically from a small empirical loss.

4.6.3. *Covering Numbers and Rademacher Complexity.* For a sample $S = (\mathbf{Z}_1, \ldots, \mathbf{Z}_N)$, define the normalized empirical metric

$$d_S(g, g\prime) = \left(\frac{1}{N} \sum_{i=1}^{N} \left| g\left(\mathbf{Z}_i\right) - g\prime\left(\mathbf{Z}_i\right) \right|^2 \right)^{1/2}. \tag{4.42}$$

Assumption 4.1 (Samplewise Lipschitz continuity from parameters to loss)**.** For a given sample S, there exists a finite constant L_S such that every $\boldsymbol{\vartheta}, \boldsymbol{\vartheta\prime} \in \Theta$ satisfies

$$d_S\left(g_{\boldsymbol{\vartheta}}, g_{\boldsymbol{\vartheta\prime}}\right) \le L_S \parallel \boldsymbol{\vartheta} - \boldsymbol{\vartheta\prime} \parallel_2 . \tag{4.43}$$

When the input domain and parameter set are bounded, the network depth is finite, and the first and second derivatives of the activation are bounded on the relevant range, this assumption can hold on a compact parameter set. In general, L_S grows with depth, width, weight bounds, and derivative bounds.

Lemma 4.1 (Finite-parameter covering number)**.** *Under Assumption 4.1, let* $\mathcal{G} = \{g_{\boldsymbol{\vartheta}} : \boldsymbol{\vartheta} \in \Theta\}$*, and let* $\mathcal{G}_S$ *denote the restriction of this function class to the sample* S*. Then*

$$\mathcal{N}(\varepsilon, \mathcal{G}_S, d_S) \leq \left(1 + \frac{2B\sqrt{P}\, L_S}{\varepsilon}\right)^P \tag{4.44}$$

Proof. On the parameter cube $[-B, B]^P$, place a grid of Euclidean radius ε/L_S, and use Assumption 4.1 to push the parameter covering forward to a covering of the loss functions. □

Because the parameter set is compact and the loss is continuous in the parameters, the supremum is attained. Define the empirical Rademacher complexity by

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_{\boldsymbol{\sigma}}\left[\max_{g \in \mathcal{G}} \frac{1}{N} \sum_{i=1}^{N} \sigma_i g(\mathbf{Z}_i)\right] \tag{4.45}$$

Let

$$R_S = \max_{g \in \mathcal{G}} d_S(g, 0) \tag{4.46}$$

Dudley's entropy integral gives[36]

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) \leq \inf_{0<\alpha<R_S}\left[4\alpha + \frac{12}{\sqrt{N}} \int_{\alpha}^{R_S} \sqrt{\log \mathcal{N}(\varepsilon, \mathcal{G}_S, d_S)}\, d\varepsilon\right] \tag{4.47}$$

A purely complex-linear network is a structurally constrained subclass of the corresponding real network. A widely linear network must instead be counted using its actual number P of real parameters and its actual Lipschitz constant L_S; "one complex parameter" must not be incorrectly counted as one real degree of freedom.

4.6.4. *From Gradient Risk to Uniform Hamiltonian Error.* Assume that the loss class is fixed before sampling. A standard uniform Rademacher generalization bound then implies that, with probability at least $1 - \delta$, simultaneously for every $\boldsymbol{\vartheta} \in \Theta$,

$$\mathbb{E}_{\mathbb{P}_Z} g_{\boldsymbol{\vartheta}} \leq \widehat{\mathcal{L}}_r(\boldsymbol{\vartheta}) + 2\widehat{\mathfrak{R}}_S(\mathcal{G}) + 3c\sqrt{\frac{\log(2/\delta)}{2N}} =: B_{N,\delta}(\boldsymbol{\vartheta}). \tag{4.48}$$

Assume further that $\mathbb{P}_Z$ has a density f_Z with respect to Lebesgue measure and that throughout Ω,

$$f_Z(\mathbf{Z}) \geq m_Z > 0. \tag{4.49}$$

Then, on the preceding uniform event,

$$\| \nabla e_{\boldsymbol{\vartheta}} \|_{L^r(\Omega;\ell^r)} \leq \left(\frac{B_{N,\delta}(\boldsymbol{\vartheta})}{m_Z}\right)^{1/r}. \tag{4.50}$$

Using the additive gauge, the Poincaré inequality, and the Morrey–Sobolev embedding for $r > 2d$,[37] one obtains:

Theorem 4.2 (Uniform generalization bound for the Hamiltonian)**.** *Under the preceding assumptions, with probability at least* $1 - \delta$*, simultaneously for every* $\boldsymbol{\vartheta} \in \Theta$*,*

$$\boxed{\| \widehat{H}_{\boldsymbol{\vartheta}} - H_\star \|_{L^\infty(\Omega)} \leq C_{\Omega,r}(1 + C_P)\left(\frac{B_{N,\delta}(\boldsymbol{\vartheta})}{m_Z}\right)^{1/r}.} \tag{4.51}$$

where C_P is the Poincaré constant. The overall $1/r$ power arises from converting the r-th-power risk back to a gradient norm and cannot be absorbed into a constant independent of the training error and sample size.

4.6.5. *Higher-Order Sobolev or Analytic Error Certificate.* Theorem 4.2 provides only a C^0-level uniform error. A high-frequency, small-amplitude function may converge to zero in C^0 while its higher derivatives diverge. Therefore, neither a first-derivative risk nor proximity of Hamiltonian values alone is sufficient to claim the KAM smallness condition.

Assume that Section 3 has constructed an integrable reference and coordinate transformation on the local action–angle domain

$$\mathcal{A} = \mathbb{T}^M \times U \tag{4.52}$$

If the original training data are represented in physical canonical coordinates, the learning error must be pulled back to $\mathcal{A}$, and bounds on the higher derivatives of the coordinate transformation and its inverse must be included in the constants. Choose integers s, ℓ and a statistical exponent r satisfying

$$s > \ell + \frac{2M}{r}. \tag{4.53}$$

Let the training algorithm output $\widehat{\boldsymbol{\vartheta}}$. In addition to the first-order statistical certificate from Theorem 4.2, require a higher-order certificate for this specific output: on an event of probability at least $1 - \eta$,

$$\sum_{2 \le |\alpha| \le s} \| D^\alpha \left(\widehat{H}_{\widehat{\boldsymbol{\vartheta}}} - H_\star \right) \|^r_{L^r(\mathcal{A})} \le C_{s,\eta} \left(\widehat{\boldsymbol{\vartheta}} \right)^r . \tag{4.54}$$

The certificate must control higher derivatives of the true learning error, not merely the magnitude of the network's own derivatives. Periodic Sobolev embedding then gives, on the intersection of the two events,

$$\| \widehat{H}_{\widehat{\boldsymbol{\vartheta}}} - H_\star \|_{C^\ell(\mathcal{A})} \le \mathfrak{C}_{N,\delta,\eta} \left(\widehat{\boldsymbol{\vartheta}} \right), \tag{4.55}$$

where one may take

$$\mathfrak{C}_{N,\delta,\eta} = C_{s,r,\mathcal{A}} \left[(C_P^r + 1) \frac{B_{N,\delta} \left(\widehat{\boldsymbol{\vartheta}} \right)}{m_Z} + C_{s,\eta} \left(\widehat{\boldsymbol{\vartheta}} \right)^r \right]^{1/r} . \tag{4.56}$$

If the higher-order certificate is obtained by deterministic interval validation or interpolation with a rigorous remainder, set $\eta = 0$. If it is itself probabilistic, independence from the generalization event is not assumed, and the union bound yields only a joint probability of at least $1 - \delta - \eta$.

If an analytic KAM theorem is selected, the preceding finitely smooth certificate is still insufficient; the analytic supremum norm of the learning error on a specified complex neighborhood must be controlled separately. Explicit dependence of a widely linear network on $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$ does not prevent the real-variable Hamiltonian from admitting a real-analytic extension, but smallness in a real-domain C^ℓ norm does not automatically imply smallness in a complex-neighborhood analytic norm.

4.6.6. *Accounting for the Integrable Reference, Pre-Existing Physical Perturbation, and Learning Error.* Fix the integrable reference Hamiltonian $h(\mathbf{I})$ constructed and verified in Section 3. Write the true conservative system and the learned system as

$$H_\star = h + P_0, \qquad \widehat{H}_{\widehat{\boldsymbol{\vartheta}}} = H_\star + e_{\widehat{\boldsymbol{\vartheta}}} = h + P_{\widehat{\boldsymbol{\vartheta}}}, \tag{4.57}$$

where

$$P_{\widehat{\boldsymbol{\vartheta}}} = P_0 + e_{\widehat{\boldsymbol{\vartheta}}}. \tag{4.58}$$

Therefore, in the common $C^\ell(\mathcal{A})$ norm,

$$\| P_{\widehat{\vartheta}} \|_{C^\ell} \leq \| P_0 \|_{C^\ell} + \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\vartheta}\right). \tag{4.59}$$

If the true system is already only nearly integrable, P_0 consumes part of the KAM perturbation budget; the entire threshold cannot be assigned to the learning error. Coordinate-transformation, Galerkin-truncation, parameter-identification, and model-closure errors may be added through the triangle inequality only after they have been controlled in the same function-space norm used by the KAM theorem. General dissipation, friction, background-flow input, and nonpotential magnetic control are not Hamiltonian perturbations here and remain in the open-port branches of Section 4.

4.6.7. *Probabilistic Implication for the KAM Small-Perturbation Condition.* Select an external KAM theorem in advance. Let $\mathsf{A}_{KAM}(h, P)$ denote all of its structural assumptions other than perturbation smallness, including the action–angle domain, regularity order, type of nondegeneracy, Diophantine condition, distance to the boundary, and admissible perturbation class. Let ε_{KAM} be its smallness threshold and let $\mathcal{K}(h + P)$ denote the torus-persistence conclusion supplied by that theorem.

Proposition 4.2 (Probabilistic implication)**.** *Under the preceding assumptions, with probability at least* $1 - \delta - \eta$*,*

$$\boxed{\mathsf{A}_{KAM}\left(h, P_{\widehat{\vartheta}}\right) \ \wedge \ \left(\| P_0 \|_{C^\ell(\mathcal{A})} + \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\vartheta}\right) < \varepsilon_{KAM}\right) \ \Rightarrow \ \mathcal{K}\left(\widehat{H}_{\widehat{\vartheta}}\right).} \tag{4.60}$$

Proof. On the intersection of the uniform-generalization event and the higher-order-certificate event, Section 4.6.5 gives a C^ℓ bound for the learning error. The triangle inequality in Section 4.6.6 then makes the total Hamiltonian perturbation smaller than ε_{KAM}. If all remaining conditions in A_{KAM} also hold, the selected KAM theorem applies. By the union bound, the probability of the joint event is at least $1 - \delta - \eta$. □

This proposition converts the stated learning-error bound only into the small-perturbation hypothesis of an external KAM theorem. The regularity class, measure estimate, frequency properties, and geometry of the persistent tori remain those of the chosen KAM theorem and cannot be inferred from the training loss alone.

4.6.8. *Continuous and Discrete Error Budgets.* On a common analytic domain $\mathcal{D}_\rho$, define

$$\varepsilon_{model} = \| X_{cons} - X_0 \|_\rho, \qquad \varepsilon_{learn} = \| X_\lambda - X_{cons} \|_\rho . \tag{4.61}$$

Let Φ_h^0 be the exact map of X_0 over a time step h, and let $\Psi_{h,\lambda}$ be an order-s_a symplectic algorithm applied to the learned Hamiltonian. If the vector fields and their first derivatives are uniformly bounded and the trajectories remain in a smaller analytic domain $\mathcal{D}_{\rho\prime}$, then there exists a constant $C_{\rho,\rho\prime}$ such that

$$\boxed{\| \Psi_{h,\lambda} - \Phi_h^0 \|_{\rho\prime} \leq C_{\rho,\rho\prime} h \left(\varepsilon_{model} + \varepsilon_{learn} + h^{s_a}\right) =: \varepsilon_{map}.} \tag{4.62}$$

Here, h^{s_a} arises from the local algorithmic error. This budget does not assume unconditionally that a globally exact "discrete Hamiltonian" exists. Only when

$$\varepsilon_{map} < \delta_{KAM}\left(\rho\prime, \gamma, \tau, \text{nondegeneracy margin}, h\right) \tag{4.63}$$

and the analyticity, Diophantine nonresonance, and step-size nonresonance conditions are satisfied may a specific symplectic-map KAM theorem be invoked.[25–27, 33]

4.7. **Three Discrete-Dynamics Branches.**

1. **Autonomous conservative case.** Apply a generating-function method, implicit midpoint rule, or another algorithm proved to be symplectic to $\dot{\mathbf{Z}} = \mathbf{J}\nabla\widehat{H}_\lambda$, yielding F_h^{cons}. This map is used for Shang's KAM theory for symplectic algorithms and for validation by the Haro parameterization method.
2. **Exact nonautonomous Hamiltonian case.** Apply the standard nonautonomous canonical equations to $H(\mathbf{Z}, \tau)$, or introduce the extended variables (τ, p_τ) and $\mathcal{H} = H + p_\tau$ and use a symplectic algorithm in extended phase space. Explicit time dependence alone does not require a nonstandard $K(\mathbf{z}, t)$-symplectic structure.
3. **Open-port case.** Integrate the complete forced–dissipative system consistently while monitoring its energy balance. Its map is generally neither a standard symplectic map nor automatically a Birkhoff system. A $K(\mathbf{z}, t)$-symplectic formulation of the type discussed in Section 5 is applicable only if the residual vector field, after subtraction of the known Hamiltonian and port contributions, satisfies the required self-adjoint representation condition.

4.8. **Output Objects for Invariant-Torus Analysis.** The objects required by the invariant-torus analysis in Section 5 are therefore

$$\mathcal{O}_{4\to5} = (F_h^{cons}, \varepsilon_{map}, X_{res}, \mathcal{D}, \gamma, \tau, \text{nondegeneracy margin}, model_level, representation), \tag{4.64}$$

where

$$\boxed{X_{res} = X_{data} - \mathbf{J}\nabla H_\theta - \mathbf{B}\left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}}\right).} \tag{4.65}$$

If any quantity has not yet been constructed or verified numerically, subsequent sections list it only as a condition to be tested rather than as an established conclusion.

5. Symplectic Discretization, Birkhoff Structures, and Invariant-Torus Validation

5.1. **Analysis branches.** A fixed-dimensional Galerkin truncation has already been completed before the KAM analysis. The principal theorem should therefore use a finite-dimensional Hamiltonian-flow theorem or a finite-dimensional symplectic-map theorem. Kuksin's infinite-dimensional KAM theory[32] is not used as a direct theorem for the fixed truncated system; it is retained only as a theoretical extension for discussions of the limit of infinitely many modes, the PDE limit, or consistency of the truncation. Moser's invariant-curve theorem for area-preserving annulus maps[31] primarily concerns two-dimensional twist maps. It may be used for low-dimensional sections and methodological comparison, but it cannot replace the Pöschel/Shang conditions required for multi-degree-of-freedom continuous Hamiltonian systems or high-dimensional symplectic maps.[26, 27, 33]

The systems considered here do not all share the same symplectic structure. We therefore distinguish among:

- standard $\mathbf{J}$-symplectic discretization of an autonomous conservative Hamiltonian system;
- standard nonautonomous or extended-phase-space symplectic discretization of an exact nonautonomous Hamiltonian system;
- time-dependent $\mathbf{K}(\mathbf{z}, t)$-symplectic discretization of a Birkhoff system that passes the variational self-adjointness test.

The Hamiltonian Birkhoff normal form is used in Sections 3 and 4 to construct $H_0 + \varepsilon H_1$. The self-adjoint Birkhoff representation considered here is used to test the structure of a general first-order system. The names are similar, but the mathematical objects are different.

5.2. **Standard symplectic formulation and generating functions.** For the autonomous conservative CHNN

$$\dot{\mathbf{Z}} = \mathbf{J}\nabla\widehat{H}_\lambda(\mathbf{Z}), \tag{5.1}$$

the exact flow satisfies

$$(D\Phi_t)^T \mathbf{J}(D\Phi_t) = \mathbf{J}. \tag{5.2}$$

If the one-step map F_h^{cons} is an exact symplectic map sufficiently close to the identity and the relevant Jacobian block of the chosen projection is nondegenerate, a local generating function exists. Retaining the leading term of a center-type generating function gives the implicit midpoint rule

$$\frac{\mathbf{Z}_{n+1} - \mathbf{Z}_n}{h} = \mathbf{J}\nabla\widehat{H}_\lambda\left(\frac{\mathbf{Z}_{n+1} + \mathbf{Z}_n}{2}\right), \tag{5.3}$$

which is a second-order symplectic method. Higher-order methods may be constructed through generating-function expansions, composition methods, or other proven symplectic algorithms[25].

For a nonautonomous Hamiltonian $H(\mathbf{Z}, t)$, introduce the extended phase space

$$\mathcal{H}(\mathbf{Z}, t, p_t) = H(\mathbf{Z}, t) + p_t, \tag{5.4}$$

and integrate with respect to the extended canonical symplectic form. A transition to the formulation of the next subsection is needed only when the variables are noncanonical or when the system possesses a special Birkhoff representation.

The second-order BDF scheme used to generate canonical data is not treated as a symplectic integrator. It is used only to obtain time-consistent COMSOL state samples and vector-field labels. The three-level convergence study assesses the temporal consistency of this data mapping; neither the observed energy drift nor the second-order convergence of BDF is used to infer symplecticity. Subsequent numerical KAM calculations must employ a separately justified symplectic map and quantify its discretization error.

5.3. **Self-adjoint representation of a Birkhoff system.** Let $\mathbf{z} \in \mathbb{R}^{2n}$, let $F_i(\mathbf{z}, t)$ be Birkhoff functions, let $B(\mathbf{z}, t)$ be the Birkhoffian, and define

$$K_{ij}(\mathbf{z}, t) = \frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}. \tag{5.5}$$

The Birkhoff equations are

$$\boxed{K_{ij}(\mathbf{z}, t)\,\dot{z}_j - \frac{\partial B}{\partial z_i} - \frac{\partial F_i}{\partial t} = 0.} \tag{5.6}$$

Consider a general system written as

$$K_{ij}\dot{z}_j + D_i = 0. \tag{5.7}$$

On a local star-shaped domain, if $\mathbf{K}$ is nondegenerate and satisfies skew symmetry, the spatial closure condition, and the temporal compatibility condition, then the system admits a regular self-adjoint Birkhoff representation. Equivalently, functions F_i and B can be found such that the preceding equation holds[8, Chap. 15]. Define the time-dependent two-form

$$\Omega_K = \frac{1}{2}K_{ij}(\mathbf{z}, t)\; dz_i \wedge dz_j. \tag{5.8}$$

Its phase flow satisfies the time-dependent structure-preservation relation

$$(D\Phi_{t,t_0})^T \mathbf{K}(\Phi_{t,t_0}, t)\, D\Phi_{t,t_0} = \mathbf{K}(\mathbf{z}_0, t_0). \tag{5.9}$$

When $\mathbf{K} = \mathbf{J}^{-1}$ is constant, this condition reduces to the canonical symplectic condition.

5.4. **Identification of a Birkhoff structure for the residual vector field.** The analysis is applied only to X_{res}, as defined in Section 4. Parameterize the one-form directly as

$$\mathcal{F}_\rho = \sum_{i=1}^{2n} F_{\rho,i}\left(\mathbf{z},t\right)\, dz_i, \qquad K_{\rho,ij} = \partial_{z_i} F_{\rho,j} - \partial_{z_j} F_{\rho,i}, \tag{5.10}$$

introduce B_β, and define the candidate vector field

$$\widetilde{X}_{\rho,\beta} = \mathbf{K}_\rho^{-1}\left(\nabla B_\beta + \partial_t \mathbf{F}_\rho\right). \tag{5.11}$$

The training loss is

$$\mathcal{L}_{Birk} = \frac{1}{N}\sum_{\ell=1}^{N} \parallel \mathbf{K}_\rho X_{res} - \nabla B_\beta - \partial_t \mathbf{F}_\rho \parallel_2^2 + \mu_{nd}\frac{1}{N}\sum_{\ell=1}^{N}\left[\max\{0, \kappa - \sigma_{min}\left(\mathbf{K}_\rho\right)\}\right]^2. \tag{5.12}$$

On the domain $\mathcal{D}$, define the defect of the self-adjoint Birkhoff representation by

$$\boxed{\delta_{Birk} = \inf_{F,B,\ \inf_{\mathcal{D}} \sigma_{min}(K_F)\geq\kappa} \parallel K_F X_{res} - \nabla B - \partial_t F \parallel_{\mathcal{X}(\mathcal{D})}.} \tag{5.13}$$

If $\delta_{Birk} = 0$, an exact representation is obtained. If $\delta_{Birk} > 0$, only the projected vector field $\widetilde{X}$ has an exact K-symplectic structure, and the remaining error must be controlled separately. The complete open system containing a nonstructured residual cannot therefore be described as a strictly $K\left(\mathbf{z},t\right)$-symplectic system.

5.5. **A $K\left(\mathbf{z},t\right)$-symplectic discretization for Birkhoff systems.** Under regularity, self-adjointness, analyticity, and nondegeneracy conditions, the Birkhoff phase flow admits a gradient map and a scalar generating function under an appropriate local projection. If the generating function is analytic in the step size, it can be expanded as

$$S\left(\mathbf{Z},h\right) = \sum_{r=1}^{\infty} S_r\left(\mathbf{Z}\right) h^r. \tag{5.14}$$

Truncation at order m gives an implicit order-m $K\left(\mathbf{z},t\right)$-symplectic method satisfying

$$\left(D\Psi_h\right)^T \mathbf{K}\left(\Psi_h\left(\mathbf{z}\right), t+h\right) D\Psi_h = \mathbf{K}\left(\mathbf{z},t\right) + O\left(h^{m+1}\right). \tag{5.15}$$

This result guarantees structure preservation and the stated order when its hypotheses hold; it does not by itself imply persistence of classical Hamiltonian KAM tori. If a nonstructured residual is superposed, the resulting structural defect must enter the error budget.

5.6. **Local quartic Birkhoff normal form and qualitative KAM existence.** For the selected material-scalar conservative reference, the equilibrium is first recentered in the common modal coordinates. The four linear frequencies of the reconstructed local Hamiltonian are

$$\frac{\boldsymbol{\omega}}{2\pi} = (5.56299685,\ 5.60626502,\ 34.66176454,\ 35.09209478)\ \text{Hz}. \tag{5.16}$$

The cubic and quartic Hamiltonian terms are then reconstructed and a fourth-order nonresonant Birkhoff normalization is performed. After the local canonical Lie transformation, the fourth-order jet of the selected exact quartic conservative approximation satisfies

$$J_4 K = \boldsymbol{\omega}\cdot\mathbf{I} + Z_4(\mathbf{I}), \qquad R_{\text{exact}} = K - J_4 K = O(\|W\|^5). \tag{5.17}$$

The polynomial-algebra check gives a quartic homological residual of 2.58368×10^{-80}. For the primary J_{ABC} reconstruction, the twist matrix

$$T = \nabla_{\mathbf{I}}^2 Z_4(\mathbf{I}) \tag{5.18}$$

has full rank, $\operatorname{rank} T = 4$, with dimensionless minimum singular value 6.35822×10^{-11}. The three auxiliary matrices J_{AB}, J_{AC}, and J_{BC}, each omitting one modal pack, also have rank four. Because a rigorous bound on model-identification error is not yet available, these values establish

local nondegeneracy only for the selected quartic approximation and are not interpreted as a uniform robust twist bound for the material-reference system.

With $W = \varepsilon w$, the Hamiltonian normalized by the quartic homogeneous scale is

$$K_\varepsilon(w) = \varepsilon^{-4} K(\varepsilon w) = \varepsilon^{-2} H_2(w) + Z_4(\mathbf{J}) + \varepsilon R_{5,\varepsilon}(w). \tag{5.19}$$

Because the selected exact quartic conservative approximation and the generators χ_3 and χ_4 are polynomial, the local canonical transformations are analytic. On a fixed small complex neighborhood there is a finite constant C such that

$$\|P_\varepsilon\| \le C\varepsilon \longrightarrow 0. \tag{5.20}$$

The normal-form frequency map is

$$\boldsymbol{\Omega}_\varepsilon(\mathbf{J}) = \varepsilon^{-2}\boldsymbol{\omega} + T\mathbf{J}, \qquad D\boldsymbol{\Omega}_\varepsilon = T, \tag{5.21}$$

so $\det T \neq 0$ yields a locally open frequency image.

Proposition 5.1 (Local qualitative KAM existence for the exact quartic conservative approximation)**.** *For the exact quartic conservative approximation defined above, there exists an unspecified $\varepsilon_* > 0$ such that, for sufficiently small nonzero amplitudes $0 < \varepsilon < \varepsilon_*$, a positive-measure Cantor family of Lagrangian invariant tori exists near and approaches the elliptic equilibrium.*

The proposition follows the classical Kolmogorov KAM route: analyticity follows from the finite polynomial Hamiltonians and local canonical transformations, Equation (5.20) provides the small-perturbation condition, and $\operatorname{rank} T = 4$ gives Kolmogorov nondegeneracy. Removing resonant frequencies from the open frequency image leaves a Diophantine Cantor subset[33, 38]. This local statement does not require the entire prescribed action region to be resonance free. In fact, the finite-order K32 scan detects a resonance crossing within that region, so no numerical KAM conclusion is drawn for the full region. The proposition provides no explicit ε_*, Diophantine constant, torus measure fraction, or specific torus embedding.

The result is restricted to the selected J_{ABC} quartic conservative approximation. On the sampled local8 set, its maximum relative error with respect to the full material-reference Hamiltonian is 4.43237×10^{-8}; this number is a finite-sample error estimate rather than a supremum bound on a neighborhood. Because no rigorous $O(5+)$ neighborhood bound is available, Proposition 5.1 cannot be extended to a KAM theorem for the full material-reference Hamiltonian.

5.7. **Invariant-torus computation by the parameterization method.** For the exact symplectic one-step map F_h^{cons} of the autonomous conservative system, seek an embedding

$$K : \mathbb{T}^d \to \mathcal{M} \tag{5.22}$$

satisfying

$$\boxed{F_h^{cons}\left(K\left(\boldsymbol{\theta}\right)\right) = K\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right).} \tag{5.23}$$

Use the Fourier truncation

$$K_N\left(\boldsymbol{\theta}\right) = \sum_{|k| \le N} \widehat{K}_k e^{2\pi i k \cdot \boldsymbol{\theta}}, \qquad \widehat{K}_{-k} = \overline{\widehat{K}_k}, \tag{5.24}$$

and define the invariance residual

$$E_N\left(\boldsymbol{\theta}\right) = F_h^{cons}\left(K_N\left(\boldsymbol{\theta}\right)\right) - K_N\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right). \tag{5.25}$$

FFT analysis, Poincaré sections, and long trajectories are used only to initialize K_N and $\boldsymbol{\omega}$; a Newton-like method is then used for correction. The a posteriori KAM theorems of Haro et al. require the map to be analytic and exact symplectic, the frequency to satisfy a Diophantine condition, the dynamics to be nondegenerate, and $\| E_N \|$ to satisfy an explicit smallness condition. Existence of a true invariant torus can be certified only after all constants have been verified. A small residual alone supports only the statement that a numerical approximate torus has been obtained[28].

5.8. **Parameter continuation and safe operating domain.** Continuation is performed with respect to magnetic-field strength, insertion speed, flow velocity, contact clearance, and material parameters. At each parameter point, record:

- ε_{map} and the Hamiltonian learning error;
- the symplectic or K-symplectic structural defect;
- the invariance residual $\| E_N \|$;
- the minimum singular value of the twist matrix/frequency map;
- the finite-order Diophantine margin and decay of the Fourier coefficients;
- the energy balance, port power, peak contact response, and trajectory error;
- the discrepancy between the Kirchhoff model and the high-fidelity model.

Define the KAM-theoretic guarantee domain as

$$\mathcal{S}_{KAM} = \{\mu : \text{the symplecticity, nondegeneracy, nonresonance, and a posteriori smallness conditions all hold}\}. \tag{5.26}$$

The complete engineering safe domain is defined as

$$\boxed{\mathcal{S}_{safe} = \mathcal{S}_{KAM} \cap \mathcal{S}_{energy} \cap \mathcal{S}_{contact} \cap \mathcal{S}_{model} \cap \mathcal{S}_{control}.} \tag{5.27}$$

Here, KAM theory provides only a conditional long-time stability guarantee for the conservative backbone. Open ports, dissipation, contact switching, and high-fidelity model discrepancy are addressed by additional indicators. The safe domain is neither a single "chaos threshold" nor evidence of clinical efficacy.

At present, Proposition 5.1 establishes local qualitative KAM existence for the selected quartic conservative approximation at sufficiently small amplitude, but $\mathcal{S}_{KAM}$, $\mathcal{S}_{safe}$, a specific invariant-torus embedding K_N, and its a posteriori constants have not been instantiated numerically. The result therefore does not constitute finite-domain numerical KAM certification, a rigorous KAM result for the material-reference system, or an engineering safe domain.

6. Numerical Experiments and Results

6.1. **Experimental design and evaluation.** The numerical study is organized at four levels. First, P0 compares a generic vector-field network, Real HNN, standard complex-linear CHNN, and widely linear CHNN on a four-dimensional nonlinear coupled oscillator to examine the basic behavior of the different Hamiltonian parameterizations. Second, P1, P1.5, and P1.6 use three representative trajectories (mode1, mode2, and mixed12) in the fixed $N = 2$ representation to study derivative supervision, gray-box residual correction, and hard equilibrium constraints under limited data. The COMSOL trajectories are then used to assess the mapping from field variables to canonical states, momenta, and generalized forces, after which the 96-trajectory P2 data set is generated. Finally, the open-port calculations examine whether the discrete inertia represented by the engineering finite-element solver is mechanically consistent with the canonical inertia used in this work; this question is separate from predictive accuracy on P2.

The evaluation metrics include vector-field or one-step state error, multistep rollout error, linearization error, energy drift, canonical-dynamics closure error, and observed order of temporal convergence. Perturbation-energy drift and generalized-force closure measure different physical quantities and are therefore evaluated separately. The P2 data are partitioned in advance by whole trajectory into 64 training, 16 validation, and 16 independent-test trajectories; no trajectory is shared between subsets, model selection uses validation data only, and the test set is reserved for the final evaluation. The local quartic conservative approximation is used separately for Birkhoff-normal-form, twist-nondegeneracy, and qualitative KAM analysis and is not part of the predictive-model comparison.

6.2. **Consistency of the COMSOL–Galerkin mapping.** The numerical assessment of the COMSOL–Galerkin mapping includes stationary energy and field exports, modal consistency on two meshes, the first 12 computed modes, and three 1 s trajectories. Peak-to-peak perturbation-energy drift is assessed primarily against a 0.5% criterion: only mode2 satisfies this condition,

whereas all three trajectories satisfy an exploratory 1% threshold. Generalized-force consistency is evaluated independently by a joint relative-L2 error on a common temporal support with a common initialization convention, and all three trajectories remain below 2%. Table 6.1 and Fig. 6.1 report the two quantities separately. Energy drift characterizes an integrated energetic effect, whereas generalized-force closure compares the dynamical right-hand side directly; the two tests are therefore complementary rather than interchangeable.

Table 6.1. Trajectory-level comparison of perturbation-energy drift and generalized-force closure error for the COMSOL–Galerkin mapping. The two quantities assess energy consistency and force consistency, respectively.

Trajectory	Energy drift (%)	$< 0.5\%$	$< 1\%$	Generalized-force joint L2 (%)	$< 2\%$
mode1	0.815090	fail	pass	1.513677	pass
mode2	0.329369	pass	pass	0.062551	pass
mixed12	0.523328	fail	pass	1.080748	pass

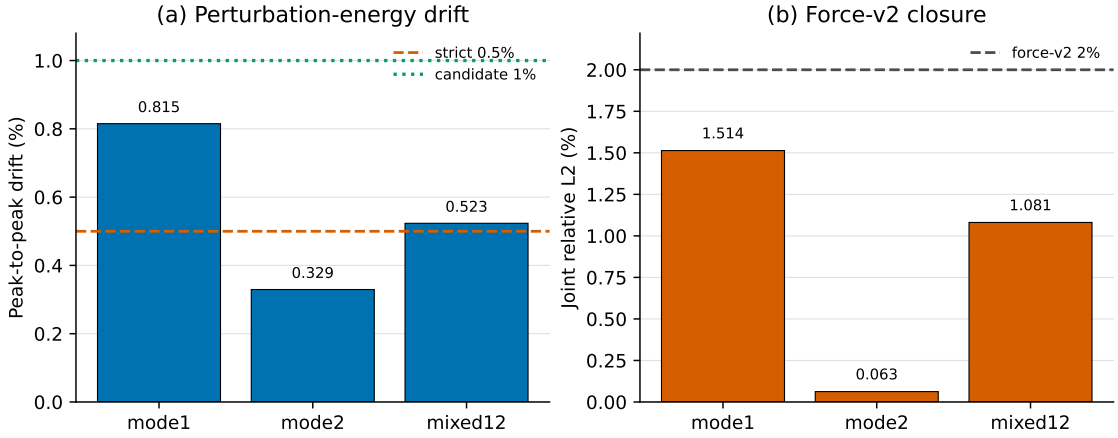


Figure 6.1. Perturbation-energy error and generalized-force closure error in the COMSOL–Galerkin mapping. The left panel shows the primary 0.5% criterion and the exploratory 1% reference level; the right panel shows the joint relative-L2 generalized-force error and the 2% criterion.

6.3. **Identification results for the conservative system.**

6.3.1. *P0 low-dimensional architecture comparison.* P0 uses 80 training, 16 validation, and 20 test trajectories of the same four-dimensional coupled oscillator. Each trajectory contains 80 time steps with $\Delta t = 0.025$; every model is trained for 300 epochs with five random-seed repetitions. With a quadratic physical baseline, standard CHNN gives a vector-field relative-L2 error of 0.0109144 and a multistep-rollout relative-L2 error of 0.0351959, compared with 0.0122422 and 0.0444270 for Real HNN. Its energy drift, 0.00455759, is slightly larger than the Real-HNN value of 0.00402084. P0 therefore serves as a low-dimensional implementation and parameterization comparison rather than as a substitute for the COMSOL-based P2 experiment.

6.3.2. *Structural ablations in P1, P1.5, and P1.6.* P1 contains 60 training runs. On the trajectory excluded from each leave-one-trajectory-out fold, standard CHNN yields joint vector-field relative errors of 0.1417%–0.2444%, multistep-rollout errors of 0.2131%–1.5237%, and linearization errors of 0.3125%–0.3286%. Across the 15 matched comparisons formed by the five random seeds, standard CHNN has lower errors than all three baselines for each of these metrics. This observation is restricted to the three fixed trajectories and the derivative-supervised objective.

P1.5 also contains 60 training runs. Relative to the physical baseline without residual correction, adding a learned residual reduces the mean joint vector-field error by 90.37%–97.29%

Table 6.2. Model comparison on the P0 quadratic-physics benchmark. The vector-field network does not use a Hamiltonian physics baseline.

Model	Vector-field relative L2	Rollout relative L2	Energy drift
Vector-field network	0.0200837	0.103076	0.0245256
Real HNN	0.0122422	0.0444270	0.00402084
Standard CHNN	**0.0109144**	**0.0351959**	0.00455759
Widely linear CHNN	0.0121700	0.0458872	0.00403167

and the mean multistep-rollout error by 53.40%–95.14%. Compared with gray-box Real HNN, standard CHNN has the lower joint vector-field error in 11 of 15 comparisons, but its mean relative reduction is −2.195%. Its rollout error is lower in 12 of 15 comparisons, with a mean reduction of 20.915%, while training and evaluation require approximately 3.04× and 2.54× the corresponding time. The benefit of standard CHNN in this experiment is therefore metric dependent and comes at a higher computational cost; it does not establish a consistent one-step vector-field advantage.

P1.6 imposes $f(0) = 0$ and the zero-state Jacobian as hard constraints on Real HNN and standard CHNN. All 30 constrained models satisfy these conditions to numerical precision, but predictive errors increase substantially. For Real HNN, the mean joint-field/rollout errors increase from $1.832 \times 10^{-3}/1.439 \times 10^{-2}$ to $2.872 \times 10^{-2}/1.011 \times 10^{-1}$; for standard CHNN, they increase from $1.819 \times 10^{-3}/1.195 \times 10^{-2}$ to $2.623 \times 10^{-2}/7.594 \times 10^{-2}$. Thus, exact enforcement of the local equilibrium structure does not necessarily improve finite-sample predictive accuracy. Table 6.3 summarizes the three ablations.

Table 6.3. Principal results and domains of validity for the P1 small-sample experiments. Each experiment uses five random-seed repetitions while retaining the same three physical trajectories.

Experiment	Numerical result	Domain of validity
P1	Standard-CHNN joint-field, multistep-rollout, and linearization errors are 0.1417%–0.2444%, 0.2131%–1.5237%, and 0.3125%–0.3286%.	Restricted to a derivative-supervised small-sample comparison on three physical trajectories.
P1.5	Mean reductions of the residual-corrected model relative to the physics-only model are 90.37%–97.29% for the field and 53.40%–95.14% for multistep rollout; the mean field-error reduction of CHNN relative to the gray-box Real HNN is −2.195%.	The improvement depends on the metric and comes at higher computational cost; it does not establish a consistent one-step vector-field advantage.
P1.6	All 30/30 constrained models satisfy the equilibrium constraints exactly, while the mean Real/CHNN joint-field errors worsen to $2.872 \times 10^{-2}/2.623 \times 10^{-2}$.	Constraint satisfaction does not imply improved accuracy.

6.3.3. *Canonical-variable mapping and time-step convergence.* The canonical-variable mapping is examined with respect to four independent issues: whether compared quantities belong to the same physical trajectory, whether momentum is defined by the appropriate cotangent projection, whether states and generalized forces correspond to the same solver time level, and whether the fixed-basis normalization and sign convention are applied consistently to both initial displacement and velocity. Same-trajectory comparisons first eliminate false closure caused by trajectory mismatch. A repeated mode1 case can still exhibit closure error, showing that modal mixing is not required for the anomaly. Further comparisons indicate that insufficient $N = 2$ truncation is not the dominant source of error. The momentum obtained by cotangent

projection of the full-order momentum is also more consistent with the canonical definition used here than $M_N \dot{q}_{\text{native}}$, which is constructed directly from the reduced mass matrix and the solver velocity.

For temporal discretization, the internal stage state of the generalized-α method is not uniquely accessible through the public interface, making it difficult to ensure that state, momentum, and generalized-force quantities refer to the same internal stage. Second-order BDF can store the required quantities at a common solver time step and is therefore used only to obtain temporally consistent states and vector-field quantities; no claim of symplecticity is made. The initialization study additionally shows that the same fixed-basis normalization and sign transformation must be applied to both displacement and velocity. After this correction, all 12 offline initialization checks and the short-time canonical closure of the mixed case satisfy the prescribed conditions.

A two-level time-step comparison changes the joint relative-L2 closure error from 0.00147337 to 0.00155680, which is insufficient to establish monotone convergence and motivates a third level. The three-level calculation compares q, the cotangent-projected momentum p, $M_N^{-1}p$, $B^T L$, and total energy directly, giving observed convergence orders between 1.64186 and 2.00088. On the extended 1 s trajectory, the joint, pointwise, and integrated relative-L2 closure errors are 0.00146226, 0.000221162, and 0.0000563606, respectively, and the peak-to-peak perturbation-energy drift is $0.00307057 < 0.005$. These results establish the temporal consistency of the mapping used to generate P2 states, momenta, generalized forces, and sampling times. The numerical values are summarized in Table 6.4 and Fig. 6.2.

Table 6.4. Three-level time-step convergence of the canonical-variable mapping and errors on the extended 1 s trajectory.

Calculation	Metric	Value	Criterion
Three-level	Observed order of q	1.9978492	pass
Three-level	Observed order of cotangent-projected momentum p	2.0008768	pass
Three-level	Observed order of $M_N^{-1}p$	2.0008769	pass
Three-level	Observed order of $B^T L$	1.9976914	pass
Three-level	Observed order of total energy	1.6418624	pass
1 s	Joint relative-L2 closure error	0.00146226	< 0.02
1 s	Pointwise relative-L2 closure error	0.000221162	< 0.02
1 s	Integrated relative-L2 closure error	0.0000563606	< 0.02
1 s	Peak-to-peak perturbation-energy drift	0.00307057	< 0.005

6.3.4. *P2: training and independent testing on 96 trajectories.* P2 contains 96 prospective COMSOL trajectories. Each trajectory spans 0.2 s with $\Delta t = 2.5 \times 10^{-4}$ s, yielding 801 sampled states and 800 one-step transitions. After the prescribed trajectory-level quality checks, the data are partitioned by whole trajectory into 64 training, 16 validation, and 16 independent-test trajectories, with no time samples shared across subsets. Table 6.5 summarizes the data partition and training settings.

The four trainable models are Residual MLP, Real HNN, Standard CHNN, and Gray CHNN; the latter uses the gray-box Hamiltonian parameterization $H_2 + \Delta H_\theta$. Each model is trained for 300 epochs with five random seeds using Adam, batch size 256, learning rate 2×10^{-3}, weight decay 10^{-6}, a global gradient-norm clipping threshold of 10, and no early stopping. For each architecture, the final model is selected solely by the primary validation metric; the independent test set is not used for model selection or hyperparameter tuning.

Table 6.6 gives the results on the 16 independent-test trajectories. Gray CHNN attains a normalized full-state one-step MSE of 1.62675×10^{-3}. At step 800, its full-state, q, and p rollout errors are 4.57008, 8.89004, and 0.250117, respectively. Among the remaining models, Real HNN has the lowest one-step MSE and full-state rollout error at step 800. Relative to this baseline, Gray CHNN reduces these two errors by approximately 98.20% and 99.07%; its q- and

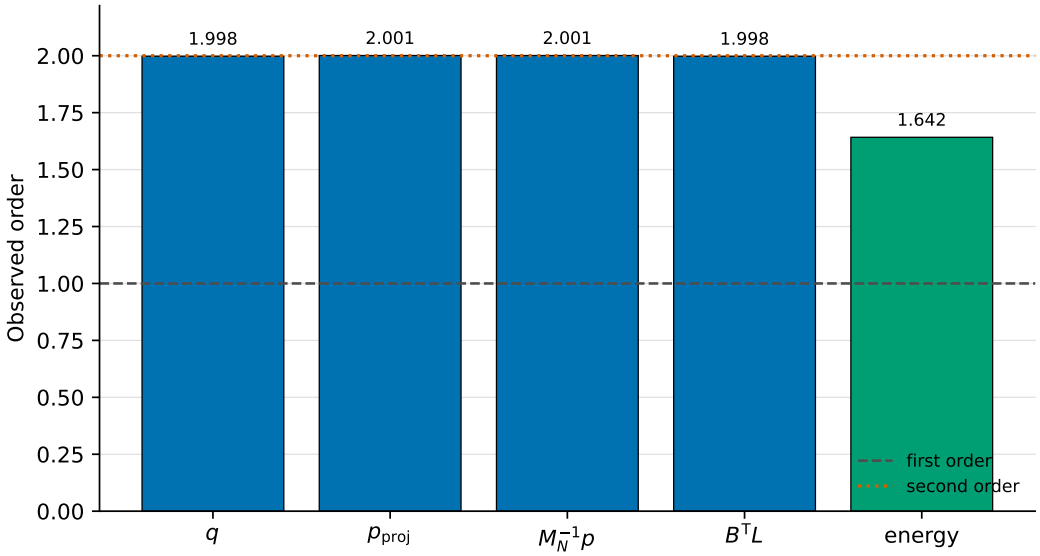


Figure 6.2. Observed convergence orders of the canonical-variable mapping under three time-step levels. The values range from 1.64186 to 2.00088, all above first order. This comparison assesses temporal consistency of the mapping and is not a test of BDF symplecticity.

Table 6.5. Data partition, training settings, and independent-test protocol for the 96 COMSOL trajectories. Model selection uses validation data only, and the test set is evaluated once after model selection is complete.

Item	Setting or result
COMSOL trajectories	All 96 trajectories satisfied the trajectory-level quality checks; each spans 0.2 s with $\Delta t = 2.5 \times 10^{-4}$ s and 801 sampled states.
Whole-trajectory partition	64 training, 16 validation, and 16 independent-test trajectories; no trajectory appears in more than one subset.
Trainable models	Residual MLP, Real HNN, Standard CHNN, and Gray CHNN.
Training repetitions	Five random seeds per model and 300 epochs per run, for 20 completed training runs.
Optimization	Adam optimizer, batch size 256, learning rate 2×10^{-3}, weight decay 10^{-6}, gradient-clipping threshold 10, and no early stopping.
Model selection	One seed per architecture selected solely by the primary validation metric; test data are not used for model selection.
Independent test	The four selected models are evaluated once on the same 16 test trajectories, with no subsequent parameter update, model reselection, or hyperparameter tuning.

p-errors at step 800 are lower by approximately 98.13% and 99.95%. Residual MLP develops errors several orders of magnitude larger under long-horizon rollout, whereas Real HNN and Standard CHNN remain at comparable overall error levels.

Thus, for the conservative canonical representation, the 96-trajectory data distribution, and validation-based model selection used here, Gray CHNN clearly outperforms the three baselines in both one-step and long-horizon prediction. This conclusion is specific to the P2 conservative task and does not establish general superiority at other modal orders, under open-port conditions, or for other magnetically actuated catheter problems; it also does not constitute numerical KAM certification.

Table 6.6. Model comparison on the 16 independent-test trajectories. All models use the same normalization and multistep-error definitions, and the test data are not used for model selection.

Model	One-step MSE	State error at step 800	q@800	p@800
Residual MLP	3.39838×10^{6}	1.42659×10^{7}	2.80846×10^{7}	4.47319×10^{5}
Real HNN	9.01604×10^{-2}	4.90473×10^{2}	4.75301×10^{2}	5.05645×10^{2}
Standard CHNN	9.01604×10^{-2}	4.90474×10^{2}	4.75304×10^{2}	5.05644×10^{2}
Gray CHNN	$\mathbf{1.62675 \times 10^{-3}}$	**4.57008**	**8.89004**	**0.250117**

6.4. **Relation between solver-discrete inertia and canonical inertia under open-port conditions.** The open-port analysis addresses a question that is independent of predictive error: whether the discrete mechanical quantities represented by the engineering finite-element solver admit the same state, momentum, and inertia interpretation as the canonical Hamiltonian model. The results below are organized around this mechanical-equivalence question; detailed extraction settings and the complete numerical evidence table are given in Section D and Table 4.1.

6.4.1. *Trajectory reproducibility and interpretation of generalized-force labels.* The open-port operating conditions produce solver-admissible transient trajectories whose state evolution is reproducible across repeated calculations. This supports extraction of energies, residuals, and tangent quantities at the same states, but it establishes only the reliability of the trajectories themselves. It does not by itself show that generalized-force labels constructed from the solver outputs have the correct discrete mechanical interpretation. Accordingly, the early open-port learning calculations are used only to demonstrate executability of the learning workflow and are not treated as final predictive-performance evidence.

6.4.2. *Physical Hamiltonian defined from scalar kinetic energy.* The raw `E/Ec` matrices assembled by COMSOL are treated here as discrete operators of the DAE equations rather than directly as a Hamiltonian kinetic metric. The physical inertia metric is instead defined as the Hessian of the scalar kinetic energy `beam.Wk_tot` with respect to the retained velocity coordinates and is denoted by $M_{\mathrm{energy}}(q)$. It is symmetric positive definite and satisfies

$$p = M_{\mathrm{energy}}(q)\dot{q}, \qquad \frac{\partial T}{\partial p} = \dot{q}.$$

Together with the elastic and magnetic potential energies, this defines

$$H_{\mathrm{phys}}(q,p) = T(q,p) + V_{\mathrm{elastic}}(q) + V_{\mathrm{mag}}(q).$$

Reassembling residual and tangent quantities while preserving the solver state reproduces these energy and weak-form contributions. The elastic, magnetic, and remaining non-inertial weak-form terms agree with their corresponding energy or virtual-work constructions, with a relative error of approximately 1.17×10^{-11} after superposition. The inertia discrepancy discussed below therefore cannot be attributed simply to these non-inertial terms or to an internal inconsistency of H_{phys}.

6.4.3. *Internal consistency of the solver inertial operator.* To distinguish reproducibility of the solver inertia from equivalence with the canonical inertia, the inertial contribution to the residual is first extracted in independent ways. The response obtained by differentiating the mass contribution agrees with an independent tangent representation to relative error below 4×10^{-17}, while a separate residual-based calculation closes to approximately 5.02×10^{-5}; the zero-rate contribution is negligible relative to the tolerance used here. Thus, the solver inertial response can be identified stably and reproducibly, but this internal consistency alone does not imply equality with the inertia derived from scalar kinetic energy.

6.4.4. *Persistent discrepancy between the two inertia representations.* On the two reduced bases used in the open-port analysis, denoted N4 and N6, the reduced inertial operator obtained from the assembled DAE equations is compared directly with M_{energy}. The relative Frobenius discrepancies are

$$\delta_M^{\mathrm{N4}} = 0.022142497926477312, \qquad \delta_M^{\mathrm{N6}} = 0.023078143780171353,$$

and the corresponding total canonical-inertia errors are

$$\delta_I^{\mathrm{N4}} = 0.038512760543410185, \qquad \delta_I^{\mathrm{N6}} = 0.042711940525142079.$$

All four values exceed the relative tolerance of 0.005 and remain of comparable magnitude for the two bases. The assembled-equation inertia and the kinetic-energy metric therefore cannot be regarded as the same discrete object in the present representation. This observation does not invalidate the internal consistency of M_{energy}, H_{phys}, or the solver-inertia extraction; it identifies a persistent discrepancy between two otherwise internally consistent representations.

6.4.5. *Origin of the discrepancy and observability of the discrete mapping.* Two direct explanations are insufficient. First, the kinetic-energy Hessian already uses a velocity representation compatible with the solver shape functions, and repeated reconstruction gives the same M_{energy}; the tested simple coordinate/rate-carrier alternatives therefore do not account for the approximately 2.2%–2.3% discrepancy. Second, the assembled inertial operator is essentially invariant under the variable scalings examined here, so solver scaling is not the dominant source either.

Strict equivalence would additionally require the inertia mapping associated with the solver test space. The available interface exposes the discrete equations but does not provide enough weak-form dependency and quadrature information to reconstruct this mapping independently. Consequently, the test-space mapping J_{test} is not uniquely determined by the observable quantities, and $J_{\mathrm{test}} = J_{\mathrm{trial}}$ cannot be assumed without additional justification. We therefore do not enforce equivalence by empirical fitting or post hoc matrix modification. The present result is better interpreted as an observability limitation of the mapping between two discretizations than as an error in either COMSOL or the Hamiltonian construction.

6.4.6. *Reference variational discretization.* To determine whether the discrepancy is intrinsic to the canonical construction, a separate variational discretization is considered in which the trial functions, test functions, quadrature, and kinetic energy are all explicit. In this reference model, the relative Frobenius error between the kinetic-energy Hessian and the direct weak-form inertia is

$$1.0711869413158042 \times 10^{-16}.$$

Across 24 states, the maximum relative error between Euler–Lagrange inertia and direct weak-form inertia is

$$1.8326979829372372 \times 10^{-16},$$

and the canonical-to-weak inertia errors on the N4 and N6 bases are

$$r_{\mathrm{N4}} = 8.733849940345549 \times 10^{-16}, \qquad r_{\mathrm{N6}} = 6.28547098728893 \times 10^{-16}.$$

Thus, when all discrete objects originate from the same explicit variational construction, the inertia representations obtained from kinetic energy, the weak form, the Euler–Lagrange equations, and the canonical transformation agree to machine precision without empirical coordinate correction. The discrepancy observed for the Built-in Beam formulation is therefore not an unavoidable consequence of Hamiltonian canonicalization and is more plausibly associated with the particular finite-element representation and its incompletely exposed mapping.

6.4.7. *Relation of the reference discretization to the continuous* SO(3) *model.* The reference variational discretization is used only to isolate the source of discrete consistency and does not replace the geometrically exact rod model. The continuous theory treats $R(S,t) \in \mathrm{SO}(3)$ as an independent cross-sectional orientation, whereas the finite set of degrees of freedom in the reference model does not uniquely determine this orientation field along the rod. A strict correspondence would require additional assumptions on orientation interpolation and discretization.

Conversely, although the continuous theory specifies finite-rotation kinematics and the variational mechanical structure, it does not uniquely prescribe the interpolation functions, quadrature, cross-sectional parameters, or discretization of the Kirchhoff constraint. Additional numerical definitions are therefore required before the continuous model determines a unique reproducible discretization.

Taken together, the open-port results establish stable identifiability of the solver inertial response, internal consistency of the kinetic-energy Hamiltonian, and a persistent discrepancy between the solver and canonical inertia representations. The unresolved issue is the mapping between the two discretizations, which is not fully observable through the current interface. Until this mapping is independently validated, the present study does not construct corrected open-port force labels or report retraining and final CHNN performance based on such labels.

6.5. **Local Birkhoff normal form and KAM result.** After recentering the material-scalar conservative reference in the common modal coordinates, the four linear frequencies are 5.562997, 5.606265, 34.661765, and 35.092095 Hz. A fourth-order nonresonant Birkhoff normalization is performed for the selected quartic conservative approximation. The primary J_{ABC} twist matrix has rank four, and the three auxiliary matrices obtained by removing one modal group at a time are also full rank. Together with the analytic local canonical transformation, the exact $O(\|W\|^5)$ remainder, and the scaled $O(\varepsilon)$ perturbation, Proposition 5.1 yields a qualitative local existence result for a positive-measure Cantor family of Lagrangian invariant tori at sufficiently small nonzero amplitudes. The supporting numerical quantities and their domains of validity are summarized in Table 6.7.

Table 6.7. Birkhoff normal form and qualitative KAM analysis for the local quartic conservative approximation. The reported quantities support only the local conclusion for this approximation.

Item	Value or conclusion	Domain of validity
Low-frequency pair	5.562997, 5.606265 Hz	Recentered four-degree-of-freedom conservative reference
High-frequency pair	34.661765, 35.092095 Hz	Same conservative reference
Maximum full-Hamiltonian relative error on the local sample set	4.43237×10^{-8}	Finite-sample error estimate; not a uniform bound on a neighborhood
Quartic BNF homological residual	2.58368×10^{-80}	Numerical residual of the quartic homological equation
Primary J_{ABC} twist rank	4	Kolmogorov twist nondegeneracy
Primary dimensionless twist $\sigma_{\min}$	6.35822×10^{-11}	Magnitude of the weakest singular direction of the twist matrix
Resonance crossing within the prescribed action region	Present	No numerical KAM conclusion is drawn for the entire region
Local qualitative KAM existence	Established	Selected quartic conservative approximation only
Extension to the full material-reference Hamiltonian	Not established	Missing a uniform neighborhood bound for the $O(5+)$ remainder

This local existence result is not a numerical KAM certification over a finite action region. A finite-order resonance scan detects a resonance crossing within the region considered, so no KAM conclusion is drawn for the entire region. Moreover, no specific invariant-torus embedding, explicit ε_*, Diophantine constant, or a posteriori invariance residual is constructed here. The discrepancy between the full material-reference Hamiltonian and the quartic approximation has also been assessed only on a finite local sample set, and a uniform neighborhood bound for the $O(5+)$ remainder is not yet available. The KAM conclusion for the quartic approximation therefore cannot be promoted to a rigorous theorem for the full material-reference Hamiltonian.

6.6. **Parameter continuation, model discrepancy, and safe operation.** Future studies will continue the solutions with respect to magnetic-field strength, insertion speed, flow velocity, contact gap, and material parameters while monitoring structural response, port power, and discrepancies between the reduced and high-fidelity models. An engineering safe operating domain additionally requires contact limits, open-port power exchange, high-fidelity model discrepancy, and uncertainty propagation and should not be identified with the local theoretical domain of the conservative KAM analysis. These calculations are not yet complete, and no engineering safety domain is claimed here.

7. Discussion and scope

7.1. **Theoretical basis and methodological scope.** This study connects geometrically exact rod mechanics, finite-dimensional Hamiltonian reduction, structure-informed learning, and KAM analysis within a common modeling framework, but these theories act on different mathematical objects. The continuum model draws on geometrically exact rod theory and geometric mechanics, the numerical formulation uses concepts from geometric integration, and the learning model builds on HNNs, dissipative systems, and complex-valued networks. Continuous KAM theory, KAM theory for symplectic maps, and parameterization methods address conservative flows, symplectic discretizations, and invariant tori, respectively[1, 2, 17–29, 33, 38]. A structural property established at one level therefore does not automatically supply the assumptions required at another. In particular, a finite-dimensional Hamiltonian learning model does not by itself establish the hypotheses of KAM theory for the underlying multiphysics continuum system, and consistency of a software implementation does not replace the mathematical definitions required by continuum or variational mechanics. Moser's annulus theorem, infinite-dimensional KAM theory, metriplectic formulations, and specialized complex reaction-network results are used here to delimit related settings rather than as governing theorems for the finite-dimensional catheter model[30–34].

The COMSOL–Galerkin diagnostics show that structural consistency must be examined separately in configuration, tangent, and cotangent spaces. A small numerical residual is not meaningful if the compared quantities represent different physical objects. Momentum must be defined through the appropriate cotangent map rather than replaced by an arbitrary mass-matrix–velocity product, and states, velocities, and generalized forces must refer to the same solver time and discrete interpretation. Normalization, sign conventions, and the transformation of the initial state must also be consistent with the chosen fixed basis. Only after these representation issues have been excluded can time-step refinement be used to isolate temporal discretization error. Perturbation-energy drift and generalized-force closure therefore test different properties: the former measures energy consistency along a trajectory, whereas the latter tests force balance in the state equations.

The open-port calculations reveal a related issue at the interface between engineering finite-element discretization and canonical Hamiltonian mechanics. Reproducible transient states do not imply that canonical generalized forces or inertia reconstructed from those states are mechanically equivalent to the solver's internal discrete representation. Two independent extractions of the solver inertial operator agree closely, yet a stable discrepancy remains between that operator and the canonical inertia metric derived from scalar kinetic energy; the tested simple coordinate/rate-carrier alternatives and variable scaling do not explain the dominant difference. Strict equivalence would additionally require the inertia mapping associated with the test space, which cannot be reconstructed uniquely from the present solver interface. By contrast, a reference variational discretization with explicit trial functions, test functions, quadrature, and kinetic energy recovers agreement among the alternative inertia representations to machine precision. This comparison indicates that the Hamiltonian construction is not intrinsically inconsistent; rather, equivalence between distinct discretizations requires an explicit and verifiable mapping.

7.2. **Scope of the principal results.** The model comparisons operate at two levels. P0 and the P1/P1.5/P1.6 series examine the effects of network structure and physical constraints in

low-dimensional or small-sample settings, whereas the 96 COMSOL trajectories provide a fixed conservative task for comparing four trainable models. The trajectories are partitioned by complete trajectory into 64 training, 16 validation, and 16 independent-test cases. Each model is trained from five random initializations, and model selection is based only on validation data. Gray CHNN gives substantially lower one-step and long-horizon errors than the three baselines on this task, indicating that the combination of a physical Hamiltonian and a gray-box residual is effective for the present data distribution and reduced representation. This is a task-specific empirical result and does not establish superiority at arbitrary modal orders, excitation conditions, or open-port regimes.

The open-port result has a different interpretation. In the two reduced subspaces used in the analysis, N4 and N6, the inertia obtained from the assembled DAE equations differs from the canonical inertia metric derived from scalar kinetic energy by approximately 2.21% and 2.31%, respectively; the corresponding total canonical-inertia errors are approximately 3.85% and 4.27%. Because the solver inertia itself can be extracted reproducibly, and the tested simple coordinate/rate-carrier alternatives together with solver scaling do not explain the dominant discrepancy, the evidence points to an unresolved mapping between discrete representations rather than to simple numerical instability or a scaling error. The current interface does not uniquely expose the required test-space inertia map. Consequently, the present results do not provide a validated open-port canonical force mapping or justify retraining from such labels. The reference variational discretization shows that machine-precision consistency is attainable when the discrete ingredients are explicit, but it does not establish that the reference discretization is identical to the engineering realization of the continuous SO(3) beam model used here.

The KAM result is likewise local. A nonresonant Birkhoff normal form is constructed for the quartic conservative approximation, and the associated frequency map satisfies the Kolmogorov nondegeneracy condition. These properties support a local positive-measure Cantor family of Lagrangian invariant tori for sufficiently small nonzero amplitudes. A uniform higher-order neighborhood bound between the material-reference Hamiltonian and the quartic approximation is not yet available, so the result cannot be promoted to a rigorous KAM statement for the material-reference system. Similarly, BDF is used only to obtain temporally consistent states and vector-field data and has not been shown to be symplectic. The Lean4 formalization verifies several finite-dimensional structural identities and port-power relations but does not establish the complete physical model, dissipative-KAM persistence, or an engineering safe domain. For a general $\mathbf{D}_\theta(\mathbf{Z}) \succeq 0$, the present result guarantees only nonpositive instantaneous dissipative power; $\mathbf{D} = \gamma\mathbf{I}$ represents only one particular weak-damping model.

7.3. **Directions for further study.** Future work should first improve traceability between the continuum model and its numerical realization. Strict weak-form discretizations, Timoshenko beams, and three-dimensional solid models can be used to quantify high-fidelity model discrepancy, but a reproducible benchmark must specify the discrete representation of cross-sectional orientation, interpolation functions, quadrature, and treatment of the Kirchhoff constraint. An unobservable test-space map should not be fitted merely to force agreement with the canonical inertia. Structure-informed learning and dynamic-balance evaluation under open-port conditions become mechanically interpretable only after the mappings from solver states, inertia, and generalized forces to canonical variables have been validated independently.

The next step for numerical KAM analysis is likewise not to repeat the local normal-form calculation, but to construct specific invariant tori and provide independently checkable numerical evidence, such as frequency-diffusion measurements, Poincaré sections, a posteriori invariance residuals, and rigorous higher-order bounds for the material-reference system. The sufficiency of the principal modal order, statistical-learning error bounds, and high-fidelity model discrepancy should also be quantified over a broader range of operating conditions. These developments will determine whether the local structural results obtained here can ultimately support stability criteria for open-port prediction and engineering design[4–8, 10, 11, 30, 32].

8. Conclusions

This work develops a unified framework for the multiphysics large-deformation dynamics of magnetically actuated flexible catheters, connecting a geometrically exact continuum model, joint Galerkin reduction, a canonical Hamiltonian formulation, structured CHNNs, and KAM stability analysis. The continuum description retains independent cross-sectional orientation, torsion, and rotary inertia and separates conservative effects, passive dissipation, background flow, and control through an energy–power balance. The continuum model, finite-element data, and reduced learning system therefore share a consistent state and energy representation.

The learning formulation uses a real-valued physical Hamiltonian for the conservative dynamics and represents conservative corrections, dissipation, and external inputs separately. The low-dimensional and small-sample experiments show that the effect of structural constraints on prediction is task dependent: residual correction can markedly improve a model based only on the prescribed physical terms, whereas exact satisfaction of a local equilibrium constraint does not necessarily reduce prediction error. For the fixed conservative task constructed from 96 COMSOL trajectories, the data are partitioned by complete trajectory into 64 training, 16 validation, and 16 independent-test cases. Each of the four models is trained from five random initializations, and the final model for each architecture is selected using validation data only. Gray CHNN achieves a normalized full-state one-step MSE of 1.62675×10^{-3} and an 800-step rollout state error of 4.57008, reductions of approximately 98.20% and 99.07%, respectively, relative to the best Real HNN under the same conditions. These results establish a substantial advantage for Gray CHNN on the present reduced representation and data distribution, but they do not imply general superiority at other modal orders, excitation conditions, or open-port regimes.

The COMSOL–Galerkin diagnostics further show that a reliable data mapping must preserve consistency among configuration, tangent, and cotangent quantities while separating representation error from temporal discretization error. The three-level time-step study gives observed convergence orders of 1.64186–2.00088. In the extended 1 s case, the joint, pointwise, and integrated relative-L2 closure errors are 0.00146226, 0.000221162, and 0.0000563606, respectively, while the peak-to-peak perturbation-energy drift is $0.00307057 < 0.005$. These results provide numerical support for the state, momentum, and generalized-force mapping used in the fixed conservative task.

The open-port analysis shows that the discrete inertia represented by an engineering finite-element solver cannot be assumed a priori to coincide with the canonical inertia derived from scalar kinetic energy. Two independent extractions of the solver inertial operator agree closely, yet discrepancies of approximately 2.21% and 2.31% remain in the N4 and N6 reduced subspaces, with corresponding total canonical-inertia errors of approximately 3.85% and 4.27%. The tested simple coordinate/rate-carrier alternatives and solver scaling are insufficient to explain the dominant difference, and the test-space inertia map required for strict equivalence cannot be reconstructed uniquely from the present interface. In a reference variational discretization for which the trial functions, test functions, quadrature, and kinetic energy are explicit, the kinetic-energy Hessian, weak-form inertia, Euler–Lagrange inertia, and canonical inertia agree to approximately 10^{-16}. Thus, an explicit mapping between discrete representations is a necessary ingredient for identifying solver dynamics with canonical Hamiltonian dynamics. The present evidence is insufficient to define a validated open-port canonical force mapping, so no retraining from such labels or final open-port CHNN performance is reported.

For the KAM analysis, a fourth-order nonresonant Birkhoff normal form is constructed for the quartic conservative approximation near equilibrium, and the primary J_{ABC} twist matrix is shown to have full rank. Together with the exact $O(\|W\|^5)$ remainder and the scaled $O(\varepsilon)$ perturbation, these properties imply a local positive-measure Cantor family of Lagrangian invariant tori for sufficiently small nonzero amplitudes. This conclusion applies only to the quartic approximation. A uniform higher-order neighborhood bound relating the material-reference Hamiltonian to that approximation is not yet available, so a rigorous KAM result for the material-reference system does not follow.

Overall, the study establishes testable links among continuum mechanics, finite-dimensional canonical structure, structure-informed learning, and local KAM analysis, while identifying the additional mapping information required to connect an engineering finite-element discretization to canonical dynamics. The symplecticity of BDF, numerical or a posteriori certification of specific invariant tori, a closed high-fidelity model-discrepancy analysis, and an engineering safe operating domain remain open. The Lean4 formalization provides machine-checked support for several finite-dimensional structural identities and port-power relations but does not replace validation of the complete physical model or engineering stability. Future work will focus on traceable high-fidelity discretizations, validated canonical mappings for open-port states and generalized forces, and a posteriori certification of specific invariant tori.

Appendix A. Diagnostic Methods for the Canonical Data Contract

This appendix organizes the canonical data-contract diagnosis by error source and scientific interpretation.

Table 1.1. Diagnosis of anomalies and identification of error sources in the canonical-variable mapping.

Diagnostic step	Observation and analysis	Scientific interpretation
Same-trajectory comparison	After removing a cross-trajectory comparison, mode1 met the criterion whereas mixed did not.	The apparent positive result was caused by inconsistent comparison objects.
Repeated single-mode case	A repeated mode1 case could also exhibit closure error.	Modal mixing was not necessary for the anomaly.
Cotangent projection	Insufficient $N = 2$ truncation was not the principal cause, and the cotangent-projected momentum gave the smaller closure error.	Momentum should be projected to the cotangent space of the fixed basis.
Solver-output semantics	The internal generalized-alpha stage state was unavailable through the public API, whereas BDF `tsteps` preserved states at the same solver time step.	BDF is used only for state sampling and label generation.
Fixed-basis initialization	Applying the same normalization and sign transformation to displacement and velocity produced 12/12 consistent initial conditions and short-window closure for the mixed case.	The anomaly arose from inconsistent initialization in the fixed basis.
Two-level time-step comparison	The joint relative-L2 closure error increased from 0.00147337 to 0.00155680.	Two levels did not meet the predefined monotonic-convergence criterion.
Three levels and 1 s validation	Observed orders were 1.64186–2.00088, and the extended-case closure and energy quantities satisfied the prescribed thresholds.	The canonical-variable mapping is sufficiently consistent for the subsequent multi-amplitude data generation.

The diagnosis successively examines comparison objects, modal mixing, reduced-order truncation, the cotangent-momentum definition, solver-output semantics, and fixed-basis initialization. The false positive caused by a cross-trajectory comparison shows that closure errors must be computed on the same physical trajectory and at the same sampling times. A repeated single-mode case can still exhibit the anomaly, so the error cannot be attributed directly to modal mixing. Increasing the reduced representation did not reveal the signature of an $N = 2$-truncation error; projected momentum closed more accurately than $M_N \dot{q}_{\text{native}}$.

Because the internal generalized-alpha stage is unavailable through the public API, exported displacement, velocity, and load cannot be guaranteed to correspond to the same internal solution stage. Order-2 BDF with `tsteps` is used to obtain states and vector-field labels at the same solver step. This choice does not imply that BDF is symplectic and does not interpret the discrete trajectory as a symplectic map.

Force closure and perturbation-energy drift use independent criteria. An earlier force comparison was sensitive to endpoints and the momentum anchor. On common-support indices $2 \ldots 998$, with a fixed $p[2]$ anchor, the force-v2 joint L2 is below 2% for all three trajectories. This result does not alter the trajectory-level conclusions obtained with the original 0.5% Bridge perturbation-energy criterion.

Appendix B. Metric Definitions and Numerical Settings

B.1. **Canonical mapping and closure quantities.** For the fixed $N = 2$ basis, the mapping is

$$q = \Pi_{\text{conf}}(U), \qquad p = B^T P_{\text{full}}, \tag{2.1}$$

$$\dot{q} = M_N^{-1} p, \qquad \dot{p} = B^T L. \tag{2.2}$$

Here q is a configuration coordinate and p is its associated cotangent coordinate. Closure errors compare the exported $\dot{q}$ with $M_N^{-1}p$ and the exported $\dot{p}$ with $B^T L$, while the time-discretization study also examines q, p, and total energy.

The three-level study uses coarse, middle, and fine grids. If the errors on two adjacent levels are e_h and $e_{h/2}$, the observed order is

$$r_{\text{obs}} = \log_2\left(\frac{e_h}{e_{h/2}}\right).$$

The resulting observed orders span 1.64186–2.00088. In the 1 s case, the thresholds for G2, pointwise G3, and impulse G3 are all 0.02, and the peak-to-peak perturbation-energy-drift threshold is 0.005.

B.2. **Bridge and low-dimensional network experiments.** Bridge perturbation-energy drift is compared trajectory by trajectory with the 0.5% criterion and the exploratory 1% criterion. Force-v2 uses an independent 2% joint-L2 criterion. The energy and force quantities are not combined into a single conclusion.

P0 uses a four-dimensional nonlinear coupled oscillator. The training, validation, and test sets contain 80, 16, and 20 trajectories, respectively; each trajectory has 80 steps, $\Delta t = 0.025$, and the models are trained for 300 epochs. Five random seeds estimate algorithmic repeatability.

P1 and P1.5 each comprise 60/60 completed training-and-evaluation runs, and P1.6 comprises 30/30 hard-constrained models. The three physical trajectories—mode1, mode2, and mixed12—are partitioned by leave-one-whole-trajectory-out. Different random seeds share the same physical trajectory set and therefore do not increase the physical sample size. P1.5 compares physics-only and residual learning, whereas P1.6 compares predictive accuracy and structural residuals before and after imposing the hard equilibrium constraint.

B.3. **Formal 96-trajectory learning protocol and reproducibility boundary.** The formal data set contains 96 prospective COMSOL trajectories, each covering 0.2 s with $\Delta t = 2.5 \times 10^{-4}$ s and 801 sampled states. Before any formal training, the split is frozen by whole trajectory as 64/16/16. Test numerics are first loaded only after the four candidate models have been frozen from the validation primary metric. Each of the four trainable architectures uses five seeds and 300 epochs, giving 20/20 completed formal training runs. Final testing performs only forward/rollout evaluation and metric aggregation; it performs no `backward` call, optimizer update, checkpoint modification, post-test model reselection, or tuning.

The formal reproducibility package retains the data-set manifest, split, trajectory-level QA, training run registry, candidate-checkpoint SHA-256 identities, the pre-test selection freeze, aggregate and per-case final-test metrics, parameter hashes before and after testing, formal code provenance, and the runtime environment. The large raw COMSOL/MAT data and the complete historical checkpoint collection are stored as separate primary archives; their omission from the compact supplement does not replace the retained identity and hash evidence.

The Open-Port branch has established trajectory-level energy/power QA, R3 passive-dissipation-sign evidence, solver-level exact weak-branch qualification, and independent solver-native inertial differential qualification. Exact equivalence between the current canonical energy metric and the Built-in Beam solver-discrete inertial operator, however, has not been established within the frozen qualification gate. Complete D006 dynamic balance and a final corrected canonical-label release therefore remain unqualified; any future corrected-label release must also report complete port-leakage and power-closure metrics. High-fidelity studies must quantify the discrepancy of the Kirchhoff model relative to Timoshenko-beam and solid models.

The quartic BNF/twist and qualitative KAM existence for the local exact quartic surrogate have now been backfilled. Further numerical KAM certification would still require specific-torus Diophantine/a posteriori constants, a high-order material-transfer error bound, symplectic-map defects, and invariance residuals in a common norm.

Appendix C. Open-Port Lean4 Structural-Theorem Matrix

This appendix records only the structural results directly supported by the present Lean4 audit and their boundaries. Lean4 is used as a theory-audit tool; it is not a claim that the complete physical model, complete CHNN training pipeline, dissipative KAM problem, or material-reference system has been formally proved. Statuses follow the frozen R4 Open-Port Lean4 audit archived with this release under `evidence/r4_open_port_audit/`, together with the formalization release under `evidence/lean_formalization_20260812/`.

Table 3.1: Open-Port Lean4 structural-theorem matrix.

Claim	Lean status	Scope	Boundary
Hamiltonian internal power cancellation	PROVED / PROVED_CONDITIONAL	finite-dimensional canonical structure; gradient witness for paper bridge	not full physical provenance
Canonical port power	PROVED_CONDITIONAL	canonical momentum injection and named ports	derivative/vector-field witnesses required
Factorized dissipative power non-positive	PROVED / PROVED_CONDITIONAL	$\mathbf{D} = \mathbf{L}\mathbf{L}^T \succeq 0$ branch	instantaneous power only; not dissipative KAM
Named open-port power split	PROVED_CONDITIONAL	autonomous six-port balance with stated regularity	interval integrability and ODE semantics remain explicit
Complex-to-real Hamiltonian readout	PROVED / contract-closed	frozen CHNN representation	not training or nonlinear-source proof
Magnetic no-double-counting	PROVED	exclusive potential/port constructors	induced-force witness on potential branch
Perturbation type separation	PROVED	Hamiltonian, dissipative, external and residual channels	no KAM norm bridge
General PSD damping and weak branch	PROVED_CONDITIONAL	general PSD sign; designated $\mathbf{D} = \gamma\mathbf{I}$, $\gamma \geq 0$ branch	no conformally-symplectic identity
Quartic BNF and twist rank 4	NUMERICAL_CERTIFICATE_REQUIRED	R3 exact quartic conservative surrogate	uncertainty and all-order bounds remain
Classical KAM theorem	EXTERNAL_THEOREM	exact quartic surrogate under external hypotheses	not material/open-system KAM
Dissipative KAM	NOT YET CERTIFIED	future weak-damping branch	no current Lean certification
Material-reference KAM transfer	BLOCKED / NOT QUALIFIED	full material/open system	missing rigorous neighborhood and norm transfer

Here, `PROVED_CONDITIONAL` means that the Lean theorem holds while the paper-level bridge still requires explicit gradient, derivative, integrability, or physical-provenance witnesses. `NUMERICAL_CERTIFICATE_REQUIRED` and `EXTERNAL_THEOREM` are not Lean proofs, and `BLOCKED`

denotes a missing paper-level norm or transfer certificate. This matrix does not alter the R3 quartic-surrogate, BNF, twist-rank-4, or existing numerical results.

Appendix D. Open-Port Numerical Qualification Matrix

This appendix records the Open-Port numerical qualification status used by the second formal delta writeback. The scientific narrative in Section 6.4 is organized by qualification object; stage identifiers are retained here only as provenance indices. The previous R92B cutoff and all historical final reports, hashes, manifests, and statuses remain immutable. The present evidence boundary is extended only through G1, covering the R92C–G1 delta; no claim beyond G1 is admitted.

Table 4.1: Open-Port numerical qualification matrix through the second-writeback G1 cutoff.

Qualification object	Evidence stage	Status	Paper interpretation
Trajectory-native generator	R26/R43/ R57/R63–R65	QUALIFIED	solver-admissible trajectory generation; not force-label qualification
60-case production QA	R63–R64	QUALIFIED	60/60 production freeze and dependency/re-hash audit
Candidate release integrity	R65	QUALIFIED	split 36/9/6/3/6, no trajectory leakage; historical candidate DB
Canonical force-label semantics at release	R65	NOT QUALIFIED	required subsequent solver-level semantic audit
T1–T4 execution/evaluation integrity	T1–T4	QUALIFIED	frozen protocol and one-time evaluation; diagnostic evidence
T1–T4 final model performance	T1–T4	NOT QUALIFIED	not final Open-Port scientific model evidence
Physical kinetic metric M_{energy}	R80	QUALIFIED	scalar kinetic-energy Hessian; raw `E/Ec` remain DAE operators
Physical Hamiltonian H_{phys}	R81	QUALIFIED	$T + V_{\text{elastic}} + V_{\text{mag}}$ reconstruction
Residual/`Reacf` observability	R86R	QUALIFIED	stored solver evidence recovered and reopened
BDF2 state reconstruction	R87	QUALIFIED	stored $\dot{U}$ reconstructed from accepted BDF history
Official state-preserving Assemble route	R90A	QUALIFIED	formal extraction route
Elastic/Magnetic/R3 weak semantics	R90E/R91	QUALIFIED	exact branch and additivity audits
Inertia sign-domain mapping $S_I = -1$	R92B	QUALIFIED	historical direction ambiguity resolved; R92B status immutable
Solver-native inertia differential branch	R92F	QUALIFIED	differential native branch closes; zero-rate offset impact bounded
Official DAE vs current energy metric	R92G	NOT QUALIFIED	N4/N6 metric mismatch $0.0221425/0.0230781 > 0.005$; terminal R92 attribution
Current canonical vs solver inertia equivalence	R92G	NOT QUALIFIED	N4/N6 inertia mismatch $0.0385128/0.0427119 > 0.005$
Simple coordinate/rate-carrier explanation	C1	DIAGNOSTIC	excluded; R80 metric already uses solver-shape-native carrier
Solver-scaling explanation	S1	DIAGNOSTIC	excluded; official E invariant under audited scaling controls

Qualification object	Evidence stage	Status	Paper interpretation
Native test-side mapping	W1	NOT OBSERVABLE / TERMINATED	exact mapping unavailable; no fitted/empirical repair permitted
Transparent variational inertia control	B1	QUALIFIED	energy/weak/EL/canonical inertia close at numerical floor; control only
Transparent conservative manuscript extension	B2	NOT QUALIFIED	theory compatibility requires new orientation/discretization assumptions; B3 not allowed
Manuscript-native transparent benchmark	G1	UNDERSPECIFIED / NOT QUALIFIED	continuous theory does not uniquely freeze reproducible numerical discretization; G2 not allowed
Complete D006 dynamic balance	through G1	NOT QUALIFIED	no complete full-balance claim
11-case formal qualification / R93	through G1	TERMINATED / NOT ALLOWED	prerequisites not met; no R93
Corrected-label retraining / repaired TEST/KAM	through G1	NOT QUALIFIED	not executed; no final Open-Port model-performance claim

The R65 status is intentionally layered: trajectory-generation/production QA and candidate-release integrity are qualified, whereas canonical force-label semantics were not qualified at release time. Likewise, T1–T4 execution/evaluation integrity is qualified, but those model results are retained as diagnostic rather than final Open-Port performance evidence.

R92B evidence note. The authoritative inertia interpretation uses the frozen numerical data and final status: the N4 and N6 sign-mapped frozen-dual relative residuals are 0.0192631 and 0.0213627, respectively, both above the 0.005 gate. A historical R92B final-report item contains the contradictory wording "not exceeding 5×10^{-3}"; this is treated as a non-authoritative wording typo. The historical evidence file is not modified, and the manuscript uses the numerical values and FAIL/UNRESOLVED status only.

Second-delta evidence note. The second writeback preserves the earlier qualified R80/R81/R91 and R92B statuses while adding independent solver-native inertia, attribution, exclusion, observability, and transparent-control evidence. R92F qualifies the solver-native inertial differential branch and bounds the zero-rate offset. R92G terminates the R92 series by showing that the official DAE metric is not equivalent to the current energy metric within the frozen gate; no R92H is created. C1 and S1 exclude simple coordinate/rate-carrier and solver-scaling explanations. W1 terminates the Built-in Beam canonical-repair line because the exact native test-side mapping is not observable through the audited interface, with fitted or empirical recovery explicitly disallowed. B1 is a positive transparent variational inertia control with numerical-floor closure, but B2 prevents its conservative extension from being identified with the manuscript theory without new assumptions. G1 then records that the manuscript continuous SO(3) theory does not uniquely freeze the numerical interpolation, quadrature, constraint-discretization, and benchmark data needed for a no-new-assumption transparent implementation. These are qualification and reproducibility boundaries; they do not modify the continuous theory or Lean proof surface.

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